NAME OF WORK : REPAIR /MAINT OF LT PANELS, SWITCHGEARS, LT CABLES ALONGWITH CONNECTED ITEMS OF DIFFERENT OTM ACCNS AND REPAIR MAINT OF CERTAIN MISC B/R AND E/M WORKS OF DIFFERENT ELECTRIC SUPPLY INSTALLATIONS AT INSIDE FORT WILLIAM UNDER GE (FW) KOLKATA

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Drawings :- Nil- Sheets Total _____ Pages

SIGNATURE OF CONTRACTOR
Dated :

AGE (CONTRACTS)
FOR ACCEPTING OFFICER

SPEED POST
Garrison Engineer (FW) Kolkata
Military Engineer Services
D-9, Commissariat Road
Hastings, Kolkata - 700 022.

8373/ /E8 Aug 2025

M/s/Shri _____

_____

_____

NAME OF WORK : REPAIR /MAINT OF LT PANELS, SWITCHGEARS, LT CABLES ALONGWITH CONNECTED ITEMS OF DIFFERENT OTM ACCNS AND REPAIR MAINT OF CERTAIN MISC B/R AND E/M WORKS OF DIFFERENT ELECTRIC SUPPLY INSTALLATIONS AT INSIDE FORT WILLIAM UNDER GE (FW) KOLKATA

Dear Sir (s),

1. Tender documents in respect of above work are uploaded on the site **www.defproc.gov.in**. The tender is based on single stage two cover/two stage three cover e-tendering system. The contents of Cover I & Cover II are specified in NOTICE OF TENDER.

2. Bids will be received online by ACCEPTING OFFICER upto the date and time mentioned in the **NOTICE INVITING TENDER (NIT)**. No tender/bid will be received in physical form and any tender/bid received in such manner will be treated as non bona fide.

3. Bid will be opened on line on a due date fixed for opening as per critical dates given in the portal. Cover 1 will be opened first. Tenderers may see the result of the opening of Cover 1 on the tender portal. Date of opening of Cover 2 shall be decided after technical evaluation of Cover 1 keeping in view the various eligibility criteria given in the NIT and the same will be intimated to the tenderers while uploading the technical evaluation on the tender portal.

4. Your attention is also drawn to instruction on filling and submission of tender attached herewith. You may forward your points on tender documents through e-mail and/or depute your technical representative for discussion on tender/drawings and to clarify doubts, if any, before the bid submission starts. You are requested not to write piece meal points and forward your points duly consolidated in one go. You may also attend the pre bid meeting on the date given in the tender.

5. Unenlisted contractors are required to submit the scanned copies (in pdf file) of documents required as per eligibility criteria mentioned in instructions for filling the tender documents and Appendix 'A' to NIT alongwith EARNEST MONEY DEPOSIT (EMD) and tender fee on procurement portal mentioned above and submit the physical documents in the office of **Garrison Engineer (fort William), Kolkata** within time limit specified in NIT. Inadequacy /deficiency of documents shall make the bid liable for rejection resulting in disqualification for opening of finance bid.

6. Enlisted contractors of MES shall submit the scanned copies (pdf file) of enlistment letter, tender fee and such other documents as mentioned in Appx 'A' to NIT on e-procurement portal and submit physical documents in the office of **Garrison Engineer (Fort William), Kolkata** before date & time fixed for this purpose.

Contd.....

7. The contractor must ensure that the tender/bid is uploaded on the tender portal using the authorised DSC in time as the Accepting Officer will take no cognizance of any quotations/offer received in any other electronic or physical form like email/fax/by hand/through post from tenderer/bidder.

8. Keeping in view delays due to system failure or other communication related failures, it is suggested that tender/bid may be uploaded sufficiently in advance of the last due date and time fixed on the portal.

9. General Conditions of Contracts (IAFW-2249) (1989 Print) and errata and amendments thereto, Schedule of Minimum Fair Wages and MES SSR (Part-I and Part-II) are not enclosed with these documents. These are available for perusal in the Office of GE/AGE concerned and this office. Tenderer is also requested to keep the copy of these documents with him as same are available in the market.

10. PERFORMANCE SECURITY DEPOSIT After acceptance of the Tender, the contractor will be required to lodge with the Accepting Officer PERFORMANCE SECURITY DEPOSIT @ 5 % of contract amount. The amount is required to be lodged within 28 (Twenty eight) days of the receipt by the contractor of notification of acceptance of tender/bid, failing which action as stipulated in Condition 19.3 of GCC shall be taken.

11. All bidders shall be required to sign the Bid Securing Declaration as below:-

BID SECURING DECLARATION

I/We hereby understand and accept that if I/We withdraw or modify my/our bids during the period of validity, or if I/We are awarded the contracts and on being called upon to submit the Performance Security/Security Deposit, fail to submit the Performance Security/Security Deposit before the deadline defined in the request for bid documents/Notice Inviting Tender, I/We shall be debarred from exemption of submitting Bid Security/Earnest Money Deposit for a period of 6 (six) months, from the date I/We are declared disqualified from exemption from submission of Earnest Money Deposit/Security Deposit, for all tenders issued by MES during this period.

Signature of Contractor **AGE (Contracts)**
Date: **For Accepting Officer**

12. ANY TENDERER, WHICH PROPOSES ALTERATIONS TO ANY OF THE CONDITIONS, SPECIFICATIONS LAID DOWN IN THE TENDER DOCUMENTS OR ANY NEW CONDITION, WHATSOEVER, IS LIABLE TO BE REJECTED.

Yours faithfully,

SIGNATURE OF CONTRACTOR **AGE (CONTRACTS)**
Dated : **FOR ACCEPTING OFFICER**

INSTRUCTIONS FOR COMPLETION OF TENDER DOCUMENTS TO BE COMPLIED WITH BY THE TENDERER(S)

1. **EARNEST MONEY DEPOSIT (EMD)**

Contractor(s) who are not enlisted with MES/who are enlisted but have not executed the Standing Security Bond shall submit Earnest Money Deposit as detailed in Notice of Tender in one of the following forms, along with their tender/bid:-

(a) Deposit at Call Receipt from a Scheduled Bank in favour of GE/AGE(I) concerned.

(b) Receipted Treasury Challan, the amount being credited to the Revenue Deposit of GE/AGE(I). It is advisable that Earnest Money is deposited in the form of deposit call receipt from an approved Schedule Bank for easy refund. In case the tenderer/bidder wants to lodge 'Earnest Money Deposit' in any other form allowed by MES, a confirmation about its acceptability will be obtained from the Accepting Officer well in advance of the bid submission end date and time. Earnest Money Deposit shall be submitted in the name of concerned GE/AGE(I).

NOTES:- Earnest Money Deposit (EMD) in the form of cheque/Bank Guarantee etc will not be accepted. **Non-Submission of Earnest Money Deposit (EMD) (scanned copy alongwith Technical Bid & hard copy before the date & time fixed for opening of BOQ) will render the bid disqualified for opening of Cover -II (finance bid).**

2. **GENERAL INSTRUCTIONS FOR COMPLIANCE**

2.1 The bids received only in the electronic form will be considered. All bids shall be submitted on defproc.gov.in portal. Documents should be scanned and forwarded in 'pdf' format and 'xls' form as indicated. Bids shall be uploaded on 'defproc.gov.in' portal on or before the bid closing date mentioned in the tender. No tender/bid in any other electronic or physical form like date email/fax/by hand/through post will be considered.

2.2 The bid shall be DIGITALLY signed using authorised DSC. All pages of tender documents, drawings, corrections/alterations shall be signed/initialed by the lowest bidder after acceptance of tender for making original and CTC of the contract.

2.3 Digital signatory of the bid/tender shall ensure that he is competent to bind the contractor (through partnership deed, general power of attorney or Memorandum and Article of Association of the Company) in all the matters pertaining to the contract with Union of India including arbitration clause. A scanned copy of the documents in confirmation of such authority shall be attached with the tender/bid in 'pdf' form in Cover 1. It shall be ensured that power of attorney shall be executed in accordance with the constitution of the company as laid down in its Memorandum & Article of Association.

2.4 Drawings, if issued in physical form, must be returned duly initialled by the tenderer/bidder in separate envelope indicating his name and address.

2.5 The tender shall be signed, dated and witnessed at all places provided for in the documents after acceptance. All corrections shall be initialled. The Contractor shall initial every page of tender and shall sign all drawings forming part of the tender. Any tender/bid, which proposes alterations to any of the conditions whatsoever, is liable to be rejected.

Contd….

INSTRUCTIONS FOR COMPLETION OF TENDER DOCUMENTS TO BE COMPLIED WITH BY THE TENDERER(S) (Contd...)

3. **BOQ**

3.1 The tenderer shall quote his rates on the BOQ EXCEL FILE only as per guide line of e-procure web portal. No alteration to the format will be accepted and such bid will be disqualified.

3.2 In case any tenderer wishes to revise/modify the rates quoted in the BOQ file, he can do so only in the BOQ files before uploading the tender through **https://defproc.gov.in** site only before closing date & time.

3.3 After the uploading of tender, Department may upload the errata/ amendment through corrigendum. The tenderer/bidder should submit their offer considering the errata/amendment carried out through corrigendum issued from time to time.

3.4 While uploading the bid, the tenderers/bidders should specifically check whether any **revised BOQ** has been uploaded by department through corrigendum prior to **Bid submission start date**. Tenderers/bidders attention is specifically drawn to the fact that they should submit their offer on revised BOQ only. In case any tenderer/bidder submits offer on pre-revised BOQ in lieu of Revised BOQ, **it will be considered as a willful negligence by the tenderer/bidder and quotation shall be considered non-bonafide**.

4. Tenders/bidders who uploaded their priced tenders/bids and are desirous of being present at the time of opening of the tenders/bids, may do so at the appointed time.

5. Hard copies of all above documents shall be sent by the contractor to the Tender issuing authority well in advance to be received before the date & time fixed for the same.

6. The contractor shall employ Indian Nationals after verifying their antecedents and loyalty. Attention is also drawn to relevant provisions in Special Condition referred here-in-after and also Conditions 24 & 25 of IAFW-2249 (General Conditions of Contract).

7. These instructions shall form part of the contract documents.

SIGNATURE OF CONTRACTOR **AGE (CONTRACTS)**
FOR ACCEPTING OFFICER

Dated :

MILITARY ENGINEER SERVICES
NOTICE OF TENDER IAFW-2162

APPLICABLE FOR MEASUREMENT CONTRACTS

1. A tender is invited for the work as mentioned in Appendix `A' to this NOTICE INVITING TENDER (NIT).

2. The work is estimated to cost as indicated in the aforesaid Appendix 'A'. This estimate, however, is not a guarantee and is merely given as a rough guide and if, the work costs more or less, a tenderer/bidder will have no claim on this account.

3. The work is to be completed within the period as indicated in aforesaid Appendix `A' in accordance with the phasing, if any, indicated in the tender from the date of handing over site, which will be on or about two weeks after the date of Acceptance of tender.

4. Contractors whose names are on the MES approved list and within whose financial category the estimated amount would fall and unenlisted contractors may submit tender/bid subject to other criteria mentioned in Appendix A. However in case of term contracts, enlisted contractors of Class SS to E may submit tender. Not more than one tender shall be submitted/uploaded by one contractor/ firm. Under no circumstances will a father and his son(s) or other close relations who have business dealing with one another be allowed to tender/bid for the same contract as separate competitors. Two firms shall be deemed to have business dealing if any of the partners/proprietor/director is common among both of them. A breach of this condition will render the tenders/bids of both the parties liable for rejection.

5. The office of Garrison Engineer (Fort William), Kolkata will be the Accepting Officer here-in-after referred to as such for the purpose of this contract.

6. Not more than one tender/bid shall be submitted/uploaded by one bidder firm. Under no circumstances will a father and his son(s) or other close relations who have business dealing with one another be allowed to tender/bid for the same tender as separate competitors. A breach of this condition will render the tenders/bids of both the parties liable for rejection.

7. The Technical Bid and Financial Bid (Cover-1 and Cover-2) shall be uploaded by the contractor on or before the date & time mentioned in **NIT**. A scanned copy of DD with enlistment details and other documents as specified in Appendix A shall be uploaded as Cover-1 (Technical bid) of the tender on e-tendering portal. DD is refundable in case the contractor is not considered eligible in technical evaluation of Cover 1 resulting in non-opening of Cover1. The applicant contractor shall bear the cost of bank charges for procuring and encashing the DD including revalidation of DDs and shall not have any claim from Government whatsoever on this account.

8. Tender form and conditions of contract and other necessary documents shall be available on eprocuremes.gov.in/eprocurement.gov.in site for download and shall form part of contract agreement in case the tender/bid is accepted.

9. In case of MES enlisted contractor who has not executed the Standing Security Bond and un enlisted contractor, the Cover-I shall be accompanied by Earnest Money for the amount mentioned in Appendix 'A' in the form of deposit at call receipt in favour of concerned CCE/GE/GE (I)/AGE (I) (see Appendix 'A') by a Scheduled Bank or in received treasury Challan the amount being credited to the revenue deposit of the concerned CCE/GE/GE(I)/AGE(I) (see Appendix 'A'). The CCE/GE/GE (I)/AGE (I) will return the Earnest Money, wherever applicable, to all unsuccessful tenderers/bidders by endorsing an authority on the deposit at call receipt for it's refund, on receipt of intimation from the Accepting Officer to do that.

Contd…

MILITARY ENGINEER SERVICES
NOTICE OF TENDER IAFW-2162 (Contd...)

10. In case of successful contractor i.e the lowest contractor having submitted EMD, he shall have the option of converting the EMD instrument into part of the Performance Security to be deposited by him within 28 days from the receipt of intimation of acceptance of tender from Accepting Officer.

11. Sample of materials and stores to be supplied by the contractor will also be available for inspection by the bidder at the office of concerned GE/GE (I)/AGE (I)/Project Manager during working hours. The bidder is advised to visit the site of work by making prior appointment with GE/GE (I)/AGE (I)/CCE/Project Manager, who is the Executing Agency of the work (see Appendix 'A'). The bidder shall be deemed to have full knowledge of all relevant documents, samples, site etc whether he has inspected them or not.

12. Any bid which proposes any alteration to any of the conditions laid down or proposes any other new condition whatsoever, is liable to be rejected.

13. The uploading of bid by a bidder implies that bidder has read this notice and the conditions of contract and has made himself aware of the scope and specification of work to be done and of the conditions and rates at which stores (as applicable) etc will be issued to him and local conditions and other factors having bearing on the execution of the work.

14. The bidder must be in possession of a copy of the MES Schedule (SSR) (Part-I & Part-II of latest edition) including amendments and errata thereto.

15. Accepting Officer does not bind himself to accept the lowest or any tender/bid or to give any reason for not doing so.

16. The Accepting Officer reserves the right to accept a tender submitted by a Public Undertaking/Small & Medium Enterprises (SMEs), giving a price preference/purchase preference over other tender(s)/bids which may be lower, as are admissible under the Government Policy. No claim for any compensation or otherwise shall be admissible for such tenderer/bidder whose tender/bid is rejected.

17. The **Notice Inviting Tender (NIT)** including Appendix 'A' and Annexures thereto, if any, shall form part of the contract agreement.

SIGNATURE OF CONTRACTOR

AGE (CONTRACTS)
FOR ACCEPTING OFFICER

Dated :

APPENDIX 'A' TO NOTICE INVITING TENDER (NIT)

1	Name of Work	:	**REPAIR /MAINT OF LT PANELS, SWITCHGEARS, LT CABLES ALONGWITH CONNECTED ITEMS OF DIFFERENT OTM ACCNS AND REPAIR MAINT OF CERTAIN MISC B/R AND E/M WORKS OF DIFFERENT ELECTRIC SUPPLY INSTALLATIONS AT INSIDE FORT WILLIAM UNDER GE (FW) KOLKATA**
2	Estimated cost of work	:	**Rs 37.00 Lakhs**
3	Period of Completion	:	**270 Days**
4	Cost of tender documents	:	**Rs 500.00** in the shape of DD/Bankers cheque from any Scheduled Bank in favour of **GE (FW) Kolkata** and payable at **Kolkata**. (Note: In case of re-tendering, the contractor who had quoted in the previous call is not required to submit the cost of tender)
5	Website/portal address	:	www.defproc.gov.in
6	Type of contract	:	The tender shall be Item Rate Contract based on IAFW-1779A & General conditions of Contracts (IAFW-2249) with Schedule 'A' (list of items of works) to be priced by contractor.
7	Timeline Details :-		
	(a) Bid Submission Start Date	:	Refer critical dates on the website
	(b) Bid submission end date	:	
	(c) Date of bid opening	:	
8	**Eligibility Criteria** (a) For MES enlisted contractors	:	Contractor shall be enlisted with MES in **Class 'D'** and above and category "**b(i)**" subject to satisfactory remarks wrt performance in respects of works in hand as reflected in in Work Load Return (WLR) or any other report circulated by competent engineer authority.
	(b) For contractors not enlisted with MES	:	(a) Contractor not enlisted with MES should meet the enlistment criteria of **'D'** class and "**b(i)**" category contractor with regard to satisfactory completion of requisite value works with Central/State Government/ Central/State PSUs/AWHO/AFNHB/CGEWHO/DGMAP, annual turnover, bank solvency, working capital and other requirements given in Para 1.4 & 1.5 of Section 1 of MES Manual of Contracts 2020 as available in all MES formations as well as MES website (www.mes.gov.in) (b) Not carrying adverse remarks in Work Load Report (WLR) or any other similar report circulated by any competent authority, if already working in MES. (c) Not suspended/debarred/blacklisted (either permanently or temporarily) from participating in any bid or for business dealings by any Central/State Government Department or any Central/State Government PSU or any Autonomous Body under Central/State Government or any Local Body as on the bid submission end date. (d) Details of works completed and under progress in MES be submitted in the following format:- Srl No | CA No. & Name of Work | Value of CA | Date of Comme-ncement | Date of Completion | Extended date of completion (e) un-enlisted contractor who have secured two works in MES should get themselves registered in the appropriate designated class with any registering Authority, else the firm will not be eligible for participation in the tender unless until the firm is enlisted with the MES.
9	Tender issuing and Accepting Officer	:	**Name** : Garrison Engineer (Fort William) **Address** : D-9, Commissariat Road, Hastings, Kolkata – 22 **Contact Details of concerned officer** : **Phone No** : 033-22482824 **Email Id** : garrison.engineer2@gmail.com
10	Executing Agency	:	Garrison Engineer (Fort William) Kolkata
11	Earnest Money	:	**Rs 74,000/-** in favour of GE (FW) Kolkata in form of DD/Banker's cheque. FDR not acceptable. MES enlisted contractor are exempted from submission of EMD.

Contd…

APPENDIX 'A' TO NOTICE INVITING TENDER (Contd...)

NOTES :-

1. In case after opening of Cover 1, the number of MES enlisted contractors of eligible class as well as eligible un-enlisted contractors, if any, fulfilling the other eligibility criteria given in NIT is **less than 7 (seven),** applications in respect of MES contractors of one class or two classes (in case of remote and difficult areas to be decided as per list circulated by CE command/ADG) below the eligible class shall also be considered subject to fulfillment of other eligibility criteria given in the NIT. Therefore, MES contractors one class below (two classes below in case of remote and difficult areas) may also bid for this tender. **Such contractors (contractors of one/two classes below the eligible class) shall not be considered in case their present residual work in hand is more than FIVE TIMES their present tendering limit.** However, in case such contractors fulfil the criteria of upgradation to the stipulated eligible class based on past experience of completed works (individual work experience and/or average annual turnover, as applicable) and financial soundness (solvency/financial soundness and working capital), the ceiling of present residual work will not apply and they will be considered for issue of tender. Such bidders shall upload in their Cover-1, bid details related to residual work in hand, like details of works in hand showing names of work, names of Accepting Officers, Contract amounts, dates of commencement and completion (stipulated) and progress as on bid submission date. Such contractors, if claim to fulfil the criteria of upgradation shall also upload the requisite information/documents in support of upgradation. These details shall be verified by the Tender Issuing Authority from concerned formations in case bids of such contractors are considered for evaluation.

2. In case after opening of Cover 1, the number of MES enlisted contractors of eligible class as well as un-enlisted contractors, if any, fulfilling the other eligibility criteria given in NIT, are 07 (Seven) or more, applications of only those one class below the eligible class bidders shall be considered, who have previously completed similar works satisfactorily and are meeting the criteria of up gradation in respect of past experience of completed works (individual work experience and/or average turnover as applicable) and financial soundness (solvency/financial soundness and working capital) as per details given in Manual on Contracts. Therefore, such contractors shall upload the requisite information/documents in the Cover-1.

3. "E" Class contractors may also submit their bids for the subject work. However, their bids shall be evaluated as per Note 01 & 02 above. Therefore, 'E' Class contractors are required to upload the following details/documents in Cover-1 alongwith their bids :-

(A) DETAILS OF WORKS IN HAND

SER NO	CA NO & NAME OF WORK	ACCEPTING OFFICER	CA AMOUNT	DATE OF COMMENCEMENT	DATE OF COMPLETION	% PROGRESS OF WORK
1)						
2)						
3)						

(B) DETAILS OF MEETING ELIGIBILITY CRITERIA OF 'D' CLASS CONTRACTOR

WORK EXPERIENCE	SOLVENCY LIMIT	WORKING CAPITAL AMOUNT
Copies of Work Order and Completion certificate of works executed during the previous two years for more than Rs 25.00 (two works) or Rs 40.00 lakhs (one work) or CA report showing Rs 50.00 lakhs average annual turnover for immediate last two consecutive financial years	Certificate in attached format as Appendix 'A' to be obtained from bank and uploaded with bid	Certificate in attached format as Annexure 'A' to be obtained from bank and uploaded with bid. Otherwise, contractor may also upload copy of balance sheet/bank statement for last 6 months/ overdraft or cash credit facility from bank/fixed deposit receipts with banks

Contd…

APPENDIX 'A' TO NOTICE INVITING TENDER (Contd...)

4. Un-enlisted contractor shall be considered provided he meets the criteria. Foreign firms shall not be eligible for this tender. However, Indian Firms having foreign national/Indian nationals staying abroad/Indian national having taken foreign citizenship, as director(s) shall be considered subject to security clearance from the concerned authorities.

5. Contractors enlisted with MES will upload following documents in Cover 1 for checking eligibility :-

(i) Application for tender on firm's letterhead.
(ii) Enlistment letter issued by the Registering Authority duly renewed for the cycle period in vogue.
(iii) Scanned copy of DD/Bankers Cheque toward cost of tender and EMD instrument in case SSD bond is not signed at the time of registration.
(iv) GST registration certificate
(v) EPF registration certificate
(vi) Any other document required as described in this Appendix

6. Contractors not enlisted with MES will be required to upload following documents in Cover 1 for checking eligibility: -

(i) Application for tender on firm's letterhead.
(ii) Scanned copy of DD/Bankers Cheque toward cost of tender and Earnest Money Deposit(EMD) instrument
(iii) Copy of Police Verification Report/Police Clearance Certificate/ Character Certificate from the Police Authority of the area where the registered office of the firm is located/notarized copy of valid passport of Proprietor/each Partner/Director.
(iv) All documents required for enlistment in MES for the class mentioned in Para 8(b) above as per Para 1.5 of Section 1 of MES Manual on Contracts 2020.
(v) Details of works being executed in MES, if any.
(vi) GST registration certificate.
(vii) EPF registration certificate.
(viii) Any other document required as described in this Appendix.

7. Tenders not accompanied by scanned copies of requisite DD/Bankers Cheque towards cost of tender and earnest money (as applicable) in Cover v1 shall not be considered for validation of 'T' bid and their financial bids will not be opened.

8. Contractors should ensure that their original physical DDs and Earnest Money Deposit (EMD) instruments (as applicable) reach the office of the Accepting Officer within 05 (five) days of bid submission end date failing which the following action shall be taken: -

(i) In case of tenders from an enlisted contractor of MES, where scanned copies of requisite DD/Bankers Cheque towards cost of tender have been uploaded in Cover 1 but physical copies are not received within the stipulated period, their financial bids will be opened. However, non submission of physical copies of cost of tender shall be considered as willful negligence of the tenderer with ulterior motives and such bidder shall be banned from bidding for a period of six months commencing from the date of opening of financial bid (Cover 2). Nevertheless, the amount of cost of tender shall be recovered from any amount due to the MES enlisted contractor.

(ii) In case of tenders from un-enlisted contractor, where scanned copies of requisite DD/Banker's cheque towards cost of tender has been uploaded in Cover 1 but physical copies are not received within the stipulated period, their financial bids (Cover 2) will not be opened. Name of such contractors alongwith complete address shall be circulated for not opening of their bids for a period of six months commencing from the date of opening of financial bid (Cover 2).

(iii) In case of tenders from enlisted and un-enlisted contractors, where scanned copies of instruments for Earnest Money Deposit (as applicable) have been uploaded in Cover 1 but the same are not received in physical form within stipulated period, such tenders shall not qualify for opening of financial bid (Cover 2).

Contd…

APPENDIX 'A' TO NOTICE INVITING e-TENDER (Contd...)

9. Contractor will not be allowed to execute the work by subletting or through power of attorney to a third party/another firm on his behalf. However, a contractor can execute the work through power of attorney to sons/daughters/spouse of Proprietor/Partner/Director and firm's own employees, Director, Project Manager provided they are not having a separate enlisted firm in MES in their name as Proprietor/Partner/Director.

10. After opening of Cover 1 and during its technical evaluation, in case of any deficiency is noticed in the documents required to be uploaded by the tenderer as per NIT, a communication in the form of e-mail/SMS/Speed Post etc shall be sent to the contractor to rectify the deficiency within a period of seven days from the date of communication failing which their financial bid (Cover 2) shall not be opened and contractor shall not have any claim on the same.

11. Invitation for e-tender does not constitute any guarantee for validation of technical bid and subsequent opening of financial bid of any applicant/bidder merely by enclosing DD. Accepting Officer reserves the right to reject the technical bid and not to open the financial bid of any applicant/bidder. Technical bid validation shall be decided by the Accepting Officer based on eligibility of the firm as per criteria given in this Appendix. Tenderer/bidder will be informed regarding non-validation of his technical bid assigning reasons therefore through tender evaluation report which shall be uploaded on the website. Such tenderer, if desires, may appeal to the next higher Engineer Authority (NHEA) viz HQ Commander Works Engineers, Kolkata on email id cwekol-ec-army@nic.in with copy to the Accepting Officer on email before the scheduled date of opening of Cover 2. NHEA shall decide the matter within a period of seven working days from the date of receipt of appeal. The decision of NHEA shall be final and binding. The tenderer/bidder shall not be entitled for any compensation whatsoever for rejection of his bid.

12. In case an un-enlisted contractor is already executing work in MES, he shall not be considered eligible for the subject tender if the total value of such works is more than twice the tendering limit of the MES Class of contractor for which it is eligible. For this purpose, details of the works being executed by such a contractor shall be uploaded in Cover 1 of the bid and shall be checked/verified by the Accepting Officer.

13. In case the BOQ is revised through corrigendum and the bidder has failed to quote on revised BOQ (i.e. he has quoted on pre-revised BOQ), such bid shall be treated as willful negligence by the bidder and his quotation shall be considered non-bona-fide. In such cases, the lowest tender shall be determined from amongst the valid/bona-fide bids only. Accepting Officer may decide whether to retender or consider the lowest bona-fide tender for acceptance.

14. Revoking the offer or revising the rates upward or offering voluntary reduction by the lowest tenderer after opening of Cover 2 shall be considered as willful default. For this default, a penalty of an amount equal to the Earnest Money shall be levied. In case of an un-enlisted tenderer, Earnest Money deposited by him shall be forfeited. In case of MES enlisted tenderer having deposited the Standing Security Bond, an amount equal to the Earnest Money stipulated in the NIT, shall be notified to the tenderer for depositing through MRO and consideration of such tenderer in tender evaluation for future works shall remain suspended till the aforesaid amount is deposited in the Government Treasury. No other disciplinary/administrative action shall be taken against such tenderers. In such a situation, the next lowest offer shall not be considered for acceptance. Instead, retendering shall be resorted to in a transparent and fair manner and the defaulting tender and his related firm, if any, shall not be eligible for this tender in second call or subsequent calls.

15. Tender to related firms shall not be issued simultaneously. Firms shall be termed as related if Proprietor/one or more Partners/Directors are common. Decision of Accepting Officer on issue/deny the tender to any one of the related firms shall be final and binding.

8373/ /E8

Garrison Engineer (FW) Kolkata
Military Engineer Services
D-9, Commissariat Road
Hastings, Kolkata - 700 022

AGE (Contracts)
For Accepting Officer

Dated : Aug 2025

ANNEXURE 'A'

FORM OF SOLVENCY CERTIFICATE FROM NATIONALISED/SCHEDULED BANK

This is certified that to the best of our knowledge and information M/s Shri/Smt. _____ having address _____, a customer of our bank are/is respectable and can be considered solvent upto Rs. _____ (Rupees _____)/financially sound for any engagement upto Rs _____ (Rupees _____). This certificate is issued without any guarantee or responsibility on the bank or any of the officers.

(Signature)
Name, Designation and Personal Code No. Of Signatory & Seal of Bank

Note : In case of partnership firm, certificate to include names of all partners as recorded with the bank.

FORM OF WORKING CAPITAL CERTIFICATE FROM NATIONALISED/SCHEDULED BANK

This is certified that M/s/Shri/Smt. _____ having address _____ has/have been maintain a saving bank account/current account/fixed deposit account with this branch of bank been maintaining a saving bank account/current account/fixed deposit account with this branch of bank since _____ and the firm is having working capital of approximately Rs. _____ and/or the firm is enjoying overdraft/credit facilities upto limit of Rs. _____ and/or the firm is enjoying overdraft/credit facilities upto limit of Rs. _____. This certificate is issued without any guarantee or responsibility on the bank of any or the officers.

(Signature)
Name, Designation and Personal Code No. of Signatory & Seal of Bank

MILITARY ENGINEER SERVICES

(In lieu of IAFW-1779A
(Revised) & 1955)

TO BE READ IN CONJUNCTION WITH GENERAL CONDITIONS OF CONTRACTS IAFW-2249(1989 PRINT)

Garrison Engineer (FW) Kolkata
Military Engineer Services
D-9, Commissariat Road
Hastings, Kolkata - 700 022.

8373/ /E8 Aug 2025

TENDER AND ITEM RATE CONTRACT FOR WORKS BY MEASUREMENT

NAME OF WORK : REPAIR /MAINT OF LT PANELS, SWITCHGEARS, LT CABLES ALONGWITH CONNECTED ITEMS OF DIFFERENT OTM ACCNS AND REPAIR MAINT OF CERTAIN MISC B/R AND E/M WORKS OF DIFFERENT ELECTRIC SUPPLY INSTALLATIONS AT INSIDE FORT WILLIAM UNDER GE (FW) KOLKATA

1. M/s/Shri _____ of _____ is/are hereby authorized to tender for the above work. The BOQ is to be uploaded in the www.defproc.gov.in site (single stage two cover system) on or after the bid submission start date.

2. Any correspondence concerning this tender should be online and no offline correspondence is permitted.

3. Drawings and specification if not included with tender documents may be seen on application at the office of the Garrison Engineer, D-9, Commissariat Road, Hastings, Kolkata – 700022.

THE PRESIDENT OF INDIA DOES NOT BIND HIMSELF TO ACCEPT THE LOWEST OR ANY TENDER.

Signature of Officer issuing the Tender documents
Appointment: AGE (Contracts)

Dated: _____

SCHEDULE – 'A'
IAFW-1779-A (REVISED 1955)
(LIST OF ITEMS OF WORK)

NAME OF WORK : REPAIR /MAINT OF LT PANELS, SWITCHGEARS, LT CABLES ALONGWITH CONNECTED ITEMS OF DIFFERENT OTM ACCNS AND REPAIR MAINT OF CERTAIN MISC B/R AND E/M WORKS OF DIFFERENT ELECTRIC SUPPLY INSTALLATIONS AT INSIDE FORT WILLIAM UNDER GE (FW) KOLKATA

(A) NOTES (GENERAL) APPLICABLE TO ALL SECTIONS :

1. The tender comprises of the works as described in BOQ and specified in particular specifications.

2. The quantities shown in column 3 of BOQ are approximate and are inserted as a guide only. However, these shall not be varied beyond the limits laid down in condition 7 of IAFW-2249 General Condition of contracts.

3. Tenderer is to quote his unit rates under column 5.

4. The rates shall be quoted in figure only. Software system will automatically derive the quoted amount (in figure and words) on the basis of rate inserted by the bidder/tenderer.

5. Total amount under column 7 is not firm but will be treated as "CONTRACT SUM" for all the purpose as referred to inIAFW-2249.

6. **UNIT RATE :** The tenderer shall calculate his own details and insert unit rates in Column 5 against each item of Schedule 'A'. it is an express condition of his contract that the rates quoted by the tenderer in the tender for various items shall be deemed to include for the entire and final completion of the items of work in accordance with provision of the tender. Government shall not entertain any claim whatsoever on account of inaccuracies/misunderstanding, if any in the aforesaid rates.

7. The under mentioned remarks shall be deemed to have been inserted against each item of BOQ:-
   (a) Refer list of drawing if applicable
   (b) Refer note No : 8 of Schedule 'A' Notes
   (c) Refer notes to Schedule 'A'.

8. **PERIOD OF COMPLETION** : Total work under this contract shall be completed within **270 days** from the date of handing over of site.

9. Tenderers shall be deemed to include in his unit rate for each item for the provision of all materials, labour, operations, tools/plants as required and other requirements as detailed in the Particular Specifications irrespective of whether these appear as specific item or mentioned in the description of items in Schedule 'A' or not and the tenderer shall not be paid any extra for the same.

10. Unless otherwise specified method of measurement of various items of work shall be as stipulated in MES standard Schedule of Rates (Part I - 2009 and Part II - 2020).

Contd….

SCHEDULE – 'A' NOTES (Contd....)

11. The description of items given in Schedule 'A' is in brief. This shall deemed to be amplified and read in conjunction with special conditions, particular specification, drawings (including notes thereof), specifications for materials and workmanship given in MES Schedule Part – I, special conditions and preambles given in relevant trade sections of MES Schedule Part – II.

12. The rates quoted by the tenderer shall be deemed to include any minor details/items of work and/or construction which are obviously and fairly intended and which may not have been included in these documents but which are essential for the execution and entire completion of the work and decision of the Accepting Officer on this account shall be final, conclusive and binding.

13. Site for execution of works will be available as soon as the work order No 1 is placed by the GE. In case it is not possible for the Department to make the entire site available immediately after the work order No 1 is placed, the contractor will have to arrange his working programme accordingly. No claim what so ever for not giving the entire site on placement of work order No 1 and or for giving the site gradually will be tenable.

14. **TAXES** : -

14.1. Consequent to implementation of Goods and Service Tax (GST), the contractor's quotation shall be deemed to include Tax under GST, as applicable to the goods procured for execution of contract and services rendered under the contract as works contract. This provision shall supersede all other provisions mentioned elsewhere in the tender documents and General Conditions of Contracts (IAFW-2249).

14.2. In addition to GST, as brought out above, the contractor's quotation shall be deemed to include labour welfare cess, Seignioroge for mining, if not included in GST.

14.3. Any other taxes, not brought out above but are applicable as on date of submission of tender, shall be deemed to be included in the tender. In this connection, the provisions as brought out under condition 51 of IAFW 2249 shall be applicable.

14.4. The increase/ decrease in any taxes/ duties, statutory or otherwise after submission of tender and during the execution of the works shall be adjusted as provided in relevant clause of Special Conditions, herein after. The various taxes mentioned in Special Conditions, Condition 10 of IAFW – 2249, shall be deemed to be amended, as applicable as on date of submission of tender.

14.5. Tax deduction at source on works contracts shall be deducted at source, on payments made to the contractor as specified in the GST Act.

15.0 **Cost of Testing including Transportation of Materials or Equipments**

15.1 Provision of all facilities for Testing in accordance with Specifications and/or IS shall be borne by the Contractor unless otherwise mentioned.

16.0 **SAFETY MEASURES, PRECAUTION'S, RISKS ETC.,**

16.1 The Work shall be carried out with utmost care to ensure that no damage to Existing/Adjoining Work is done failing which the damage, if any shall be rectified by the Contractor to match with the Existing/Adjoining Work to the entire satisfaction of Engineer-in-Charge under Contractor's own arrangement and at his own expenses.

16.2 Suitable Tools, Plants, Equipments, Mechanism, etc., as considered necessary shall be adopted during execution of the Work. The Contractor shall take all Precautions, Safety Measures, etc., to avoid any Damage, Miss-happening, Accident, etc., to the Workmen engaged by him to carry out the Work. The UNIT RATES quoted by Contractor shall be deemed to have included the Element of adopting Safety Measures, Precautions and also the Risks, etc. involved in Work and nothing EXTRA shall be admissible on this account.

Contd….

SCHEDULE – 'A' NOTES (Contd....)

17. **METHOD OF MEASUREMENTS**

17.1 The method of Measurements to be adopted in the Contract shall be as laid down in MES STANDARD SCHEDULE OF RATES PART-I & II, **IS: 1200** and other relevant BIS standard. The General Rules, Special Conditions, Preambles and Other Provisions in MES STANDARD SCHEDULE OF RATES PART-I & II and relevant BIS standard shall be deemed to apply to the Work under this Contract unless otherwise mentioned in the Tender documents. However, where the UNITS of Work given in Column 4 against Items of SCHEDULE 'A' are at variance with that mentioned against respective Items in MES STANDARD SCHEDULE OF RATES, the Measurements of Work against such Items of SCHEDULE 'A' shall be done as per the UNITS given in SCHEDULE 'A'.

18. The tenderers are required to visit the respective site of work and make themselves familiar with the contract condition which may affect the completion of work. Irrespective of the fact whether the contractor visits the site or not, it will be deemed that he has physically inspected the site and the quoted rates are inclusive of any/all such factors which may affects extension and completion of work. No claim whatsoever will be entertained on this account.

19. The contractor shall provide all watchmen necessary for protection of site, the works and the materials etc, during the progress of work and shall be solely responsible for protecting, securing, lighting and watching all places as required in terms of condition 36 of General Condition of Contracts (IAFW-2219).

20. The tenderers are requested to note that the scope of work includes execution of various job works. In the schedule of works, the item of work are given job wise and the item of works might have been repeated under various jobs. The tenderers are requested to quote uniform unit rates against similar items of work under various jobs. Please note, if, different unit rates are quoted, against similar items of the work, the lowest rate quoted shall only be considered and accordingly, the variable rates quoted against similar items of work shall be amended to the lowest unit rate and the tender amount shall be amended accordingly.

21. In the Schedule 'A', model No in respect of certain fittings has been specified with reference to one make. If the Contractor proposes equivalent model, from other makes listed in list of makes, necessary information/ technical brouchers, to Prove 'equivalent', to the model specified, shall be submitted through Engineer-in-charge, for approval of Accepting Officer.

22. Description of certain items in BOQ, may contain the words "in repair or fixed in repair", which means that the item is inclusive of the cost of taking down or dismantling of the existing provision/item. No separate measurement of dismantling/taking down shall be taken.

23. The words "M&L" or "S&F" in BOQ denotes "Material and labour" and "Supply and Fixing" respectively.

24. **EMPLOYEES PROVIDENT FUND & MISCELLANEOUS PROVISIONS (EPF & MO) ACT 1952- REG.**

24.1 The contractor shall have invariably Provident Fund Code Number and GST Registration Number and also ensure that all the workers deployed by him are enrolled as members of provident Fund and should be given the Universal Account Number (UAN). While clearing the bills of the contractor, certificate be obtained that all workers employing directly or indirectly by him are registered for EPF and the due contribution have been credited into their account.

SCHEDULE 'B'

ISSUE OF MATERIALS TO THE CONTRACTOR
(SEE CONDITION 10 of IAFW-2249)

Ser No	Particulars	Unit	Rate at which materials will be issued to the contractor	Place of issue	Remarks
1	2	3	4	5	6
----- NIL -----					

SCHEDULE - 'C'

LIST OF TOOLS AND PLANTS (OTHER THAN TRANSPORT) WHICH WILL BE HIRED TO THE CONTRACTOR

(See Conditon-16 , 34 & 55 of IAFW-2249)

Ser No	Quantity	Particulars	Details of MES crew supplied	Hire charges per unit for working	Stand by charges per unit per off day	Place of issue (by name)	Remarks
1	2	3	4	5	6	7	8
----- NIL -----							

SCHEDULE - 'D'

TRANSPORT TO BE HIRED TO THE CONTACTOR
(REFER CONDITION 16 AND 35 OF IAFW-2249)

Ser No	Quantity	Particulars	Per unit /per working day	Place of Issue (be name)	Remarks
1	2	3	4	5	6
----- NIL -----					

SIGNATURE OF CONTRACTOR
DATED : _____

AGE (CONTRACTS)
FOR GARRISON ENGINEER

TENDER

TO

THE PRESIDENT OF INDIA

Having examined and perused the following documents: -

1. Specifications signed by AGE (Contracts)

2. BLANK

3. Schedule 'A', 'B', 'C' & 'D' attached hereto.

4. MES Standard Schedule of rates 2009 Part-I & 2020 Part-II, rates (here-in-after referred to as the MES Schedule) together with amendment Nos 1 & 2 for Part I 2009 and errata/amendment Nos ____ for Part-II (2020)

5. General Condition of contracts, IAFW-2249 (1989 Print) together with errata Nos 1 to 20 and amendments 1 to 49.

6. Water :- Condition 31 of IAFW-2249, General Condition of contracts. **Water will be supplied by MES on payment at the Rate of Rs 3.75 per every Rs. 1000.00 worth of work done priced at contracts rates.**

Should this tender be accepted, I/We agree: -

*(a) that the sum of Rs _____/- (Rupees _____only) forwarded as earnest money shall either be returned as a part of security deposit or refunded by the Government on receipt of the appropriate amount of security deposit as per condition - 22 of IAFW-2249.

(b) To execute all the works referred to the said documents upon the terms and conditions contained or referred to therein at the item rates contained in the aforesaid Sch'A' or at such other rates as may be fixed under the provisions of Condition 62 and to carry out such Deviations as may be ordered vide condition 7 of IAFW-2249 up to a maximum of **10 % (Ten percentage)** and further agree to refer all disputes, as required by condition 70 to the sole arbitration of an Serving Officer, having degree in engineering or equivalent or having passed final/direct final examination of subdivision II of institution of surveyors (India) recognized by the Govt of India, to be appointed by the Chief Engineer Kolkata Zone or in his absence the officer officiating as Chief Engineer Kolkata Zone or specifically delegated in writing by Chief Engineer Kolkata Zone whose decision shall be final, conclusive and binding.

* Delete if not applicable.

Contd…

TENDER (CONTD . . . .)

Signature_____ Name _____

In the capacity of _____ duly authorized to sign the tender for and
(IN BLOCK CAPITALS)

Behalf of _____ _____

Witness _____ Address _____

_____ _____

_____ _____

A C C E P T A N C E

_____ alterations have been made in these documents and as evidence that these alterations were made before the execution of the Contract Agreement, they have been initialed by the contractor and _____ AGE (Contracts).

The said officer (s) is/are hereby authorized to sign and initial on my behalf, the documents forming part of this Contract.

The above tender was accepted by me on behalf of the President of India at the item rates contained in Sch 'A' . on _____day of _____2025.

Signature _____Dated _____day of _____2025.

APPOINTMENT: GARRISON ENGINEER, FORT WILLIAM KOLKATA
(FOR AND ON BEHALF OF THE PRESIDENT OF INDIA)

GENERAL CONDITION OF CONTRACTS IAFW-2249. (1989 PRINT)
FOR MEASUEREMENT CONTRACTS (IAFW-1779A)

A copy of the GENERAL CONDITIONS OF CONTRACTS (IAFW-2249, 1989 Print) with Errata 1 to 20 and amendments No 1 to 40 has been supplied to me/us and is in my/our possession. I/we have already read and understood the provisions contained in the aforesaid GENERAL CONDITIONS OF CONTRACTS before submission of this tender and I/We Agree that I/We shall abide by the terms and conditions thereof, as modified if any, elsewhere in these tender documents.

It is hereby further agreed and declared by me/us, that the GENERAL CONDITIONS OF CONTRACTS (IAFW-2249, 1989 Print) including condition 70 there of pertaining to settlement of disputes by Arbitration containing 33 pages (**Serial Page No. 35 to 68**) including Errata 1 to 20 and amendments No 01 to 49 form Part of these tender documents.

SIGNATURE OF CONTRACTOR

AGE (CONTRACTS)
FOR ACCEPTING OFFICER

Dated :

NOTE :

In case the tenderer is not in possession of any of the documents referred above he should obtain a copy from the Office of the Garrison Engineer, Military Engineer Services, D-9, Commissariat Road, Hastings, Kolkata – 700 022 before submission of his tender.

SCHEDULE OF MINIMUM WAGES/FAIR WAGES
(SEE CONDITION 58 OF IAFW-2249)

1. It is an express condition of contract that the schedule of minimum wages form part of this contract agreement and the contractor is liable to pay minimum wages to labour/workers engaged in this contract as per notification issued by Central Govt. or State Govt. whichever is higher.

2. It is also an agreed condition of this contract that minimum wages Act and provision contained therein form part of this contract agreement.

SIGNATURE OF CONTRACTOR

AGE (CONTRACTS)
FOR ACCEPTING OFFICER

DATED :

SPECIAL CONDITIONS

NAME OF WORK : REPAIR /MAINT OF LT PANELS, SWITCHGEARS, LT CABLES ALONGWITH CONNECTED ITEMS OF DIFFERENT OTM ACCNS AND REPAIR MAINT OF CERTAIN MISC B/R AND E/M WORKS OF DIFFERENT ELECTRIC SUPPLY INSTALLATIONS AT INSIDE FORT WILLIAM UNDER GE (FW) KOLKATA

1.0 **GENERAL**. The following special conditions shall be read in conjunction with General Conditions of Contracts IAFW-2249 including errata/amendments thereto. If any provision in these special conditions is at variance with that of the aforesaid documents, the provision in these special conditions shall be deemed to take precedence.

1.2 The work under this contract shall be carried out in accordance with schedule 'A' particular specifications, drawing and other provisions made in MES schedule.

1.3 The term 'General Specifications' referred to herein before/here in after as well as referred to in IAFW-2249 (General Condition of the Contracts), shall mean the specifications contained in MES schedule.

1.4 General rules, specifications, special conditions and all preambles in MES schedule shall be deemed to apply to the works under this contract unless specified otherwise; in these tender documents , in which cases the provisions in these tender documents shall be deemed to take precedence over the provisions made in MES Schedule..

2.0 **INSPECTION OF SITE**. For the purpose of inspection of site the tenderer is advised to contract for Garrison Engineer who will give reasonable facilities therefore. The tenderer shall also make himself familiar with the working conditions accessibility to site of works, availability of labour and materials and such other relevant conditions which affect the execution and the entire completion of works. The tenderer shall be deemed to have inspected the site and made himself familiar with various factors which may affect his quotation, irrespective of whether he actually inspects the site or not.

3.0 **SECURITY AND PASSES**.

3.1 Contractor's attention is invited to Condition-25 of IAFW-2249. He shall employ only Indian Nationals after verifying their antecedent and loyalty. The Contractor shall on demand by the Engineer-in-Charge, submit list of his agents, employees and work people concerned and shall satisfy the Engineer-in-Charge as to the bonafides of such people.

3.2 The Engineer-in-Charge shall at his discretion have the right to issue passes as per rules/regulations of the installation area in force to control the admission of the contractor, his agent employees and work people to site of work or any part thereof passes should be returned at the time of demand by the engineer-in-charge or the authorities concerned and in any case on completion of work.

4.0 **CONDITIONS OF WORKING AREA**:-

4.1 The works under this contract lies in Restricted Areas with restricted working hours. However the GE/Engineer-in-charge shall have the right at his discretion to issue passes to the contractor, list agents servants works people and vehicles and to control their admission to the site of work or any part thereof. The contractor shall on demand by the Engineer-in-charge, submit a list of personnel etc. Concerned and shall satisfy the Engineer-in-charge as to the bonafides of such people passes shall be returned at any time on demand by the Engineer-in-charge and in any case on completion of work.

4.2 The contractor and his work people shall observe all the rules promulgated from time to time by the authority controlling the area in which the work is to be executed viz prohibition of smoking and lighting fire precautions search of persons on entry and exit and keeping to specified routes restricted hrs of working etc. Any person found violating the security rules laid down by authority, shall be immediately expelled from the area without assigning any reasons whatsoever and no payment shall be admissible for any man hours lost on this account.

Contd . . . .

SPECIAL CONDITIONS

4.3 The contractor, his agents and representative are required individually to be in possession of identity card or pass duly verified by the Garrison engineer. The identity card or pass will be examined by the security staff at the time of entry into or exit from the restricted area and also at any time or number of times inside the restricted area. Every workman shall be in possession of an identity card which shall be issued after a thorough investigation of the antecedents of the labour employees by the contractor and attested by engineer-in-charge in accordance with the standing rules and regulations of the unit. Contractor shall be responsible for the conduct and action of workmen, agents or representative. Please shall be returned at any time on demand by engineer-in-charge and n any case on completion of work.

4.4 **WORK ON HOLIDAYS** :- The contractor shall not carry out any work on gazetted holidays, weekly holidays and other non working days except when he is authorised in working to do so by the GE. The GE may at his sole discretion declare any day as holiday or non-working day without assigning any reason for such declaration. Nothing extra shall be admissible on this accounted for any man hours lost.

4.5 **SEARCH.**
Through search of all personnel and transport as required by the authority will be carried out at any time or any number of times at work site. If contractor desires to employ female labour for works inside factory, Depot, Park etc and female searcher is not none on authorizes strength of the Depot, Park etc at the time of submission of tender he shall deemed to have allowed in his tender for pay and allowances etc for a female searcher (class IV/Group D) calculation for the period.

4.6 **FIRE PRECAUTIONS**.
The contractor, his agents, representatives, workman etc shall strictly observe the order pertaining to fire precautions prevailing within the restricted area/place of execution of work.

5.0 **NET WORK ANALYSIS**

5.1 The time and progress chart to be prepared as per condition 11 of General Conditions of Contracts (IAFW-2249) shall consist of detailed network analysis and time Schedule. The critical path not work will be drawn jointly by the GE and contractor soon after acceptance of tender. The time Scheduling of the activities will be done by the contractor so as to finish the work within the stipulated time. Completion of the time schedule a firm calendar date schedule will be prepared and submitted by the contractor to the GE who will approved it after due scrutiny. The schedule will be submitted in four copies within two weeks from the date of handing over the site.

5.2 During the currency of the works, the contractor is executed to adhere to the time schedule and this adherence will be a part of the contractor's performance under the contract. During the execution of the work, the contractor is executed to participate in the reviews and updating of the net work under taken by the GE. These reviews may be undertaken at the discretion of the GE or either as periodical appraisal measure or when the quantum of work ordered on the contractor is substantially changed through orders or amendments. Any revision by the contractor shall be informed to the GE within a week for his approval after due scrutiny.

5.3 The contractor shall adhere to the revised time schedule thereafter. In case of a contractor disagreeing with the revised schedule, the same will be referred to the Accepting Officer, whose decision shall be final, conclusive and binding. **GE's** approval to the revised schedule resulting in a completion date beyond the stipulated date of completion shall not automatically amount to a grant of extension of time. Extension of time shall be considered and decided by the appropriate authority mentioned in condition 11 of IAFW-2249 and separately regulated.

5.4 Contractor shall mobilize employment of sufficient resources to achieve the detailed schedule within the broad frame work of the accepted method of working and safety. No additional payment will be made to the contractor for any multiple shift work or other incentive method contemplated by him in his work schedule, even though the time schedule is approved by the Department.

Contd….

SPECIAL CONDITIONS (CONTD….)

6.0 **SECURITY OF CLASSIFIED DOCUMENTS**. Contractor's special attention is drawn to conditions 2A and 3 of IAFW-2249 General Conditions of contracts. The contractor shall not communicate any classified information regarding works either to such contractors or others without prior approval of the Engineer-in-Charge. The contractor shall also not make copied of the design drawings and other documents furnished to him in respect of works and he shall return all documents on completion of the works and or on earlier determination of the contract. The contractor shall alongwith final bill attach a receipt form the Engineer-in-Charge in respect of his having returned the classified documents as per condition of General Conditions contracts.

7.0 **CONTRACTOR'S REPRESENTATIVE AND WORKMEN**. The Contractor shall employ only Indian National as representative agents and workman after clarifying their antecedents and loyalty. He shall ensure that no person of doubtful and amendments rationality, is in any way associated with the works. If for reasons technical collaboration consideration, the employment of any foreigner National is unavoidable, the contractor shall furnish full particulars to this effect to the Accepting Officer at the time of submission of the tender. The GE shall order the contractor to cease to employ in connection with the contract any representative agents and workman or employees those continue employees in his opinion is undesirable. The contractor shall not be entitled for any compensation on this account.

8.0 **MINIMUM WAGES PAYABLE**

8.1 Refer condition 58 of IAFW-2249. The "Schedule of Minimum Fair Wages" as published vide Govt. of India Notification date 10 Mar 92 (revised upto date) forms part of the tender documents and is given in the tender documents here in before. The contractor shall not pay wages lower than minimum wages for labour as fixed by the Govt. of India/State Govt. under Minimum Wages Act of Contract Labour (Abolition and Regulation Act), whichever is higher.

8.2 The fair wage referred to in condition 58 of IAFW-2249 will be deemed to be the same as the minimum wages payable as referred to above.

8.3 The contractor shall have no claim whatsoever, if on account of local factors and or regulations, he is required to pay the wages in excess of minimum wages as described above during the execution of work.

9.0 **ROYALTIES**. Reference Condition 14 of General Conditions of contract IAFW-2249, No quarry on Defence land are available and quarrying is not permitted in defence land.

10.0 **LAND FOR LABOUR ACCOMMODATION, TEMPORARY WORKSHOP AND STORES ETC**.

10.1 Delete the following from lines 5 to 9 of Sub Para 1 condition 24 of IAFW-2249 reading. 'In the event of areas of land. Land allotted to him' and in lieu insert as under:-

"The contractor shall be allotted the area as marked on the site plan/layout plan for the purpose of storage of materials, erection of temporary workshop stores etc. The contractor shall pay normal rent of Rs 1/- per year or part of a year in respect of each and every separate area land allotted to him. No defence land is available for accommodation of labour for which the contractor shall make his own arrangement at his own expense.

11.0 **WATER**.

Refer condition 31 of general condition of contract IAFW-2249 and clause 1.13 of MES schedule Part I. Water will be supplied by MES at the points marked on site plan or as decided by GE @ Rs 3.75 per every Rs 1,000.00 worth of work done. As the water supply by MES is likely to be intermittent/short supply, the contractor shall make his own arrangements for storing the water required for the works, labour and workman etc at his own expense. The contractor shall not have any claim on account of short/intermittent supply and shall make his own arrangements to supplements the requisite quantity of water.

Contd….

SPECIAL CONDITIONS (CONTD….)

12.0 **CO-OPERATION WITH OTHER AGENCIES**.

The contractor shall permit free access and generally afford reasonable facilities to other agencies of departmental workmen engaged by the Government to carry out their part work, if any, under separate arrangements. The contractor shall not be entitled for any extra payment in this account.

13.0 **ELECTRIC SUPPLY**

13.1 Electricity will be supplied by MES. The contractor will be charged for electric energy consumed at the following rates:

(i) At rupees seven and paise seventy eight for lighting per unit.

(ii) At rupees seven and paise seventy eight for power per unit.

Electric supply required for works shall be made available by the MES at the incoming terminal of the main switch marked on the layout plan. The main switch and KWH meter to register the electricity supplied shall be provided and installed by MES. The contractor shall provide all necessary connection, cable, fittings etc and main switch in order to ensure proper and suitable supply of electricity.

13.2 MES do not guarantee continuity of supply and no compensation whatsoever shall be allowed for supply becoming intermittent or for break down in the system or any reason whatsoever.

13.3 GE or his representative shall be free to inspect all the power consuming devices and electric lines provided by the contractor and disconnected any device r electric line which is not to the satisfaction of the GE.

14.0 **RECORD OF CONSUMPTION OF CEMENT**

14.1 For the purpose of keeping and record of cement consumed in works, the contractor shall maintain pucca bound register in the form approved by the Engineer-in-Charge showing interalia cement received, quantity used in works and the balance in hand. The register shall be signed daily by the contractor's representative and the Engineer-in-Charge in token of their verification of its correctness. The check will not, however, absolve the contractor of his responsibility to justify the consumption of cement at the rate of utilization of his account.

14.2 The register shall be kept at site in the custody of the contract during progress of the work and shall on demand be produced a for verification of inspecting officers. On completion of the work the register shall be handed over to the engineer-in-charge for record.

15.0 **PERIOD FOR KEEPING THE TENDER OPEN**

The tender shall remain open for acceptance for a period 60 (Sixty) days from the date on which the tenders are due to submitted. For the purpose of calculating this period, the day on which the tenders are due to submitted shall be excluded.

16. **OCTROI SALES TAX AND OTHER DUTIES**

The tenderer's rates shall be deemed to include all mandatory charges and taxes such as octroi, sales tax, excise etc. as referred in Condition 10 of IAFW-2249 and include State Govt. sales tax on works contracts payable under respective states pursuant to the constitution (Forty Sixth Amendment, Act 1982). Any other conditions stipulated by the tenderer regarding sales tax on works contracts will not be considered and such tender shall be liable for rejection. The rates shall also be inclusive of 1% cess on gross amount of work done to be recovered from RAR/Final bill on account of Building and Other Construction Workers (Rec) Act 1996 and Building and Other Construction Works Welfare Cess Act 1996. The amount so collected shall be deposited by GE to "WB Building and other construction works welfare fund" payable at Kolkata.

Contd…

SPECIAL CONDITIONS (CONTD....)

17.0 **CONCILIATION**

17.1 **SCOPE OF CONCILIATION**

17.1.1 The Scope of conciliation shall be restricted to the following types of disputes with financial limits as indicated therein:-

(a) Disputes relating to levy of compensation for delay in completion-actual amount of compensation.
(b) Disputes relating to technical examination of works.
(c) Disputes relating to interpretation of the provisions of the contract with reference to the application to parties.
(d) Disputes relating to non-return of Schedule 'B' stores over-issued to the contractor.
(e) Any other dispute having fair chances of being resolved by conciliation and considered fit to be referred to conciliation by the parties.
(f) For item (b), (c), (d) and (e) each as stated above, the financial limit shall be Rupees two lakhs or one percent of the contract amount whichever is less.

17.2 **COMMENCEMENT OF CONCILIATION PROCEEDINGS**.

17.2.1 The party initiating conciliation shall send to the other party a written invitation to conciliate, briefly identifying the subject of the dispute.

17.2.2 Conciliation proceedings shall commence when the other party accepts in writing the invitation to conciliate.

17.2.3 If the other party rejects the invitation, there will be no conciliation proceedings. If the party initiating conciliation does not receive a reply within 30 days from the date on which he sends or within such other periods of time as specified in the invitation, he shall inform in writing the other party accordingly.

17.3 **NUMBER OF CONCILIATORS**

There shall be a Sole Conciliator.

17.4 **APPOINTMENT OF CONCILIATOR**:- All disputes brought out in 14.1.1 (a) to (e) above shall be referred to the sole conciliator viz. Serving Officer not blow the rank of Superintending Engineer/Superintending Engineer (QS&C) having degree in Engineering or equivalent or having passed final/direct final examination of Sub Division-II of Institution of Surveyors (India) to be appointed of sole conciliator by the Chief Engineer Kolkata Zone or in his absence the Officer Officiating as Chief Engineer Kolkata Zone, Eastern Command, Kolkata specifically delegated by the Engineer-in-Chief in writing.

18.0 **ACCEPTANCE QUALITY OF WORK AND FINISHES**

18.1 To determine the acceptable standard of materials and workmanship, one portion (to be decided by GE) shall be completed by the contractor well in advance (minimum two stages ahead of remaining work) as directed by GE under close supervision of the Engineer-in-Charge and shall be got approved by the GE. The workmanship of various trades and finishes of this portion shall serve as guiding sample for the work in remaining portion.

18.2 Approval of the stages and workmanship of ample portion shall be separately entered and approved in stage passing register giving reference to location for easy identification even at a later date.

19.0 **DAMAGE TO EXISTING STRUCTURE/BUILDING** Any, damage done by the existing structural during the execution of work shall e made good by the contractor at his own cost and the site of work left clean and tidy on completion . Rectification/reinstatement/making good, etc shall conform to the standard materials originally y used in the work and finished work shall match with existing work in all respects to the entire satisfaction of the GE. In case of any dispute on this account, the matter shall be referred to the CWE whose decision ion writing shall be final conclusive and binding.

Contd...

SPECIAL CONDITIONS (CONTD....)

20.0 **CLEANING DOWN** :(Refer Condition 49 of IAFW-2249, General Conditions of Contracts) :-
The contractor shall clean all floors, walls, remove cement/lime/paint marks/drops, etc, of work in connection therewith and leave the whole premises in clean and tidy condition before handing over the building.

21.0 **Re-imbursement/Refund on variation in "Taxes directly related to Contract value : -**
(a) The rates quoted by the Contractor shall be deemed to be inclusive of all taxes (including Sales Tax/VAT on materials, Sales Tax/VAT or Works contracts Turnover Tax, Service Tax , Labour Welfare Cess/tax etc.) duties, Royalties Octroi & other levies payable under the respective Statutes. No reimbursement/refund for variation in rates of taxes, duties, Royalties, Octroi & other levies, and/or imposition/abolition of any new/existing taxes, duties Royalties, Octroi & other levies shall be made except as provided in Sub Para ( ) here-in-below :-

(b) (i) The taxes which are levied by Govt at certain percentage rates on Contract Sum/Amount shall be termed as "taxes directly related to contract value" such as sales Tax/VAT on Works Contracts, Turnover Tax /Labour Welfare Cess/tax and like but excluding income Tax . The tenderers rates shall be deemed to be inclusive of all "taxes directly related contract value" with existing percentage rates as prevailing on last due date for receipt of tenders. Any increase in percentage rates of "taxes directly related to Contract value" with reference to prevailing rates on last due date for receipt of tenders shall be reimbursed to the Contractor and any decrease in percentage rates of "Taxes directly related to Contract value "with reference to prevailing rates on last due date for receipt of tenders shall be refunded by the Contractor to the Govt/deducted by the Govt from any payments due to the Contractor. Similarly imposition of a new "taxes directly related to Contract value" after the last due date of receipt of tenders shall be reimbursed to the contractor and abolition of any "taxes directly related to Contract value" prevailing on last due date for receipt of tenders shall be refunded by the Contractor to the Govt/deducted by the Govt from the payments due to the contractor.

(ii) The contractor shall, within a reasonable time of his becoming aware of variation in percentage rates and/or imposition of any further "tax directly related to Contract value", give written notice thereof to the GE stating that the same is given pursuant to this Special Condition together with all information relating thereto which he may be in a position to supply. The contractors shall submit the other documentary proof information's as the GE may require.

(iii) The Contractor shall for the purpose of this condition keep such books of account and other documents as are necessary and shall allow inspection of the same by a duly authorized representative of Govt and shall further, at the request of the GE furnish, verified in such a manner by the GE may required, any documents so kept and such other information as the GE may require.

(iv) Re-imbursement for increase in percentage rates/imposition of taxes directly related to Contract value" shall be made only if the contractor necessarily & properly pays additional taxes directly related to Contract value" to the Govt, without getting the same adjusted against any other tax liability or without getting the same refunded from the concerned Govt authority and submits documentary proof for the same as the GE may require".

NOTE FOR CLAUSE NO 15.1 & 15.2. CONSEQUENT TO IMPLEMENTATIONS OF GST ACT, CERTAIN NOTES ARE GIVEN IN SCHEDULE 'A' NOTES. SPECIAL CONDITIONS CLAUSE NOS 15.1 TO 15.2 (IV) ARE TO BE READ IN CONJUNCTION WITH SCHEDULE 'A' NOTES.

Contd..../-

SPECIAL CONDITIONS (CONTD....)

22.0 **MATERIALS AND SAMPLES**

22.1 Materials to be provided by the contractor for incorporation in the works shall, unless other wise specified in the particular specifications, comply with the requirements of the relevant Indian standard (IS) of latest edition.

22.2 The contractor shall produce samples of all materials , articles, fittings, etc. that he proposes to use and get; these approvals in writing by the Garrison engineer within 3 months from the date of commencement of work as per work order. The material's, articles, etc as approved shall be labelled as such as shall be signed by the GE and contractor's representative. These samples shall be kept I the custody of the garrison Engineer / Engineer –in-Charge.

22.3 The brand of all materials, articles, fittings etc. together with the name of the manufactures and firm from which samples have been arranged shall be recorded in the works passing register.

22.4 The quoted rates for various items shall be deemed to be inclusive of all charges on account of octroi terminal or sales tax (vat) and other duties as applicable if any. No claim whatsoever shall be entertained on this Account.

23.0 **RECORD OF MATERIALS**

23.1 The quantity of materials such as paints, wager proofing compound, chemicals for antitermite treatment and the like, as directed by the Engineer-in-Charge (the quantity of which cannot be checked after incorporation in the works), shall be recorded in measurement books and signed by the contractor and the Engineer-in-Charge as a check to ensure that the required quantity has been brought to site for incorporation in the work.

23.2 Materials brought to site shall be stored as directed by the Engineer-in-Charge and those already recorded in measurement book shall be suitably marked for identification.

23.3 The contractor shall, on demand, produce to the GE, Original receipted vouchers/invoices in respect of the supplies. The vouchers/invoices shall be defaced and stamped by Engineer-in-Charge indicating contract number, name of work, under his dated signature. The contractor shall ensure that the materials are brought to site, in original sealed containers/; acking, bearing manufacturer's marking except in the case of the requirement of material (s) being less than smallest packing.

23.4 The contractor shall produce to the GE, original receipted vouchers/invoices from the manufacturers and/or their authorized agents in respect of the items for which payment is claimed in RAR as "Materials Lying at site". The vouchers/invoices shall be defaced and stamped by Engineer-in-Charge indicating contract number, name of work, under his dated signature. The contractor shall ensure that the materials are brought to site, in original sealed containers/packing, bearing manufacturer's marking. Purchase vouchers in respect of the balance items are not required to be submitted. However, contractor shall produce original purchase vouchers of Cement & Steel as specified in particular specifications here-in-after and original receipted vouchers/invoices from the manufacturers and/or their authorized dealers for the following items :-

(a)	Chemicals	(b)	Ceramic Tiles	(c)	Aluminium Section
(d)	Plumbing Items	(e)	Sanitary Items	(f)	Paint, Distemper, Primer
(g)	PVC tank	(h)	Roofing sheets	(j)	Ceiling grid & board
(k)	APP membrane	(l)	Factory made shutters	(m)	Joinery Items
(n)	PVC joinery	(o)	Steel Windows	(p)	Galvalume Sheet
(q)	Wires & cables	(r)	light Fittings	(e)	DB & MCB

24.0 **DEFECTS LIABILITY PERIOD** (Refer Condition 46 of IAFW-2249, General Conditions of Contracts) :-
As amended vide Amendment No. 48 to IAFW-2249, General Conditions of Contracts, Period of defects liability period shall be **12(Twelve) calendar months** after the works have been handed over to the Government.

Contd..../-

SPECIAL CONDITIONS (CONTD….)

25.0 **BANK GUARANTEE BOND AGAINST PERFORMANCE SECURITY** :-

25.01 Condition 19.1 of IAFW-2249 provides for submission of Performance Security by the successful contractor in the form of bank Guarantee Bonds or Govt Securities, FDR or any other form of deposit stipulated by the Accepting Officer.

25.02 Form of bank Guarantee Bond against Performance Security Deposit shall be same as Appendix 2.1 of MES Manual on Contracts 2007 (Reprint 2012) with "security deposit" mentioned in para 1, line 4 to be replaced by "Performance Security Deposit".

25.03 The performance security shall be in favour of the Accepting Officer and shall be in any of the forms mentioned above. Work Order for commencement of work shall be placed only after submission of Performance Security Deposit of adequate value by the contractor.

26.0 **REQUIREMENT OF ELECTRICAL LICENSE FOR CARRYING OUT ELECTRICAL WORKS**

26.01 For execution of electrical works/electrical component of works requiring Electrical License in accordance with Rule 29 under Part-III of Central Electricity Authority Regulation, 2010, the contractor shall have valid electrical license issued by concerned State/Union Territory in the name of the firm or shall get the work executed through an agency having valid Electrical License. Items of works under category (c) [including its sub-categories (i) to (vii)] and d(v) shall also be executed in this manner to the extent Rule 29 mentioned above is applicable to such items.

26.02 Prior to commencement of electrical work/electrical component of works and other works as mentioned under Para 23.01 above, copy of valid Electrical License in the name of the contractor or copy of the agreement with agency having valid Electrical License alongwith the agency's Electrical license shall be submitted by the contractor to the Accepting Officer through concerned Engineer-in-Charge. In case contractor, does not possesses valid Electrical License and he intends to get such work executed through agency having valid Electrical License and with whom he has agreement, prior approval of such agency in all contracts shall be got done from the Accepting Officer.

26.03 Concerned AGE/Engineer-in-Charge shall ensure that electrical work/electrical component of works and other works as mentioned in Para 23.01 above required to be executed in accordance with Rule 29 mentioned above are executed by contractor/agency having valid Electrical license, as applicable. In addition, Supervisor for execution of electrical works/electrical component of works employed by the contractor shall possess Supervisor Certificate of Competency issued by concerned State Government/Union Territory and the worker/tradesman for execution of electrical works/electrical component of works shall hold necessary permit issued by concerned State Government/Union Territory.

26.04 Reference of Electrical license and details of Supervisor's Certificate of Competency and Permit of tradesman employed for execution of electrical work/electrical component of works shall be duly incorporated in the Works Diary and relevant site documents respectively for the whole period during which such work is executed.

28. **Testing of Materials**

28.1 **'A' Level Tests for Works Costing 100 Lakhs and above**: The contractor shall set up site laboratory for testing of materials (except Sch 'B' materials) for 'A' level tests as listed in Appx 'A'. The contractor shall arrange all equipment/ machines for the tests specified in Appx 'A' as A level tests at his own cost with prior approval of GE. This cost shall be included in the rates quoted by the contractor. The contractor shall employ a competent technical representative as approved by the GE for the purpose of testing and all such tests shall be carried out in the presence of Eng-in-charge. The successful tests result thereof shall be recorded and signed jointly by the contractor and the Eng-in-charge. The charges for these tests i.e. A level tests carried out in site laboratory of the contractor shall not be recovered. In case, the contractor has not set up the site laboratory and the tests are carried out in zonal or any other laboratory set up/approved by GE, the recovery shall be made at rates applicable i.e. as given in Appx 'A'.

Contd…./-

SPECIAL CONDITIONS (CONTD....)

28.2 **'A' Level Tests for Works Costing upto 100 Lakhs**: The contractor may set up site laboratory at his option for works costing upto 100 lakhs. The other stipulations will be same as specified in preceding para 36.1. However in case the contractor has not set up the site laboratory and tests are carried out in zonal or any other laboratory approved/set up by GE, the recovery shall be made at rates applicable ie as given in Appx 'A'.

28.3 **B & C Level Tests**: For tests of 'B' & 'C' level as indicated in Appx 'A' , the contractor shall provide all facilities for testing of materials at zonal laboratory/Govt. approved laboratories or test house/Engg. Colleges at his own cost. The rates quoted by the contractor shall be deemed to be inclusive of these tests. The rates of various tests conducted in laboratory of MES are indicated in Appx 'A'. The contractor shall bear the actual charges of 'C' level tests (to be done in labs other than MES labs) irrespective of rates indicated in Appx 'A'. Wherever it is convenient to get 'B' level test done at approved test house/Engg. College, the same can be done at the cost of the contractor and no separate recoveries will be made by the Department for the same.

28.4 The recoveries on account of testing charges wherever applicable shall be effected from the running account payments due to the contractor payable after completion of the respective tests or whenever the test is due whichever is earlier.

SIGNATURE OF CONTRACTOR **FOR ACCEPTING OFFICER**

DATED : _____

APPENDIX 'A'

MATERIAL AND THEIR TESTS

Sl No	Materials	Test		Method of testing	Frequency of Tests			Level of test	Rate per test Rs. Ps.	Remarks
1	2	3		4	5			6	7	8
1.	Brick	(i)	Compressive strength	IS-3495 (Part-II)	As per IS-5454 as given under:-			'A'	330.00	Checks for visual and dimensional characteristics shall also be carried out.
		(ii)	Water absorption	-do-	Lot size	Sample size	Permissible No of defective bricks.	'A'	330.00	As per IS-5454.
		(iii)	Efflorescence	-do- (Part-I)	1001 to 10000	5	0	'A'	330.00	Legend A-Site lab (if required) B-Zonal lab/Command Test Laboratory C-National test house/SEMT wing CME Pune/ Govt Engg College / Regional Research Labs / IITs / NITs /NABL Labs / Govt approved laboratories / Command Test laboratory (if such facilities exists)
					10001 to 35000	10	0			
					35001 to 50000	15	1			
2.	Coarse aggregate	(i)	Sieve analysis	IS-2386 (Part-I)	One test for every 15CuM of aggregates or part thereof brought to site.			'A'	660.00	
		(ii)	Flakiness Index	-do-	-do-			'A'	250.00	
		(iii)	Estimation of deleterious materials	-do-	One test for every 100 CuM of aggregates or part thereof			'A'	600.00	
		(iv)	Organic impurities	-do-	One test per source of supply			'C'	275.00	
		(v)	Moisture content	IS-2386 (Part-II)	Regularly as required			'A'	330.00	
		(vi)	Specific gravity	-do-	One test for each source of supply			'B'	330.00	

MATERIAL AND THEIR TESTS (Contd...)

1	2	3		4	5	6	7	8
3.	Fine aggregate	(i)	Sieve analysis	IS-2386 (Part-I)	One test for every 15 Cum of FA or part when brought to site.	'A'	660.00	
		(ii)	Test for clay, silt and impurities	IS-2386 (Part-I)	-do-	'A'	500.00	
		(iii)	Specific gravity	IS-2386 (Part-I)	One test for each source of supply	'B'	330.00	
		(iv)	Moisture content	IS-2386 (Part-II)	Regularly as required subject to 2 tests/day when being used.	'A'	330.00	
		(v)	Test for organic impurities	IS-2386 (Part-II)	One test for each source of supply	'C'	275.00	
4.	Cement	(i)	Setting time	IS:4031-63 affirmed 1980	Once for each consignment or as and when required.	'B'	500.00	
		(ii)	Soundness	-do-	-do-	'B'	550.00	
		(iii)	Compressive strength.	-do-	-do-	'B'	550.00	
		(iv)	Fineness	-do-	-do-	'B'	275.00	
5.	Structural Concrete (M-15 grade and above)	(i)	Slump test or compacting factor test or vee-bee time.	IS-1199	The minimum frequency of sampling of concrete of each grade shall be as under :-	'A'	300.00	Random sampling shall be carried to cover all mix units.
		(ii)	Compressive strengths	IS-516	Qty of concrete in the work (Cubic Meter) — *No of samples*	'A'	900.00	As per IS-456 clause 14 for frequency of sampling.
					1-5 — 1	'A'	900.00	
					6-15 — 2			
					16-30 — 3			
					31-50 — 4			
					51 and above — 4+1 for each Addl 50 CuM or part thereof.			

MATERIAL AND THEIR TESTS (Contd...)

1	2	3	4	5	6	7	8	9
6.	(a) PCC Block for walling (Hollow Block)	(i)	Compressive strength	IS-2185	08 Blocks out of 14	'A'	900.00	Sample:- 14 blocks from consignment of every 5000 blocks or part thereof.
		(ii)	Water absorption	-do-	03 Blocks out of 14	'B'	330.00	
		(iii)	Density	-do-	03 Blocks out of 14	'B'	330.00	
	(b) PCC solid Block for walling	(i)	Compressive strength	IS-2185	12 Blocks out of 18	'A'	900.00	Sample :- 18 blocks from consignment of every 1000 blocks or part thereof.
		(ii)	Water absorption	-do-	03 Blocks out of 18	'B'	330.00	These Block to be checked for dimension and weight.
		(iii)	Density	-do-	03 Block out of 18	'B'	330.00	
7.	Cement Flooring Tiles/Terrazzo Tiles	(i)	Water absorption	IS-1237 (Appx'D')	06 tiles out of 18	'B'	330.00	Sample of 18 tiles from each source of supply selected at random.
		(ii)	Wet transverse strength	-do- (Appx'E')	-do-	'B'	660.00	
		(iii)	Resistance to wear	-do- (Appx'F')	-do-	'C'	1000.00	
8.	Burnt clay roofing tiles (hand made) as per IS-2690 (Part II) **Length** 150 mm to 250 mm Width 100mm to 200 mm Thickness 35mm to 50mm	(i)	Water absorption	IS-3495 (Part-II)	6 tiles out of 12	'B'	216.00	Samples:- 12 tiles from each source of supply selected at random.
		(ii)	Compressive strength	-do- (Part-I)	-do-	'A'	180.00	
9.	**Mangalore pattern roofing tiles**	(i)	Water Absorption	IS-654 (Appx'A')	6 Tiles out of 32	'B'	180.00	Samples :- 32 tiles from each consignment of 3000 tiles or part thereof. These tiles shall be checked for dimension.
		(ii)	Breaking Load	-do- (Appx'C')	-do-	'B'	120.00	

MATERIAL AND THEIR TESTS (Contd...)

1	2	3		4	5	6	7	8
10.	**TIMBER**	(i)	Specific gravity and weight	IS-1708-1960	Minimum '3' samples from a lot of 4 Cum or 250 pieces of seasoned timber.	'B'	120.00	
		(ii)	Moisture content	-do-		'A'	120.00	
11.	**Water for construction purposes**	(i)	Test for Acidity	IS-456 & 3015	Once at the stage of approval of source of water.	'B'	500.00	Also refer clause 4.3 of IS-456 and its subsequent sub clauses regarding suitability of water.
		(ii)	Test for Alkalinity	-do-		'B'	500.00	
		(iii)	Test for solid content	-do-		'C'	500.00	
12	**Welding of steel work**		Visual inspection test	IS-822-1970 Clause 7.1	100% by visual inspection	Work site	360.00	Specialised tests their method and frequency to be decided on consideration of their importance by the Accepting Officer.
13	**Timber panelled and glazed Door/ Window & shutters (including Factory made shutters)**	(a)	Dimensions, size, workmanship and finish.	IS-1003-1977 (Part-I)	Frequency of sampling form each lot shall be as under :- Lot size / Sample 26 to 50 / 5 51 to 100 / 8 101 to 150 / 13 151 to 300 / 20 301 to 500 / 32 501 to 1000 / 50 1001and above / 80	'A'	180.00	
		(b)	**Strength test: -**					
		(i)	Slamming	IS-1303-1990	From the each lot 5% of the factory made shutters shall be manufacturer tested for strength tests			
		(ii)	Impact indentation	-do-				
		(iii)	Shock resistance	-do-				
		(iv)	Edge loading	-do-				

MATERIAL AND THEIR TESTS (Contd...)

1	2	3	4	5	6	7	8	9
14.	Plywood (IS-303-1989)	(a)	Moisture content	IS-1734 1983(Part-)	Six tests the pieces cut from each of the boards selected as per table shall be subjected to tests.	'C'	240.00	Sampling shall be as per IS-7835-1975 Tables.
		(b)	Moisture Content	-do-	-do-	'A'&'B'	60.00	
		(c)	Water Absorption	-do- (Part-16)	-do- (Size 300 mm X 300mm)	'A'	60.00	
		(d)	Swelling due to surface Absorption	-do- (Part-17)	-do- (Size 125 mm X 100mm)	'A'	60.00	
		(e)	Swelling in water	-do-	-do- (Size 200 mm X 100mm)	'A'	60.00	
		(f)	Modulus of rupture	-do- (Part-4)	Three test specimens as per IS-2380-77	'B'	90.00	
		(g)	Screw withdrawal strength	-do- (Part-4)	-do- As per IS-2385	'C'	120.00	
16.	Ceramic Tiles	(a)	Dimensions and surface quality	IS:13630 (Part-1)	As per requirement of IS	'A'	10.00	
		(b)	Modulas of rupture	IS:13630 (Part-6)	As per requirement of IS	'C'	100.00	
		(c)	Moisture expansion using bailing water	IS:13630 (Part-3)	As per requirement of IS	'A'	40.00	
		(d)	Surface abrasions	IS:13630 (Part-11)	As per requirement of IS	'C'	100.00	
17.	Reinforcement Steel	(a)	Physical Tests upto 16 mm dia (Normal mass, tensile elongation, bend and rebend)		As per requirement of IS	'B'	2500.00	
		(b)	More than 16 mm dia		As per requirement of IS	'B'	2750.00	

MATERIAL AND THEIR TESTS (Contd...)

1	2	3	4	5	6	7	8	9
18.	Wood Particle Board (Medium Density IS : 3097)	(a)	Density	IS : 2360 (Part-III)	Three Test Specimen from each Sample (Size 150mm x 75mm)	'A'	*60.00*	*Sampling shall be as per IS:3487 with moisture metre*
		(b)	Moisture Content	IS : 2360 (Part-III)	–do– (Size 150mm x 75mm)	'A & B'	*60.00*	
		(c)	Water Absorption	IS : 2360 (Part-16)	–do– (Size 300mm x 300mm)	'A'	*60.00*	
		(d)	Swelling due to surface absorption	IS : 2360 (Part-17)	–do– (Size 125mm x 100mm)	'A'	*60.00*	
		(e)	Swelling in Water	IS : 2360 (Part-17)	–do– (Size 200mm x 100mm)	'A'	*60.00*	
		(f)	Modulus of rupture	IS : 2360 (Part-4)	Three Test Specimen as per IS : 2380	'B'	*90.00*	
		(g)	Screw withdrawal Strength	IS : 2360 (Part-4)	Three Test Specimen as per IS : 2385	'C'	*120.00*	

PARTICULAR SPECIFICATIONS

NAME OF WORK : REPAIR /MAINT OF LT PANELS, SWITCHGEARS, LT CABLES ALONGWITH CONNECTED ITEMS OF DIFFERENT OTM ACCNS AND REPAIR MAINT OF CERTAIN MISC B/R AND E/M WORKS OF DIFFERENT ELECTRIC SUPPLY INSTALLATIONS AT INSIDE FORT WILLIAM UNDER GE (FW) KOLKATA

1. **GENERAL REQUIREMENTS**

1.1 These particular specifications shall be read in conjunction with MES Schedule 2009 (Part I) specification & 2020 Part II for Rates. General rules, specifications for materials, workmanship and standard specification, referred to the MES Standard Scheduled of Rates 2009 Part I and 2020 Part II (here-in-after referred to as MES Schedule of SSR) shall be deemed to apply to this contract unless mentioned otherwise in these tender documents. Where these are at variance, the later shall take precedence.

1.2 General specifications referred to in IAFW-2249(1989 Print) and in other documents mean specifications including other provision given in MES Schedule 2009 Part-I and 2020 Part II including amendments, errata as applicable thereto.

1.3 Reference to some paragraphs of MES Schedule has been made in these particular specifications but other paragraphs and provisions as applicable are also to be followed e.g. reference to paras pertaining to general workmanship for brick work, joinery, iron and steel work etc. have not been made but provisions therein as required for work are applicable.

1.4 Materials to be provided by the contractor for incorporation in the work shall unless otherwise specified in the particular specification comply with their requirements of the relevant Indian Standard (IS) of the year of publication (Edition/Revision) as specified in MES schedule 2009 (Part I) or published on a data prior to issue of tender whichever is later. The contractor may, however, at his own discretion provide and incorporate in the works such materials complying with the requirement of any subsequently amended or revised edition of the corresponding IS without any price adjustment of the lump sum or while pricing deviation.

1.5. Particular specifications given here-in-after are in brief and intended only to particularize, amend and emphasis the specification given in MES Schedule which are not repeated here-in-after.

1.6 Where specifications provisions given in these particular specifications are at variance with the provisions / specifications given in MES Schedule, specifications / provisions given in these particular specifications shall be followed.

1.7 Specification of materials and workmanship shall be all as directed in MES Schedule except where specifically mentioned otherwise in these documents. General rules, preambles to various section/rates, special condition, method of measurement given in MES Schedule shall also apply to the contract unless otherwise mentioned in the tender documents. The workmanship must conform to the best engineering practices.

2.0 **SCOPE OF WORK**

2.1 The work under this contract shall be carried out in accordance with Schedule 'A', the particular specifications and other provisions in MES schedule. The work shall be carried out under the jurisdiction of AGE E/M (FW) Kolkata and AGE B/R-I (FW) Kolkata.

3.0 **SITE OF WORK**

3.1 The Works shall be carried out at different Locations/Site(s). The Contractor shall finalise the programme with the Engineer-in-Charge well in advance in such a way that neither the USERS feel inconvenience nor Work is delayed.

3.2 Buildings under occupation of USERS where Works to be executed, the Contractor shall made himself aware about the Nature of the Work involved for carrying out Works against the Items catered in SCHEDULE 'A'. The Contractor is deemed to have inspected the Site of Work and nothing extra shall be admissible to Contractor on this ground.

3.3 **VISIT OF SITE**

3.3.1 The Tenderers are advised to Contact GE on any Working Day during Working Hours, who will arrange for their Visit to Site of Work.

3.3.2 The Tenderer shall be deemed to have visited the Site of Work before quoting their UNIT RATES and no Claim for any EXTRA PAYMENT on account of any misunderstanding, etc. will be entertained by the Department, irrespective of the fact whether they have actually visited the Site or not.

4 **LAYOUTS**

4.1 Exact Layout of Pipe Line, Cable, etc. and Accessories, etc. shall be as directed by Engineer-in-Charge to suit/meet the Site Requirements. No Adjustment in the quoted Unit Rates, shall be made on account of such variations in Layout.

PARTICULAR SPECIFICATIONS (Contd....,)

5. **STANDARD OF QUALITY & WORKSMANSHIP**

5.1 **GENERAL**

5.1.1 The Work shall strictly comply with the provisions contained in the Latest Edition of INDIA STANDARD CODE OF PRACTICE and/or IS SPECIFICATION as applicable, Works except where such Regulations and Rules are modified by these Particular Specifications.

5.1.2 Work shall be carried out all as per provisions specified in SSR part I and IEE rules. The work shall be carried out in a workman like manner by trained personnel in the trade. The contractor is required to produce proof regarding qualification and experience of each tradesman employed by him to the Engineer-in-Charge on demand. Damages caused to any installation as a result of faulty workmanship or due to negligence, in the opinion of the Accepting Officer, the contractor shall have to make good such damages at his own cost, for which no claim what-so-ever is admissible.

5.2. **MATERIALS**

5.2.1 All materials to be supplied by contractor for incorporation in the work shall conform to relevant specifications. In case specification for any materials needed for incorporation in the work is not contained in any of the contract documents the specification of such materials proposed to be incorporated in work shall be got approved in drawing from the GE before incorporation in the work.

5.2.2 All manufactured materials/articles other than those manufactured in contractor's workshop at site of works shall bear IS certification marks. In case any such articles do not bear the IS certification marks but conform to relevant specifications and are proposed to be incorporated in the works, samples of the same shall as per samples kept in GE's officer and shall be got approved in writing from the GE before their incorporation in the works. The contractor shall submit sufficient evidence to Engineer-in-Charge to show that such articles conform to the relevant specifications.

5.2.3 The contractor shall supply samples of materials taking into account delivery period (s) after placing firm order and allowing ample time for their testing and approval in order that if the samples supplied fail to amply with the specified requirements there will be sufficient time for him to submit fresh samples well before the materials are required to be used.

5.2.4 All manufactured materials/articles shall be brought at site in original sealed containers/packings bearing manufacturer's marking unless the quantity required is a fraction of the smallest commercial packing. Sealed container shall be opened in the presence of Engineer-in-Charge as and when required and empty container should be kept in contractor's store till the completion of the project for inspection by GE/other Inspecting Officers.

5.2.5 Tenderers attention is drawn to conditions 10A of IAFW-2249 (General Conditions of contracts) as amended vide amendment No 23. The contractor shall comply submission of proof of procurement of materials, as detailed therein. However, the purchase Vouchers in respect of following items is also required to be submitted irrespective of its value:-

(i) Cement, Steel

(ii) Materials of proprietary nature such as paints, water proofing compound, bitumen for water proofing of the roof, chemicals for anti-termite treatment and the like, quantity of which can not be ascertained after their incorporation in the works, shall be measured and recorded in the measurement books as soon as the same is/are brought to site and approved for use. Such measurements will be signed by Engineer-in-Charge and the authorized representative of the contractor. The contractor is required to obtain all such materials direct from the manufacturers/ their authorized dealers (if any) and to submit to Engineer-in Charge original stamped receipted bills along with relevant invoices in support of having brought the full quantity required for incorporation in the work and get these documents duly endorsed by the Engineer-in-Charge to the effect that the same materials and quantity have been used in the works. A record of their use shall be kept date wise in works dairy. The contractor shall comply with such instructions as are given to him in writing from time to time by the Engineer-in-Charge.

(iii) The contractor is required to submit purchase vouchers in respect of all items for which payment is claimed through RARs, as materials lying at site.

5.2.6 All building materials like aggregate, sand, bricks etc shall generally conforming to the samples kept in GE's office in addition to their conforming with relevant specifications given in the tender documents. The samples of all such materials shall be got approved from the GE in writing before these materials are brought to site in bulk. In case of any difference between samples kept in GE's office and as that specified in respective IS the specification laid down in IS shall take precedence. In case any difference in option between GE and contractor in this regard, the decision of CWE shall be final and binding. The sources of some of the materials are indicated below:-

PARTICULAR SPECIFICATIONS (Contd.....)

SER NO	MATERIAL	SOURCES
(a)	Sand	Tarakeshwar / EIam Bazar
(b)	Broken/Crushed stone aggregates	Asansol /Pakur
(c)	Bricks	Best quality locally available

5.2.7 Letters conveying approval of samples of materials by GE shall interalia mention sources of supply, name of manufacturers, traders name/brand (if any) and reference to clause of tender documents containing specifications of particular materials.

5.2.8 Contractor at his own may procure materials from other sources without any extra cost to the Deptt provided that the same conform to respective IS as stated here-in-before.

5.3 **Testing of materials** :-

5.3.1 Before incorporation in the work, the building materials, concrete etc are required to be tested for quality control in material testing laboratory.

5.3.2 The contractor may be permitted to set up site Laboratory to carry out necessary tests of materials. The various tests (as applicable) and frequency thereof shall be as given in **Special Condition 28.** The type of test to be carried out shall be as directed by GE. Contractor shall be deemed to have quoted accordingly. No claim whatsoever shall be entertained at a later date for setting up site Laboratory and testing. However, contractor will be allowed to take out all the testing equipments after satisfactory completion of work and leave the site neat and tidy. In case contractor does not set up the site Laboratory, the requisite tests shall either be carried out in MES Zonal Laboratory (if such facilities exist) at the rate shown in **Special Conndition 28** or shall be carried out at the Laboratory mentioned in PS Clause 5.3.3 here-in-after at his own arrangement and expenses.

5.3.3 However, a percentage/selected tests, as directed by the Accepting Officer, shall be got done independently in Zonal Laboratory of CE Kolkata Zone and recovery charge for these tests shall be effected from the contractor as per rate shown in **Special Condition 28**. If no such tests facility exists in Zonal Laboratory, the same shall be carried out at the following Laboratories/Institutions at the contractor's own arrangement and expenses :-

a) National Test House, Alipore, Kolkata.
b) Jadavpur Engineering College, Kolkata
c) SEMT Pune.
d) Regional Testing Laboratory, 110/111 BT Road, Kolkata.
e) Bengal Engineering College, Shivpur.
f) Regional Engineering College, Durgapur.

5.3.5 In case any test is not conducted in above said test house/college, the test should be carried out in any Govt. laboratory / Govt. approved laboratory/Institution as approved by GE.

5.3.6 The contractor shall supply all the materials, moulds etc required for testing including transportation of materials without any extra cost.

5.3.7 Contractor shall employ competent technical person as approved by the GE to carry out tests in the site laboratory established by him. Contractor's quoted rate shall be deemed to have included the expenses of employment of technical person.

6 **EARTH WORK**

6.1 **GENERAL**

6.1.1 Refer to relevant Clauses of MES Schedule Part-I. Earthwork and Excavation involved in this Contract shall be in Soft/Loose Soil to the depth & gradient as required at Site and ordered by Engineer-in-Charge.

6.2 **SURFACE DRESSING/EXCAVATION**

6.2.1 Surface Dressing including Cutting of Earth Heaps (if necessary) shall be carried out upto 3.00 METRE from the Outer periphery all as specified in Clause No. 3.10 of MES Schedule Part-I and as directed by the Engineer-in-Charge. All Rubbish obtained from Site Clearance and all Spoil obtained from Surface Dressing shall be removed, spread and leveled all as directed by Engineer-in-Charge.

6.2.2 The QUOTED UNIT RATE in relevant SCHEDULE 'A' Item (wherever required), shall deemed to be included for clearing away all Grass, Vegetation, Saplings, Shrubs, Brush-wood, Undergrowth and Roots and removing upto a distance not exc. 50.00 METRE as directed by Engineer-in-Charge.

6.3 **EXCAVATION IN TRENCHES AND OVER AREAS**

6.3.1 Excavation in trenches and over areas in soft soil shall be carried out as described in Paras 1.13.1 to 3.13.4 of MES Schedule Part-I. Bailing and pumping of water, if required, will be done as described in Para 3.17 of MES Schedule Part-I. No extra payment shall be admissible for the above mentioned operation. In the event of deviation no adjustment shall be made for cost of bailing pumping and de-watering as specified here-in-before.

PARTICULAR SPECIFICATIONS (Contd.....)

6.4 **FILLING IN TRENCHES**

6.4.1 Filling excavated Earth in Foundation Trenches, in Plinth and Under Floor, etc., shall be carried out as per Clause No. 3.19 and 3.20 of MES Schedule Part–I as applicable and as directed by the Engineer-in-Charge. Earth obtained from Excavation and not approved for filling or surplus to the requirement shall be removed to a distance not exc. 50.00 METRE spread and leveled as directed by Engineer-in-Charge. If any additional Earth is required for filling, it shall be obtained from out side MD Land without any Extra Cost to the Govt. No Royalty shall be charged for earth obtained from MD Land for using in the Work.

6.5 **BAILING/PUMPING OF WATER & PROTECTION**

6.5.1 Tenderer shall ascertain himself the Level of Sub-Soil Water at Site of Work to allow in his Tender for any Bailing and Pumping Out Water, etc., as per Site Condition. Bailing/Pumping of Water and Protection, if/as required, will be done as described in Clause No. 3.17 and 3.18 of MES Schedule Part-I. No Extra Payment shall be admissible for the above mentioned Operations.

6.6 **REMOVAL OF SURPLUS SOIL**

6.6.1 Surplus Soil and/or Rubbish obtained from Excavation shall be disposed unless otherwise specified in Particular Specifications here-in-after, upto a distance of exc. 50 Metre and not exc. 100 Metre at the Location(s) as directed by Engineer-in-Charge, deposited, spread and levelled as specified in MES Schedule Part-I.

6.7 **EXCAVATION IN POST HOLES**

6.7.1 Refer to Clause No. 3.27 of MES Schedule Part-I as applicable and as directed by the Engineer-in-Charge. Any Rubbish, Spoil obtained shall be removed, spread and leveled all as directed by Engineer-in-Charge.

6.8 **HARD CORE**

6.8.1 Refer Clause No. 3.27 and subsequent Clauses thereof of MES Schedule Part-I as applicable.

6.8.2 The material for Hardcore wherever mentioned in Particular Specifications here-in-after or as indicated in Drawings shall be of Crushed/Broken Stone Aggregate (at the discretion of Contractor without any Price Adjustment) of gauge not exc. 63 mm conforming to IS-383. In case Coarse Aggregate does not conform to CA provision, the same shall be replaced by the Contractor, with out any Extra Cost to the Govt., so as to conform the Aggregate to the relevant IS and CA provision.

6.8.3 The Hardcore shall be spread, watered and rammed. The thickness of Hardcore shown in Drawings is the consolidated thickness. Where thickness is not indicated in Drawings, it shall be 150 mm thick consolidated.

7.0 **CEMENT TO BE SUPPLIED BY THE CONTRACTOR**

7.1 Full quantity of cement required for this work shall be supplied by the contractor under his own arrangement. Ordinary Portland cement of Grade 43 or higher conforming to IS:269-2015 (latest revision) or Portland Pozzolana Cement conforming to IS:1489-1991 shall be incorporated in the work. Clause-4.3 of SSR Part-I, 2009 may also please be referred.

Note: - In case more than one type of cement is used in any work, a record shall be kept showing the location and the type of cement used. Different types of cement shall not be mixed together.

7.2 **PROCUREMENT**: -

7.2.1 Cement supplied by the Contractor shall be procured from the main producers or from their authorized dealers/stockists. The following are the main approved producers of cement :-

Sl	Company Name	Brand	Type of Cement
1)	The Associated Cement Companies Ltd	ACC	-All-
2)	Ultra Tech Cement Ltd	ULTRATECH	-All-
3)	The India Cement Ltd	-	-All-
4)	M/S Dalmia Cement (Bharat) Ltd	DALMIYA / KONARK	-All-
5)	Century Cements	CENTURY	-All-
6)	Saurashtra Cement	SAURASHTRA	-All-
7)	The Ramco Cements Ltd	RAMCO	-All-
8)	Mangalam Cement Ltd	MANGALAM	-All-
9)	Birla Corporation Ltd	BIRLA	-All-
10)	Orient Cement	ORIENT	-All-
11)	Nuvoco Vistas Corporation ltd	NUVOCO	-All-
12)	Shree Cement	SHREE	-All-
13)	JK Cement	JK	-All-
14)	JK Lakshmi Cement Ltd	JK LAKSHMI	-All-

PARTICULAR SPECIFICATIONS (Contd....,)

15)	Jaypee Rewa Cement	JAYPEE	-All-
16)	Ambuja Cement Ltd	AMBUJA	-All-
17)	M/S Shree Guru Kripa Cement (Pvt) Ltd	SARTAJ	OPC 43 Gr / PPC
18)	M/S Parasakti Cements Ltd	PRASAKTI	OPC 43 Gr / PPC
19)	M/S My Home Industries	MAHA CEMENT	OPC 43 Gr / PPC/PSC
20)	M/S Chettinad Cement	CHETTINAD CEMENT	OPC 43 Gr / PPC
21)	M/S Sanghi Industries Ltd	SANGHI	OPC 53 Gr / PPC
22)	M/S Wonder Cement Ltd	WONDER CEMENT	OPC 43 Gr / OPC 53 Gr / PPC
23)	M/S Kesoram Industries Ltd	BIRLA SHAKTI	OPC 43 Gr / OPC 53 Gr / PPC
24)	M/S JSW Cement Ltd	JSW PSC	PSC

7.3 Following conditions shall have to be satisfied before PPC will be allowed to be used in work:-

a) PPC should meet the strength criteria of 43 Grade OPC as laid down in IS : 269 – 2015.

b) Stripping time shall be 14 days.

c) In cold climate regions where temperature is lower than 15° C and important structures like overhead reservoirs, underground sumps, prestigious buildings and buildings with spans 10 metres or more, only OPC shall be used.

d) Both OPC & PPC shall not be permitted for use in the same building except for plaster & mortar.

e) While procuring PPC and PPC shall, following requirements are to be ensured and certificate to the effect is to be obtained by the contractor for each batch from the manufacturer and the same shall be submitted to the GE for approval to procure PPC :-

i) The quantity of fly ash is strictly as per IS -1489 (Part-I) – 2002.

ii) Fly ash is inter ground with clinker not mixed with clinker.

iii) Dry fly ash is transported in closed containers and stored in silos. Only pneumatic pumping should be used.

iv) The fly ash received from thermal power plants using high temperature combustion above 1000 degree C should be used.

v) The fly ash content in PPC shall not exceed 25% to ensure consistency.

7.4 The particulars of the manufacturer /main producer of cement along with the date of manufacturer shall be submitted by the contractor for every lot of cement separately. The documents in support of purchase of cement will be verified by the site staff and GE.

7.5 **STORAGE OF CEMENT:**

7.5.1 Two separate godowns shall be provided by contractor at site of work exclusively for storage of cement:-

(a) " First Godown" shall have the cement lot(s) for which independent testing is to got done and/ or under process.

(b) " Second Godown" shall have the cement lot(s) whose independent testing had been got done and after successful testing stands approved by GE/AGE (I) for incorporation in the work.

7.5.2 Cement shall be stored over dry platform with gap of 20 cm room walls all-round and at least 20 cm high from floor level. The stacking of cement shall not be more than 10 Bags high. Compliance of provisions in clause 4.3.1, 4.3.2, and 4.3.3 of MES Schedule Part I shall be ensured. Godowns shall have easy access for proper inspection & cement shall be protected from dampness/moisture to minimize storage deterioration or intrusion of foreign matter. **Cement older than Six months shall not be used in RCC works.**

7.5.3 Both the Godowns shall be provided with two locks on each door. The key of one lock at each door shall remain with the Engineer-in-Charge or representative and that of the other lock with the contractors authorized agent at the site of works so that cement is removed form the godown only according to the requirements for independent testing purposes or for daily consumption purposes with the knowledge of both the parties.

7.6 **TESTING OF CEMENT:** Contractor shall submit particulars of manufacturer of cement along with the date of manufacture for every lot of cement separately. The manufacturer is to carry out inspections and testing of cement in accordance with the relevant BIS provisions. Contractor shall also submit manufacture's test certificates in original along with the test sheet or an authenticated copy thereof giving results of each physical test as applicable and the chemical composition of cement. The documents giving above particulars shall be verified by site staff and GE/AGE (I) in support of the purchases of cement and these details shall be recorded in the cement Acceptance Register as per Performa given in Appendix 'A' to the tender documents here in after.

PARTICULAR SPECIFICATIONS (Contd....,)

7.6.1 The GE shall organize independent testing of random samples of cement drawn from the various lots which shall be got tested by the contractor under his own arrangements and the cost of both materials and labour required for sampling, testing, packing, transportation of samples to the testing places and testing charges thereof, and for subsequent identification of cement sampled, shall be borne by the contractor and is deemed to be included in the rates quoted by the contractor. The independent testing shall be got done from any of the following where facilities exist for all the tests required, viz, National Test House, Calcutta, National Test House, SEMT Wing – CME, Pune, Regional Research Laboratories, Zonal Laboratories, Govt Engineering College and Govt approved laboratories.

7.6.2 The independent testing shall be got done as per: -

(a) IS-5335 of 1986 : Method of sampling hydraulic cement.
(b) IS-4031 : Methods of physical test for hydraulic cement.
(c) IS-4032 of 1985 : Method of Chemical analysis of hydraulic cement.

7.6.3 The other relevant IS shall also be followed. However the tests shall be restricted to those indicated here in below: -

Ser No	Materials	Test	Method of Testing	Frequency of tests	Remarks
1.	Cement	(a) Setting time	IS-4031-Part-5 1988 REAFFIRMED – 2000	Once for each consignment or as and where required	
		(b) Soundness	-do- Part-3	-do-	
		(c) Compressive Strength	-do- Part-6	-do-	
		(d) Fineness	IS-4031 Part-2 1999	-do-	

7.6.4 The above tests shall be independent tests. The samples shall be taken immediately within 3 weeks of delivery at the site of works and all the tests got carried out by contractor within one week of sampling. In case it is not possible to test the samples within one week, the samples shall be packed and stores in airtight containers as decided by GE/AGE (I) till such time that they are tested.

7.6.5 The entire lot(s) of cement shall stand rejected, if it does not comply with any of the requirement of IS specifications and the test results on samples are not within the acceptable limits. The rejected lot(s) shall be removed from the site of works by the contractor under their own arrangement and cost. The cost of tests shall be borne by the contractor irrespective of the results.

7.6.6 The contractor shall use only one type brand of cement for entire structural /reinforced concrete work. In the event of contractor desiring use of any different brand of cement, fresh design mix shall be carried out and submitted for approval without any extra cost to the Govt.

7.7 **DOCUMENTATION:**

The contractor shall submit original paid vouchers from the manufactures for the total quantity of cement supplied under each consignment brought to the site of work.

7.7.1 All consignments so received at the work site shall be inspected by GE along with the relevant documents for permitting storage in the 'first godown' defined here-in-before.

7.7.2 The original vouchers and the test certificates shall be kept In record in the office of GE duly authenticated and with cross reference to the Control Number recorded in the Cement Register as per format given in Appendix-A here-in after, maintained for "first godown" duly signed by JE(Civil), Engineer-in-Charge, GE and the contractor.

7.7.3 After independent testing has been got done by GE and satisfactory results obtained and kept on record the consignment so tested shall be removed from the 'first godown' and transferred to the 'second godown' as defined here-in before. Original vouchers and the test certificates submitted by the contractor along with the consignment and test certificates of independent testing shall be defaced by Engineer-in-Charge and kept on record duly authenticated and with cross reference to the control Number rendered in the cement register maintained for the "Godwon" This register shall be termed as "**Cement Acceptance Register"** which shall be signed by JE(Civil) , Engineer-in-Charge, GE and the contractor. Besides this the quantity of cement shall be also suitably recorded in the Measurement Book for record purposes before incorporation in the work and shall be signed by Engineer-in-Charge and the contractor.

7.7.4 The Accepting Officer may order a Board of Officers in which contractor shall also be a member for random check of cement and verification of connected documents.

7.7.5 Daily inspection shall be carried out by Engineer-in-Charge or his representative and contractor's authorized agent and separate register shall be maintained duly indicating on daily basis the inspection carried out and indicating holding in godown at beginning of the day, inward supplies in the day, removal in the day and balance at the close of the day duly signed by above persons. In the removal column particulars of lot removed and purpose for which removed shall be indicated.

7.7.6 The purchase voucher and test certificate of manufacturer shall be linked.

PARTICULAR SPECIFICATIONS (Contd....,)

7.8 **WEIGHING AND PAYMENT FOR CEMENT**.

7.8.1 Random sample of 5 bags per 100 bags shall be got weighed by Engineer-in-Charge in presence of Representative of contractor from each consignment to arrive at average weight per bag for calculating total quantity of cement bought for incorporation in the work. All arrangement for weighing of cement bags including labour etc. shall be provided by the contractor at their own cost.

7.8.2 The quantity of cement arrived at as aforesaid shall be considered for payment as applicable in accordance with IAFW-2249, General Conditions of Contracts.

7.8.3 Only when the rest results of independent testes carried out on the cement are found satisfactory and it is permitted to be stored in the 'Second Godown' and quantity entered in MB, then only this lot shall be considered for payment in RARs.

7.8.4 Payments shall be restricted as per provisions in clause No 64 "Advance on Account: of General Conditions of Contracts (IAFW-2249).

8.0 **CONCRETE WORK**

8.1 Coarse and fine aggregate for cement concrete: Refer paras 4.4.1 to 4.4.6 of MES Schedule Part-I. The coarse aggregate shall be crushed stone aggregate and fine aggregate shall be sand. Use of sand conforming to grading Zone-IV of IS 383 shall not be allowed in RCC/PCC work.

8.1.1 Grading of coarse aggregate for plain and reinforce concrete i.e. stone aggregate shall be as per Para 4.4.7.1 of MES Schedule Part I.

8.1.2 Fine aggregate shall be natural sand or crushed stone free from admixture of clay, loam, silt, organic matter and other impurities and shall conform to Zone II of table IV of IS-338. The aggregate shall be drawn from approved quarries.

8.1.2 All cement concrete, shall be mixed in mechanical mixer. However, in case of small quantity, the contractor may obtain written permission of Engineer-in-Charge, to adopt hand mixing over watertight platform and care shall be taken to ensure mixing is uniform in colour and consistency.

8.1.2 Samples of the aggregates proposed to be used shall be approved by the Accepting Officer, prior to bulk delivery to the site of work. Field tests for determining the contents of silt, loam, clay etc for fine aggregate shall be carried out by the Engineer-in-Charge from time to time to ensure that materials brought to site are inconformity with the samples approved by the Accepting Officer.

8.2 **Water proofing compound:** Refer Para 4.8 of MES Schedule Part I.

8.3 **Mix & grade of cement concrete**: Refer Para 4.11.1, 4.11.2 and 4.11.2.1 to 4.11.2.4 of MES Sch Part I.

8.4 **Workability of concrete**: Refer to Para 4.11.4 of MES Schedule Part I & Para 7 of IS-456-2000.

8.5 **Mixing of concrete**: Refer to para 4.11.5 and 4.11.5.1 of MES Schedule Part I.

8.6 **Transportation of concrete**: Refer to paras 4.11.9 of MES Schedule part I.

8.7 **Placing of concrete**: Refer para 4.11.10. to 4.11.10.3 of MES Schedule part I.

8.8 **Compaction of concrete**: Refer Para 4.11.11 of MES Schedule Part I. Compaction of concrete in RCC slab, beam walls and columns shall be done with approved mechanical vibrator. In other locations hand compaction may be permitted by taping or Roding as approved by the GE.

8.9 **Curing of concrete**: Refer to Para 4.11.13 of MES Schedule Part I.

8.10 **Protection and Inspection of Concrete**: Refer paras 4.11.14 and 4.11.15 of MES SSR part-I.

8.11 **Concrete covers to reinforcement**:Refer to paras 4.11.8.1 to 4.11.8.4 of MES Schedule Part-I.

8.12 **CURING & MINIMUM PERIOD FOR STRIKING OFF FORM WORK WITH POZZOLANA CEMENT USED.**

8.12.1 Where pozzalana cement is used for RCC works in various locations, the minimum period for curing and striking off formwork shall be as under in lieu of that specified in clause 4.11.6.3 of MES Schedule Part-I.

8.12.2 **Curing**: -

(i) Structural RCC Work : 21 days

(ii) PCC Work, Plastering Etc. : 14 days

8.12.3 **Striking off form work**: -

(i) Walls ; columns and vertical sides of beams : 3 to 4 days

(ii) Slabs(Props left under) : 8 Days

(iii) Beams soffits(Props left under) : 9 Days

(iv) Removal of props to slab/beam:-

(aa) Spanning up to 6M Span : 16 days

(ab) Spanning beyond 6M Span : 22 days

(v) For cantilever portion of slab, form work with support shall be retained until the completion of the casting of the entire frame work of the building.

PARTICULAR SPECIFICATIONS (Contd....)

8.12.4 In case of bad weather, periods mentioned above may be revised at the discretion of the Engineer-in - Charge. The contractor shall be deemed to have considered the above provision before tendering and quote rate accordingly. No claim will be entertained if longer periods required for striking off form work and curing and all such effected matters is looked into upon the use of pozzolana cement as a result being instructions issued to the contractor in full or part of for bad weather.

8.12.5 Curing shall be carried out using pump of suitable capacity at no extra cost to the Govt all as directed by Engineer-in-Charge.

9.0 **FORM WORK** : Refer to Para 7.15 of MES Sch Part I. Unless otherwise specified under relevant item of BOQ, form work shall be for rough finish. Camber shall be provided as per direction of Engr-in-Charge.

10.0 **FLY ASH CEMENT BRICKS**

10.1 Fly ash cement bricks (here after termed as bricks) shall conform to the requirements laid down in IS 13757-1993 and IS 12894 – 2002. Bricks shall have minimum compressive strength 75 Kg/CM2 (Class designation 7.5).

10.2 The brick shall have smooth rectangular faces with sharp corner and slight round edges, and shall be well burnt, uniform in colour, free from cracks, flaws, nodules of lime & emit clear ringing sound when struck.

10.3 Type and size of Bricks shall be as under :-

10.3.1 Type of bricks - Non modular bricks

10.3.2 Size of bricks - 250 x 125 x 75mm

10.3.3 Dimensional Tolerance for group of 20 bricks :-

Length 20 x Length of Bricks + 80mm
Width 20 x width of Bricks + 40mm
Height 20 x height of Bricks + 40mm

10.1.1.2 **SAMPLING & TESTING** – Sampling & Testing of Bricks shall be carried out by the Contractor as per IS-5454 and IS-3495 (Part-I to Part-IV) and as directed by GE without any Extra Cost to the Department.

10.1.2 **CEMENT -** Cement shall be as specified in CONCRETE WORK here-in-before.

10.1.3 **SAND**

10.3.1 Refer Clause No. 5.4 and 5.4.1 of MES Schedule Part-I. Sand for the Work shall be naturally occurring River Bed Sand and conforming to Sample kept in GE's Office and/or as approved by GE. Sand shall not contain any type of harmful Materials, shall be free from all type of Impurities, Dust, Silts, etc.

10.1.3.2 Sand shall be stored at Site in Bins or in Dumps over Brick Platform or by using any other approved method so as to avoid contamination or shoveling all as directed.

10.1.4 **WATER :-** Refer Clause No. 5.18 of MES Schedule Part-I.

10.2 **WORKMANSHIP**

10.2.1 **GENERAL** – Refer Clause No. 5.19.1, 5.19.2, 5.19.6, 5.19.7 and 5.20 to 5.32 of MES Schedule Part-I. During execution of Work, Centre Line Dimensions shall be maintained.

10.3 **HALF BRICK WALLS**

10.3.1 Refer Clause No. 5.37 of MES Schedule Part-I and shall be constructed in Stretcher Bond.

10.3.2 The Width of the Bricks for Half Brick walls shall be considered as 110mm and deviation if any shall not be made for Bricks used. Centre Line dimensions shall, however be maintained in case of variation in size of Bricks from that specified in Tender. No adjustment of Extra Cost due to increase in width of Foundation Concrete, Walls, Lintels, Band & Blocks etc. shall be admissible to Contractor on this account.

11. **PLASTERING AND POINTING**

11.1 Cement: Cement shall be as specified for concrete work.

11.2 Sand: Refer paras 14.5, 14.5.1, and 14.5.2 of MES Schedule Part-I.

11.3 Water: Water shall be as specified for concrete work.

11.4. The thickness of plaster mentioned in Sch 'A' is finished thickness exclusive of dubbing out, plastering and dubbing out shall however, be done in one operation.

11.5 **WPC:** Integral water proofing compound (WPC) shall conform to IS -2645 and the brand shall be got approved in writing from the GE. Preparation of mix & method of mixing shall be as per manufacturer's instructions. While pricing deviation for omission, qty shall be considered as 3% by weight of cement.

12. **WOOD WORK (CARPENTERS WORK AND JOINERY)**

12.1 Form work: Refer to Para 7.15 of MES Schedule Part I. The formwork shall be for rough finish. No chamber in form work as mentioned in para 7.15.7 of MES Schedule Part I need be provided.

PARTICULAR SPECIFICATIONS (Contd....,)

12.2 Timber and defects in timber: Refer to para 7.3.1,7.3.1.1,7.4.1,7.4.2,7.5.1 & 7.5.2 of MES Sch Part I.

12.3 Timber for wooden work and joinery (Except particle board and block board) shall be of second class hardwood as mentioned in Schedule 'A' and the following species: -

(i)	Timber for fillets work etc.	-	Second class hard wood of Hollock/ Panisaj / Kalasiris / Chickrashi.
(ii)	Timber for door / window frame	-	Second class hard wood Sal.
(iii)	Timber for scantling work etc.	-	Second class hard wood of Siris / Gamari/ Panisaj / Kalasiris

12.4 Workmanship General: refer para 8.15 of MES Schedule Part-I. Unless otherwise specified in Sch 'A' all exposed faces of woodwork and joinery shall be wrought. Unexposed surfaces shall be clean sawn.

12.5 **SEASONING**

12.5.1 Seasoning and moisture contents: refer to para 7.6 and 7.7 of MES Schedule Part I. Moisture contents shall be as specified for Zone IV.

12.5.2 All timbers shall be kiln season by suitable process conforming to IS-1141-1973 (Code of practice seasoning of timber) to maintain the moisture content as permissible tabulated in clause 7.7 of SSR Part-I (2009) and in IS-287-1973 (recommendations of maximum permissible moisture content for timber used for different purpose). Seasoning shall be carried out in two stages i.e. to remove partial moisture content before treatment and then to maintain desired percentage of moisture content after treatment.

12.6 **TREATMENT OF TIMBER**

12.6.1 All timber incorporated in the work to be manufactured at site shall be treated as mentioned in IS 401 by modified hot and cold treatment, the minimum quantity of chemical/impregnated in each cubic metre of timber shall be at the rate as specified in Table 2 Group 5 of the IS. Cost of treatment to woodwork shall be deemed to be included in the cost of respective item of Sch 'A'.

12.6.2 Rates of Schedule 'A' for items of woodwork and joinery shall be deemed to include above preservation treatment.

NOTES: -

1. The above treatment is not applicable to particle board/plywood and to teak wood members, which shall be treated as specified in relevant IS.
2. The composition & concentration of chemical s shall be as per Clause 3.42 & 3.43 of IS-401.
3. ASCU preservation will be accepted against copper chrome arsenic composition without any price adjustment.
4. The contractor may, however use factory made pressure treated timber with prior approval of GE in writing without any price adjustment. The contractor shall produce certificate from the factory for such treatment.

13. **JOINERY**

13.1 **PANELLED,GLAZED AND GAUZED SHUTTERS.**

13.1.1 Refer Para 8.21.1 to 8.21.7 & 8.24 of SSR Part-I. Panelled, glazed, gauged shutters shall be factory made.

13.2 **FLUSH DOOR SHUTTERS**

13.2.1 Refer para 8.25 and ,8.25.1 to 8.25.9 of MES Schedule Part-I. Flush door shutters shall be ISI Marked [IS: 2202 (Part-I) – 1999.

13.3 **PVC DOOR AND FRAMES**

13.3.1 Irrespective of whatever shown on drawings, toilet door frames shall be factory made solid PVC door frames. The same shall be of size 50 mm x 47 mm made out of 5 mm PVC sheet. The door profiles are to be reinforced with 19 x 19 mm MS square tube. EDPM rubber gasket weather seal shall be provided throughout the frame. The profile shall be mitre cut at two corners & joined by inserting 2 nos. 150 mm long brackets of 15 x 15 mm square tube inside 19 mm x 19 mm M S square tube & suitable screwed.

13.3.2 Irrespective of whatever shown on drawings, toilet door shutters shall be PVC panelled shutter with styles, top & bottom rails of size 90 x 37 mm & lock rail of size 105 x 37 mm with multi cavity hollow sections 2 mm average wall thickness all as shown at detail in drg No. AF/TD/142 sheet 1/1. The panel inserts shall be multi cavity hollow sections of size 100 x 20 mm with wall thickness of 1.2 mm. The corners of styles & rails shall be mitred cut & fixed with solid PVC 'L' shaped brackets two nos of size (200 mm x 100 mm & 100 mm x 100mm) at each corners. Two nos 6 mm dia MS rods with two coats of red oxide primer shall be inserted horizontally at appropriate levels through styles & panel inserts with check nuts and PVC washers. The door shall be as approved by GE and fixing shall be done as directed by Engineer-in-Charge.

PARTICULAR SPECIFICATIONS (Contd.....)

14. **BUILDER'S HARDWARE**

4.1 GENERAL : Articles of builder's hardware (fittings) shall be of aluminium alloy. The type and size of fittings shall also be as indicated. Fittings shall be of approved make. Refer para 9.2.1, 9.2.2, 9.2.3, 9.2.4 & 9.2.5 of SSR Part I.

14.2 Screws – Refer para 9.2.6 of SSR Part I.

14.3 Tower bolt - Refer para 9.3 of MES Schedule Part-I. The diameter of bolts shall be 10 mm and the bolts shall be barrel bolts.

14.4 Butt hinges - Refer para 9.7 of SSR Part- I. Butt hinges shall be of pressed steel/cold rolled mild steel medium weight.

14.5 Ball catch shall be of cast brass all as specified in para.9.14 of SSR Part-I.

14.6 All builder's hardware shall be ISI marked except bow handles which shall conform to relevant IS.

14.7 Mortice Lock – Refer para 9.9 and subsequent paras thereof. Lock shall be of approved make and shall be fixed as per manufacturer's instructions.

14.8 Handles - Refer para 9.11 and subsequent paras thereof . Lock shall be of approved make and shall be fixed as per manufacturer's instructions.

14.9 Floor Door stoppers - Refer para 9.15 and subsequent paras thereof . Lock shall be of approved make and shall be fixed as per manufacturer's instructions.

14.10 Door Closers - Refer para 9.16 and subsequent paras thereof. Lock shall be of approved make and shall be fixed as per manufacturer's instructions.

14.11 Wire cloth for mosquito proofing shall be stainless steel wire cloth 0.36 mm nominal dia. of wire and average width of aperture 1.40 mm fixed with stainless steel wire staples.

15.0 **STEEL**

15.1 **Type of Steel**

15.2 **Reinforcement steel for all locations and structures except sump :**

Steel bars for concrete reinforcement wherever shown on drawings shall be TMT bars produced by Thermo Mechanical Treatment Process (Grade Fe 500 or Fe 500D), meeting all requirement of IS-1786-2008. Wherever mild steel reinforcement bars are shown on drawings these shall be amended to read as TMT bars of same dia or nearest higher dia and accordingly to be provided.

15.3 **Structural Steel**

Structural steel shall be any of the following two types depending upon the use :-

15.4 **Structural Steel E 250 (Fe 410W A & B)** :

All types of structures including those subjected to dynamic loading and where fatigue, wide fluctuation of stresses, reversal of stresses and great restrain are involved e.g. residential buildings, office building, workshop building, platform roofs, foot over bridges, transmission towers, crane gantry girders, road bridges etc.

15.5 **Structural Steel E 165 (Fe 290)** :

Grills / guard bars, hold fasts, railing, fencing, steel windows and steel doors etc.

15.6 **SOURCES OF PROCUREMENT**

15.7 **Structural Steel** : The structural steel E 250 (Fe 410W A & B) conforming to IS-2062 shall be procured from the following manufacturers :-

Sl No	Company Name	Brand	Type of Steel
1)	Steel Authority of India Limited	SAIL	-do-
2)	Rashtriya Ispat Nigam Limited	RINL	Structural Steel (Angle, Bream, Column, Channel. Plate)
3)	Tata Iron & Steel Company (TISCO or TATA STEEL)	TATA	-do-
4)	M/S Jindal Steels and power Ltd	JINDAL	-do-

15.7.1 In case of non availability with primary producers, the structural steel can be procured from approved secondary producers with a reduction of 5% (five percent) of the accepted rate of structural steel. In case the desired section of structural steel is not rolled/manufactured by primary producers, there shall be no price adjustment in use of structural steel procured from approved secondary producers. Secondary producers for structural steel are approved by E-in-C's Branch from time to time.

15.7.2 Structural Steel of grade E-165 (Fe 290) conforming to IS:2062 can be procured by the contractor from main producers / approved secondary producers / BIS marked manufacturer's or their authorized dealers at the option of the contractor.

PARTICULAR SPECIFICATIONS (Contd.....)

15.8 **Reinforcement Steel** : TMT Steel Bars shall be procured from the following manufacturers :-

Sl No	Company Name	Brand	Type of Steel	Valid Upto
1)	Rashtriya Ispat Nigam Limited	RINL	-All-	-
2)	Tata Iron & Steel Company (TISCO or TATA STEEL)	TATA	-All-	-
3)	Steel Authority of India Limited	SAIL	-All-	-
4)	M/S Jai Balaji Industries Ltd	BALAJI SHAKTI	Fe 500, Fe 500D	Feb 2025
5)	M/S Shyam Steel Industries	SHYAM	Fe 500, Fe 500D, CRS	Jan 2025
6)	M/S Steel Exchange India Ltd	SIMHADRI TMT	Fe 500, Fe 500D, HSCRM	Jan 2025
7)	M/S Jindal Steels and Power Ltd	JINDAL PANTHER	Fe 500, Fe 500D, Fe 550, Fe 550D & CRS (8-40mm)	Feb 2025
8)	M/S SRMB Srijan Pvt Ltd	SRMB	Fe 500, Fe 500D, Fe 550, Fe 550D (8-32 mm dia)	Aug 2022
9)	M/S Shri Bajrang Power & Ispat Ltd	GOEL TMT	Fe 500, Fe 500D (8-32 mm dia)	07 Jan 2020
10)	M/S JSW Steel Ltd	NEOSTEEL	Fe 500, Fe 500D, Fe 550D, CRS (8-40 mm dia)	06 Feb 2021
11)	M/S Shyam Metalics & Energy Ltd	SEL	Fe 500, Fe 500D (8-32 mm dia)	Dec 2020
12)	M/S Kamachi Industries Ltd	KAMACHI	Fe 500, Fe 500D, Fe 550, Fe 550D, HCRM (8-40 mm dia)	May 2021
13)	M/S Real Ispat & Power Ltd	G K TMT	Fe 500, Fe 500D (8-36 mm dia)	Jan 2020
14)	M/S Super Smelters Ltd, Kolkata	SUPER SHAKTI	Fe 500, Fe 500D, Fe 550 (8-32 mm dia)	Feb 2020
15)	M/S Electrotherm (India) Ltd	ET TMT	Fe 500, Fe 500D & CRS (8-32 mm dia)	Jan 2021
16)	M/S Adhunik Industries Ltd	ADHUNIK Fe 500 SD	Fe 500D (8-32 mm dia)	04 Mar 2021
17)	M/S Gallantt Metal Ltd	GALLANT TMX	Fe 500, Fe 500D, CRS (8-32 mm dia)	05 Jul 2022
18)	M/S Rashmi Metaliks Ltd	RASHMI TMT	Fe 500 (8-32 mm dia), Fe 500D, Fe 550D (8-25 mm dia)	23 Aug 2022

15.9 **Galvanized steel sheets & Fabric reinforcement for concrete** :

(a) These shall be procured from main producers i.e SAIL, Rashtriya Ispat Nigam Ltd, TISCO or from BIS marked manufactures at the option of the contractor.

(b) The documents in support of purchase of steel/BIS Licence shall be verified by Engr-in-Charge & GE.

15.10 However, reinforcement steel, Structural steel and Galvanised Steel sheets and Fabric reinforcement for concrete may be procured from authorized dealers of other main producers in case of value not exceeding Rs.7.5 lakhs.

15.11 **TESTING OF REINFORCEMENT STEEL BY MANUFACTURER** :

The manufacturer shall carry out inspection & testing of steel in accordance with the relevant BIS provisions. The contractor shall submit the producer's (not from his authorized conversion agent) test certificate & Purchase Vouchers in original along with the test sheet giving the result of each mechanical test as applicable & the chemical & metallurgical composition of the steel or authenticated copy thereof duly signed by manufacturer with each consignments as per Appendix **'C'.**

15.12 **DOCUMENTATION** :

15.12.1 The contractor shall submit original purchase vouchers from the manufacturer for the total quantity of steel supplied under each consignment to be incorporated in the work. All consignments received at the work site shall be inspected by the GE along with the relevant documents before acceptance. The original purchase vouchers and the test certificates shall be defaced by the Engineer-in-Charge and kept on record in the office of the Garrison Engineer (GE) duly authenticated and with cross reference to the consignment number recorded in the Steel Acceptance Register. The Steel Acceptance Register shall be signed by the JE (Civil), Engineer-in-Charge, GE and the contractor. The contractor shall maintain schedule of supply of steel for each consignment.

PARTICULAR SPECIFICATIONS (Contd.....)

15.12.2 The Engineer-in-Charge shall record details of steel procurement and testing in "Steel Acceptance Register" and in the format as given at Appendix **'C'**, after due verification and send a certified true copy of test sheet to GE for his records.

15.12.3 GE / CWE shall organize independent testing or random samples of steel drawn from various lots from any of the organizations as listed for testing of cement as per the recommended minimum frequency. The elongation shall not be less than 18%. The cost of samples and transportation shall be borne by the contractor. The cost of testing shall be borne in accordance with Condition 10A of IAFW-2249 i.e. testing charges shall be borne by the Deptt if the test results are found to be in order, otherwise these shall be borne by the Contractor. The records of such checks would be maintained in the steel test register.
Any particular bar / steel section of any consignment not meeting the requirement shall be rejected and the same shall be removed from the site by the contractor at his own cost and the contractor shall have no claim on this account.

15.13 **TESTING OF REINFORCEMENT STEEL & STRUCTURAL STEEL E 250 (FE 410W A & B)**

15.13.1 **High Strength deformed Steel / TMT bars** :
Nominal mass test, tensile test, bend test and rebend test shall be carried out in National Test House, Govt. approved lab or any Govt Engineering colleges.

15.13.2 **Mild Steel Bars** :
Nominal Mass test, Tensile test and bend test shall be carried out as per the minimum frequency mentioned in Para 12.6.3 here-in-after all as per IS-432 (Part-I) – 1982.

15.13.3 **FREQUENCY FOR NORMAL MASS, TENSILE AND BEND TESTS**

A. **REINFORCEMENT STEEL**

NOMINAL SIZE OF BARS	QUANTITY
(i) Under 10 mm	1 sample consisting of 3 specimens for each test from each size of bars for every 25 ton or part thereof.
(ii) 10 mm to 16 mm	1 sample consisting of 3 specimens for each test (inclusive) from each size of bars for every 35 ton or part thereof.
(iii) Over 16 mm	1 sample consisting of 3 specimens for each test from each size of bars for every 45 ton or part thereof.

B. **STRUCTURAL STEEL E 250 (FE 410W A & B)**

(i) Tensile Test	One test for every 25 tonnes of steel or part thereof
(ii) Bend Test	One test for every 10 tonnes of steel or part thereof

If any test result of any particular size of bar / section of steel of any consignment is not found satisfactory as specified in relevant IS, tests may again be carried out (if desired by contractor) for that particular size of bar/section of steel. Even after carrying out tests again if that size of bar/section of steel is not found satisfactory, the contractor shall remove the same at his own cost & no claims of contractor shall be entertained on this account. The charges for testing again shall be borne by the contractor.

15.14 **STORAGE, ACCOUNTING, PRESERVATION AND MAINTENANCE OF STEEL** :
Refer Clause 4.13 of IS-4082-1996 in this respect. The storage, accounting, preservation and maintenance of steel supplied by the contractor shall be done as per standard engineering practice till same is incorporated in the work and the cost of the same shall be deemed to be included in the unit rate / amount quoted by the tenderer. The GE shall inspect at regular intervals to verify the steel being brought at site is stored, accounted, preserved and maintained as per the norms. The steel shall be stored so as to differentiate each consignment separately. If GE is not satisfied with the storage / preservation of any size of bar / section of steel, he may order for any test(s) of steel as applicable for that size of bar / section of steel and as specified in tender documents / relevant Indian Standard to re-check the acceptability criteria for the same. The contractor shall bear the cost of necessary testing in this regard and no claim whatsoever shall be entertained.

15.15 **MEASUREMENT AND PAYMENT OF STEEL** :

15.15.1 The entire quantity of all steel items shall also be suitably recorded in the measurement book for "record purposes" as "not to be abstracted" before incorporation in the work and shall be signed by the Engineer-in-Charge and the Contractor. The nominal mass conversion factors for various steel section / size of finished bars as given in relevant IS codes shall be considered standard for measurement. The contractor shall plan to procure steel considering the time required for approval and testing well in advance. No extension of time on grounds of delay in procurement or testing / retesting of steel shall be admissible.

PARTICULAR SPECIFICATIONS (Contd.....)

15.15.2 The payment of steel shall only be allowed after production of original purchase vouchers and test certificates from the main producer / secondary producer by the contractor for each consignment of steel and results of testing carried out by the Department (if any) are found satisfactory.

15.16 **ASSEMBLY AND PLACING OF REINFORCEMENT BARS** :

15.16.1 Reinforcement bars shall be TMT bars and shall be provided as indicated in drawings. For the purpose of pricing deviation involving reinforcement bars, the length of each bar shall be considered as 10 (Ten) meters for calculating laps unless otherwise shown on drawings.

15.16.2 Assembly and placing of reinforcement bars shall be as specified in Clause 12.1, 12.2, 12.3, 12.5 and 12.6 of IS-456-2000 and as shown in drawings. In case of any variation the drawings shall be followed. Cover to reinforcement bars shall be as per Clause 26.4.2.1, 26.4.2.2 and Table – 16 (for moderate exposure) of IS 456-2000.

15.16.3 In case of pricing deviations 10 mm and over dia TMT bars as stirrups shall be priced at the rate of TMT bar stirrups of 5 mm to 10 mm in MES Schedule Part – II subject to Contractor's percentage.

15.17 **MILD STEEL WIRE**

15.17.1 Mild steel wire shall be all as specified in Schedule 'A' and as per clause No. 10.31 on Serial Page No. 214 of SSR Part-I 2009.

15.18 **EXPANDED METAL STEELS**

15.18.1 Expanded metal steel shall be all as specified in Schedule 'A' and as per clause No. 10.35 on Serial Page No. 215 of SSR Part-I 2009.

15.19 **MILD STEEL GUARD BARS/OTHER STEEL ITEMS**

15.19.1 Mild steel guard bars shall be 10 mm square bars of Fe-310-0 or Fe-410-0 quality. Guard bars shall be provided to all windows & vents ass shown on drawings. The guard bars including windows shall be painted with 2 coats of synthetic enamel paints over a coat of red oxide chrome primer.

15.19.2 Other steel items like steel doors with PGI sheet covering, railing in balconies and stair cases, water meter and switch board boxes etc shall be fabricated to the dimensions as shown on drawings and as directed by Engr-in-Charge. Welding, bolting, drilling holes etc shall be as per specified in MES schedule.

15.20 **STEEL WINDOWS AND VENTILATORS**

15.20.1 Refer Paras 10.25.1 to 10.25.12 of MES Schedule Part-1 Steel window and ventilator shall be from steel roiled section and shall conform to IS-1038 and of approved manufacturers. The section for steel windows shall be as per IS-7452-1982 & as specified in SSR Part-I subject to the following :-

a) Size shall be as shown on drawings.
b) While supplying steel windows, the same shall have one shop coat of anticorrosive paint.
c) Windows shall be given another coat of anticorrosive priming coat of paint before fixing.
d) Window fittings shall be from the approved manufacturer conforming to relevant IS.
e) Sub-dividing bar of the units shall be tenoned and riveted to the frame and not welded.
f) Steel windows (side hung) shall have both peg stay (300mm long) and heavy type box hinges (75mm) long
g) Steel windows shall be fixed by securely embedding.
h) Lugs shall be fixed in cement concrete blocks of appropriate size.
j) MS grills shall be provided to all steel windows and MS guard bars shall be provided to ventilators as shown on drawings.
k) All steel windows and ventilator shall be fixed in position as per IS-1081
l) The joints of all steel windows shall be flush Butt welded.

15.20.2 Steel windows, ventilators shall be factory made & shall be procured from approved manufacturers.

16. **FABRICATION & ERECTION OF STEEL TRUSS**

16.1 **GENERAL**

16.1.1 Refer to Clause No. 10.16 of MES Sch Part-I as applicable and as directed by the Engr-in-Charge.

16.2 **COMPONENT MEMBERS :-** All Component Members shall all as specified here-in-before.

16.3 **FABRICATION :-** Fabrication of all Components shall be carried out all as specified here-in-before.

16.4 **ERECTION OF TRUSSES**

16.4.1 Erection shall be carried out as specified in Clause No. 10.16 of MES Schedule Part-I.

16.4.2 The Trusses shall be lifted only at nodes. The Trusses shall not be aligned at the Apex. It shall be lifted by slinging at Two Mid Points of Rafter, which shall be temporarily braced by suitable Section. After Trusses are placed in position, Purlins and Wind Bracing shall be fixed as soon as possible.

16.4.3 Contractor's particular attention is drawn to IS-7205 (Safety Code for erection on Structural Steel Work). Anchor bolts for fastening of steel structure shall be set in designed position and grouted alongwith foundation. Also refer Clause No. 10.16.2 and 10.16.3 of MES Sch Part-I.

PARTICULAR SPECIFICATIONS (Contd.....)

16.4.4 The erected parts of structure shall be stable during all the stage of erection and the structural elements to be erected, shall be stable and strong to bear erection load.

16.4.5 Working on the already erected structure is permitted only after they are finally fixed. Erection of structures of each tier of high structure shall be executed only after the relevant fastening of lower ties by the permanent or temporary fastening devices as clarified for safety.

17. **ROOFING**

17.1 **GALVALUME SHEET ROOFING /WALL CLADDING.**

17.1.1 Sheets: - Refer para 11.3, 11.3.1 and 11.3.2. of MES Schedule Part-I. Galvalume coloured profile roofing sheet with 0.50 mm thick (total coated thickness) of 550 MPa minimum yield strength shall be provided under the subject work. Sheet shall be provided and fixed as specified in SSR part-I.

17.1.2 Roofing Accessories: - Refer para 11.3.3 of MES Schedule Part-I. Roofing accessories i.e. corner piece apron, eaves/valley gutters, hoopers, ridges, sheet floor connectors etc. shall be of galvalume sheet and of size shape as specified and directed by Engr-in-Charge.

17.1.3 **WORKMANSHIP:-**

17.1.3.1 Fixing of galvalume roofing shall be done as per manufacturer's instructions and shall be safe against effects of wind velocity. Sheets shall be laid on to purlins with hot dipped galvanized self drilling, self tapping screws, neoprene and EPDM washers. Screws shall be of 24 TPI, self tapping, self drilling type conforming to class 3 of AS 3566. For roof sheeting, screws shall be spaced at every 500mm centre to centre in general but at 250 mm centre to centre at eaves, end laps and ridge level. Sheets shall be supplied in lengths suited to purlin spacing and as per approved sheet layout only. Precaution shall be taken during the erection of sheets to ensure that maximum weather protection against prevailing wind direction is obtained.

17.1.3.2 Roof sheets shall be laid with minimum bearing of 50 mm on purlins and end projection of 75 mm (for slopes up to and including 15 degrees) or 100 mm (for slopes below than 15 degrees) to give end-laps of 150 mm or 200 mm respectively. For side cladding manufacturer's recommendations shall be followed for hidden fixing system.

17.1.3.3 The minimum lap of the sheet with ridges, hips and valleys shall be 200 mm measured at right angles to7 the line of the ridge, hip or valley as applicable. Flashings, closures etc. on vertical surfaces shall be provided with minimum lap of 150 mm.

17.1.3.4 Flashings, hips, ridges, valleys etc. shall be made from plain sheets, in colour and shade matching to the sheets used in roof / side cladding. These shall be cut to suit the dimensions and required shape of the roof either along their length or their width as required, and, shall be so formed that their ends slant across the front of main sheet parallel to the trapezoidal or curvature form.

17.1.3.5 Penetrations and laps in sheet shall be sealed by using proper sealant profile. HDPE fillers shall be provided wherever required to close voids between sheets, sheet & fasteners.

17.1.3.5 In areas where dust proof enclosure is required, the lap joints shall be sealed using two-part polysulphide sealing compound after proper surface preparation or to be filled with silicone sealant. Nominal thickness of the sealing shall be 3 mm and width 6 mm. This sealing shall be in addition to the provision of seam bolts / rivets used for joining.

17.1.3.6 No sheeting work shall be commenced without ensuring that purlins and sheeting runners are true to plane. During erection, precaution shall be taken not to lean ladders on any side cladding without precautions against damage to the sheet. Notwithstanding the above all sheeting and / or cladding shall be laid strictly in accordance with the sheet manufacturer's instructions.

17.1.3.7 The entire sheeting work in wall and roof shall be fixed plumb and level with all fixing evenly spaced out and accurately lined.

17.1.3.8 Sheeting work, whether in roof or wall cladding, shall invariably commence from one end of the building and proceed continuously to the other end. Where fronts are not available in between, sheeting work shall not be taken up at all. Roof sheet shall be laid from eaves to ridge and secret fix wall cladding shall be laid from top to bottom.

17.1.9 Gutters shall be made of 3.00mm thick MS sheets with two coats of epoxy primer and two coats of chlorinated rubber paint, welded at all laps, joints, downtake spouts etc. Gutters shall be supported with 50mm wide, 3/4mm thick. galvanised strips designed in two pieces to impart desired slope and placed @ every 500mm c/c. The design, slope & shape shall be as approved by Engineer in Charge.

17.2 **RIDGES AND HIPS** - Ridges and hips shall be covered with ridge and hip sections with a minimum 150 mm lap on either side over the roofing sheets and shall be properly bent to shape and fixed. The end lap of the ridges the hips and between ridges and hips shall be not less than 150 mm.

PARTICULAR SPECIFICATIONS (CONTD.....)

17.2.1 Ridges and hips shall be fixed to the roof members with GI hook bolts and nuts and bitumen and GI limpet washer, which fix the sheets to the purlins. At least one of the fixing bolts shall pass through the end laps of ridges and hips, on one-side laps shall be set in red lead. The edges of the ridges and hip shall be straight when fixed and to end planes of the roof. The ridges and hips shall fit squarely on the sheets.

17.3 **ASBESTOS CEMENT ROOFING**

17.3.1 Asbestos cement corrugated and semi-corrugated sheets shall comply with IS 459 of 1992 and shall be as specified in Para 11.2.2 of SSR Part I. Fixing of asbestos cement corrugated and semi-corrugated sheet roofing shall be carried out all as specified in Para 11.24 of SSR Part I under close supervision of the Engineer-in-Charge.

17.4 **SKYLIGHT (POLYCARBONATE SHEETS)**

17.4.1 Polycarbonate sheet shall be 2 mm thick clear embossed sheet (polycarbonate profile to match with standing seam profile) with side laps to be clamped with galvalume sheets of 0.55mm (tct), which can be further seamed with metal roof sheets. Product required is clear embossed with both side UV stabilizer, having at least 60-80% transparency, conforming to the relevant IS code including cutting to shape and size as required including necessary sealants (silicon sealants/butyl sealants for transverse laps) including all materials, tools & tackles, labours, scaffolding, safety net etc. requires for completing the work in all respect as per direction of Engr-in-charge.

18.1 **PCC FLOORS**

18.1.1 **GENERAL**: Refer paras 13.25, 13.27 and 13.32 of MES Schedule Part-I.

18.1.2 PCC floors unless indicated, shall be considered as 40mm thick PCC of grade M-15 (Nominal Mix), laid in panels with dividing strips and shall be finished even and smooth using extra cement laid over 75mm thick PCC 1:4:8 type D-2 over 200 micron polythene film over 75 mm thick sand cushioning over rammed earth.

18.1.3 **PCC SKIRTING**: Unless otherwise indicated on drawings height of skirting shall be 10cm. Cement plaster skirting shall be provided wherever shown on drawings. It shall be 15 mm thick in CM (1:3) finished even and smooth with using extra cement, tapered on 45/60 degree with plastered surface.

18.1.4 All PCC floors shall be finished even and smooth using extra cement/without using extra cement as mentioned in Schedule 'A' before placing the concrete mix as floor toping coat cement slurry @ 3 Kg/Sq. M shall be thoroughly brushed into the prepared, surfaces of base concrete just ahead of the finish. The coat for the same shall be deemed to be included in the rate for floor topping.

18.2 **NON SKID CERAMIC TILES**

18.2.1 Non skid ceramic tiles where ever indicated in schedule of finishes shall be 1st class superior/premium quality confirming to grade B-II(A) as per IS- 13712-1993. The size and thickness of tiles shall be as indicated in Sch of finishes. In case the same is not indicated the tiles shall be of 605 x 605 x 7 to 8 mm thick. Tiles shall be light coloured plain shade as approved by GE. Tiles shall be laid set and jointed in neat cement slurry and pointed with white cement (mixed with colour pigment to match colour of tile) all as shown in BOQ/Sch of Finishes. Irrespective of whatever shown on drawings, tiles shall be laid 20mm thick screed in cement mortar 1:6 over 75mm thick sub-base with PCC 1:4:8 type D-2 over 200 micron polythene film over 75 mm thick sand cushioning over rammed earth in ground floor.

18.3 **Skirting/Dado**

18.3.1 Wherever skirting as shown on (sch of finishes) drawing, it should be of same material as that of floor. The size, shade and colour of tiles shall match with the floor finish all as directed by GE. The finished surfaces of tiles shall be in line with the plastering surfaces and the height of skirting and dado shall be as per Sch of finishes drawings. Height of skirting shall be 100 mm. The tiles shall be laid over 10 mm thick cement mortar (1:3) jointed in neat cement paste, pointed with white cement mixed with pigment. Dado shall be provided with Glazed ceramic tiles of size as specified here-in-after. Unless otherwise specified, dado shall be laid over 10 mm thick cement mortar (1:3) jointed in mixed cement paste, pointed with white cement with pigment. The height of dado shall be all as specified and shown on drawings. In case height of dado is not shown on drawings the same shall be upto cill level at WC area and upto top of door level in bath area.

18.4 **GLAZED CERAMIC TILES**

18.4.1 Glazed ceramic tiles wherever specified shall be laid with coloured tiles of 1st quality. Size of tiles shall be as mentioned in BOQ. The tiles shall be conforming to Group BIII as per IS 13753. The tiles shall conform to the specifications laid down in clause 13.15 & shall be laid as per clause 13.40 of SSR Pt – I.

PARTICULAR SPECIFICATIONS (Contd.....)

18.5 **PRECAST PAVER BLOCKS/INTERLOCKING PAVER BLOCK/COBBLE BLOCKS**

18.5.1 Precast concrete Paver blocks shall conform to IS 15658:2006, Specification for Precast concrete blocks for paving. Paver blocks shall be sound and free from cracks or other visual defects.

18.5.2 Paver blocks shall be machine pressed precast concrete interlocking paver block of triangular, zigzag, hexagonal or other shape as approved and size of 60 mm thickness, M-35 grade or 80 mm thick, M-40 grade as mentioned in Schedule 'A'/shown in drawings.

18.5.3 **SAND FILLING**

18.5.3.1 Sand for cushioning & jointing of paver blocks shall be as specified for concrete here-in-before.

18.5..4 **WORKMANSHIP**

18.5.4.1 **LAYOUT :-** The area of pavement to be laid shall be properly marked. Excavation or surface dressing, as required shall be carried out with proper slope or gradient. The area shall be properly compacted prior to laying subsequent layers.

18.5.5 **PREPARATION OF SUB BASE:** Sub base shall be of hardcore or lean cement concrete as directed by Engineer-in-Charge or as considered in BOQ.

18.5.6 **EDGE RESTRAINTS :-** Edge restraints shall be installed/placed in position on that part of the pavement edge which is not restrained by an existing structure such as building, concrete curb or concrete slab. Edge restraints are typically placed before installing the bedding sand and pavers. However, troweled concrete edge restraint shall be placed after the pavers have been placed. Backfill outside edge restraint shall be carried out as soon as possible to prevent sand from escaping under the edge restraint.

18.5.7 **SAND SETTING BED:-** Sand shall be loosely spread to an uniform thickness as directed over the compacted base course. In no case should the sand be greater than 50 mm thick.

18.5.8 **LAYING OF PAVERS :** Paver blocks shall be laid as per approved design and as per manufacturer's instructions to achieve the desired pattern. When placing the pavers, consistent joints of required thickness shall be maintained as per manufacture's recommendation or as directed by Engineer-in-Charge.

18.5.9 **CUTTING OF PAVERS :-** Pavers are sometimes, required to be cut to fit and fill against the edge restraint. Pavers may be cut with any of the three basic pieces of equipments, which are :- (a)Mechanical or guillotine splitter (b) Masonry saw (c) Hand held cut off saw. Proper safety accessories shall be used while cutting pavers.

18.5.10 **PAVEMENT COMPACTION AND JOINT SANDING :-** After placing of the pavers as per required design and pattern, the complete area shall be compacted using vibratory plate compactor. All pavers will be needed to be exposed to at least two passes of the compactor. Any broken pavers, observed after compaction shall be extracted and replaced with new paver.

After compaction of paver and replacement and re-compaction of replaced pavers, dry sand shall be spread over for filling the joints. The sand shall be swept into the joints and filled by alternating sweeping and vibrating. After completion, excess sand shall be swept off.

19.5.11 **COMPLETION OF JOB :-** After completion of paver work, the work shall be inspected after 2-3 weeks, for any unevenness or requirement of filling of sand in joints. Any defects, settlement or undulation observed shall be rectified by the contractor without any extra cost.

18.6 **PRE MOULDED PRECAST KERB STONES**

18.6.1 Precast concrete kerb stones shall be of M 35 grade concrete, factory made of size as mentioned in BOQ or as directed by Engineer-in-Charge.

18.6.2 **WORKMANSHIP**

18.6.2.1 Trenches shall first be made along the edge of the wearing course of the road to receive the kerb and the channel stones. The bed of the trenches shall be compacted manually with steel rammers to a firm and even surface and then stones shall be set in cement mortar of specified proportion. The kerb stones with top 20 cm wide shall be laid with their length running parallel to the road edge, true in line and gradient at a distance of 30 cm from the road edge to allow for the channel and shall project about 12.5 cm above the latter. The channel stones with top 30 cm wide shall be laid in position in camber with finished road surface and with sufficient slope towards the road gully chamber. The joint of kerb and channel stones shall be staggered and shall be not more than 10 mm. Wherever specified all joints shall be filled with cement mortar 1:6 and pointed with cement mortar 1:2.

18.6.3 **Finishing**

18.6.3.1 Berms and road edges shall be restored and all surplus earth including rubbish etc; disposed off as directed by Engineer-in-Charge.

PARTICULAR SPECIFICATIONS (Contd.....)

18.7 **PCC PLINTH PROTECTION**

18.7.1 Unless otherwise directed, PCC plinth protection shall be provided with 50mm thick PCC (1:3:6) type C-1 over 75 mm thick (consolidated thickness) broken stone or boulder hard core of broken stone over rammed earth. The surface of plinth protection shall be finished even and smooth without using extra cement. The concrete in plinth protection shall be laid in alternate bays of length not exceeding 3m and shall be laid in slope 1 in 12. The joints shall be 12 mm thick filled with bitumen mastic.

19. **GLAZING**

19.1 Sheet Glass – Refer para 16.2 of MES Schedule Part-I. The sheet glass shall be of `B' quality, ie ordinary for glazing purpose. The thickness for glass pane for area exceeding 0.5 Sq M shall be 5.5mm thick and 3 mm thick for area not exceeding 0.5 SqM irrespective of what is shown on drawings.

19.2 Putty – Refer Para 16.5 of MES Schedule Part-I.

19.3 Preparation of Glazing – Refer para 16.7.1 to 16.7.4 of MES Schedule Part-I.

19.4 Glazing in steel surrounds – Refer Para 16.10.1 & 16.10.2 of MES Schedule Part-I.

19.5 Glazing to aluminium windows & doors shall be done with tinted sheet glass 5.5mm thick and 6.3 mm thick respectively and shall be fixed as per manufacturer's instructions.

20. **PLASTERING AND POINTING**

20.1 Cement: Cement shall be as specified for concrete work.

20.2 Sand: Refer paras 14.5, 14.5.1, and 14.5.2 of MES Schedule Part-I.

20.3 Water: Water shall be as specified for concrete work.

20.4. The thickness of plaster mentioned in Sch 'A' is finished thickness exclusive of dubbing out, plastering and dubbing out shall however, be done in one operation.

20.5 **WPC:-** Integral water proofing compound (WPC) shall conform to IS -2645 and the brand shall be got approved in writing from the GE. Preparation of mix & method of mixing shall be as per manufacturer's instructions. While pricing deviation for omission, qty shall be considered as 3% by weight of cement.

21. **WHITE WASHING, COLOUR WASHING, DISTEMPERING AND CEMENT BASE PAINT.**

21.1 Lime: refer para 15.2. of MES Schedule Part-I.

21.2 Yellow and Red Ochre and Blue Vitriol: Refer paras 15.3 and 15.4 of MES Schedule Part-I.

21.3 White washing and colour washing: Refer para 15.12 of MES Schedule Part-I.

21.4 Distemper: Refer para 15.6 of MES Schedule Part-I.

21.5 Cement base paint: Refer para 15.7 of MES Schedule Part I

21.6 Distempering and cement base painting: Refer para 15.14 and 15.15 of MES Schedule Part I.

21.7 Newly plastered surfaces shall be thoroughly clean of all durt, dust , mortar drops, efflorescence, chalking, grease and other foreign matter before application of white wash. Old surfaces already white washed or colour washed, shall be broomed to remove all dust, dirt and all loose scales of lime wash and other foreign matter. Where heavy scaling has taken place, the entire surface shall be scrapped clean. Old colour wash on surfaces where white wash or different colour wash is to be applied, shall be entirely removed, before white wash or different colour wash is applied. All unnecessary nails shall be removed and the holes, minor pitting & cracks filled with lime putty or plaster of Paris to make the surface smooth.

21.8 Preparation and application of white wash & colour wash shall be done in accordance to the relevant items of Schedule 'A' and as specified in Paras 15.12.3 to 15.12.6.2 of SSR Part I, 2009 under close supervision of the Engineer-in-Charge.

21.9 **CEMENT BASE PAINTING**

21.9.1 Cement base paint shall comply with IS: 5410-1969, Specification for cement base paint, colour as required. The material shall be in the powder form, free from lumps that are not friable and when mixed with required volume of water shall be suitable for use on porous surfaces of masonry, concrete, bricks and rough plaster work. Colour of cement base paint shall be as approved by the Engr-in-Charge.

21.9.2 Preparation and application of cement base paint shall be done in accordance to the relevant items of Schedule 'A' and as specified in Paras 15.15.1 to 15.15.12 of SSR Part I, 2009 under close supervision of the Engineer-in-Charge.

21.10 **DISTEMPERING WITH ACRYLIC DISTEMPER**

21.10.1 Acrylic distemper shall comply with IS : 428 of 1969. Colour of distemper shall be as approved by the Engineer-in-Charge. Preparation and application of distemper shall be done in accordance to the relevant items of Schedule 'A' and as specified in Paras 15.14.1 to 15.14.5.3 of SSR Part I, 2009 under close supervision of the Engineer-in-Charge. Distemper shall be applied with brush/roller as directed by Engineer-in-Charge to achieve a smooth finish. No claim shall be entertained in this regard during currency of work.

PARTICULAR SPECIFICATIONS (Contd.....)

21.11 **PROTECTIVE MEASURE**

21.11.1 Surfaces of doors, windows, floors, articles of furniture etc., and such other parts of the building not to be treated shall be protected from being splashed upon. Such surfaces shall be cleaned of splashes of white wash, colour wash, distemper, etc. The Contractor shall be responsible for any damage to the fittings, fixture and furniture.

21.11.2 Wherever scaffolding is necessary, it shall be erected in such a way that as far as possible, no part of scaffolding shall rest against the surface to be treated. A properly secured and well tied and suspended platform (JHOOLA) may also be used. Where ladders are used, pieces of old gunny bags shall be tied at top and bottom to prevent scratches to the walls and floors. For work in ceiling, proper stage scaffolding may be erected, where necessary.

22. **PAINTING**

22.1 Paints and allied materials: Refer Para 17.2 of MES Schedule Part-I.

22.2 Painting work Generally: Refer para 17.3 of MES Schedule Part-I. Paint shall be applied with brush/roller as directed by Engineer-in-Charge to achieve a smooth finish. No claim shall be entertained in this regard during currency of work.

22.2 Painting work Generally: Refer para 17.3 of MES Schedule Part-I.

22.3 Painting woodwork: Refer para 17.6 of MES Schedule Part-I. Filler coat is not required to be applied. Where painting is indicated in BOQ, it shall be done in 3 coats viz. primer, under coat & finishing coat.

22.4 Paints shall be of Ist grade quality of the approved makes and shall be procured by the contractor in sealed containers.

22.5 Primer for different surfaces i.e. wooden / steel / plastered /concrete surfaces shall be of the appropriate type and shall be of same make as the paint to be applied over respective surfaces

22.6 Primer for different surfaces i.e. wooden / steel / plastered /concrete surfaces shall be of the appropriate type and shall be of same make as the paint to be applied over respective surfaces

22.7 The contractor shall inform the Accepting Officer, within three weeks of the acceptance of the tender, the brand names of the manufacturers of paint proposed to be used in the works and submit sample thereof and obtain prior written approval of the Accepting Officer before their use in work. Paints for priming coat under coat and finishing coat shall be of the same manufacturer. Tint of paint will be approved by Accepting Officer.

22.8 The paints shall be procured by contractor direct from the manufacturers of their authorized dealers and original paid vouchers bearing machine number for the full quantity showing the name of manufacturer and brand shall be produced to Engineer-in-Charge.

22.9 Preparation of old painted surfaces shall be done in accordance to para 17.10 of SSR Part I 2009. Application of synthetic enamel paint to wooden and steel surfaces shall be done in accordance to the relevant items of Sch 'A' and as specified in Paras 17.6 & 17.8 of SSR Part I, 2009 respectively, under close supervision of the Engineer-in-Charge.

22.10 French polishing shall be done in accordance to para 17.17.4 of SSR Part I 2009.

22.11 Varnishing shall be done in accordance to para 17.17.3 to 17.7.3.2 of SSR Part I 2009.

22.12 **MODE OF MEASUREMENT (Applicable for Periodical Services Only)**

22.12.1 The quantities of various items of work as catered under Schedule 'A', shall be considered directly from audited PSMB(s). Where PSMB (s) are not available or any addition / alteration are observed at site, measurement of various services shall be taken jointly at site. The tenderer is requested to inspect the sites and ascertain the nature of works before quoting of the tender. No further claim will be entertained at a later date whether the tenderer had inspected the sites etc before quoting of the tender or not.

23 **SCAFFOLDING TO THE BUILDINGS:-**

23.1 The scaffolding to the building shall be made of MS pipe of required dia duly cross braced and shall be properly designed for the functional aspects and safety of the work. The design and arrangements of scaffolding shall be got approved by the GE before commencement of the work. The contractor shall submit the design of scaffolding to the GE within two weeks from the date of placing Work Order NO-1 for approval of the same.

24. **PLUMBING**

24.1 **MILD STEEL GALVANISED TUBES (PIPES) AND FITTINGS**:- Refer para 18.4 of MES Sch Part – I. The pipe & fittings shall be ISI marked & shall be of incorporated in work duly approved by GE.

24.2 **BIB TAPS, PILLAR TAPS, STOP VALVE AND GUN METAL GATE VALVE**: - Refer para 18.4 of MES Sch Part – I. Screw down bib taps & stop cocks shall be of cast copper alloy body bright finished, ISI marked.

PARTICULAR SPECIFICATIONS (Contd....,)

24.3 **WORKMANSHIP**

24.3.1 **GENERAL REQUIREMENT**: - Refer paras 18.41 of MES Schedule Part – I. Generally layout of water supply work is shown on drawings. The exact position of water supply and fittings shall be as directed by the Engr-in-Charge. The mounting height of various fittings shall be as directed by Engr-in-Charge.

24.3.2 **LAYING OF GI PIPE**:- Refer para 18.46 of MES Schedule Part – I. Union shall be provided at suitable places as directed by the Engr-in-Charge so that in case of repairs etc. long length of pipes are not disturbed.

24.3.2.1 Uncovered threaded portion of pipe shall be painted with approved paint. GI tubing in chases shall be painted with two coats of anti corrosive bituminous paint.

24.3.2.2 The contractor shall use proper bends/elbows/tees at turning/corners. Bending of pipe is not permitted.

24.3.2.3 The contractor shall provide screwed plug to all open bends of fittings at the end or days work and at time when the work is stopped.

24.3.3 **FIXING OF WATER SUPPLY FITTINGS**:- Refer paras 18.62, 18.63, 18.64 and 18.66 of MES Schedule Part – I. Fixing shall be done as specified in SSR Part – 2009 and as directed by Engineer-in-Charge.

24.4 **TESTING**:- Refer clause 18.55 of MES Schedule Part – I. The water supply installation shall be tested hydraulically at a pressure of 2.5 Kg/sqcm in the presence of Engineer-in-Charge for at least 24 hours continuously to ensure leak proof connections. All arrangements for testing shall be made by the contractor at his own cost. Any leakage/defects noticed shall be rectified to entire satisfaction of Engineer-in-Charge

24.6 **PVC WATER TANKS**

24.6.1 Rotational moulded water storage tank 2 layered wall construction, ISI marked, seamless perfection, made of stabilized food grade polyethylene to ensure resistance to weather, temperature, corrosion and leakage including ball valve and shall be fixed as per the directions of the Engineer-in-Charge. The water tank shall be provided and fixed all as per manufacturer's instructions. The overflow pipe shall have perforated polythene cap screwed to one end. Washout pipe shall be provided with GI plug. Ball valve shall be of polythene 25 mm bore, high pressure with polythene float and fly nut. The inlet, outlet, over flow and wash out pipe shall be measured and paid under respective items of Schedule 'A'.

25. **SANITARY APPLIANCES**

25.1 Refer paras 18.32.1 and 18.32.1.1 of MES Sch Part I. These shall be 1st quality vitreous china ISI marked.

25.2 **WATER CLOSET (SQUAT PATTERN)**

25.2.1 Unless otherwise directed by the Engr-in-Charge, water closet shall be of squatting pattern of size 580 mm x 410 mm. The flushing inlet shall be at the narrow end. Each pan shall be provided with a 100 mm dia "P" or "S" type trap, as directed by the Engr-in-Charge, with an inspection vent. The trap shall be glazed inside. Foot Rest of squatting pan shall be of size 250 mm x 125 mm and shall be fixed as specified in Para 18.86.3 of SSR Part I. Fixing of the water closet to be done in accordance with Para 18.86 of SSR Part I.

25.3 **WASH DOWN WATER CLOSET (PEDESTAL PATTERN)**

25.3.1 Refer Para 18.32.2 and 18.87.1 of MES Schedule Part I. Water closet shall be of pattern I glazed vitreous china with integral trap either 'P' of 'S' outlet and of height 390 mm. Size of pan shall be 48 cm x 34 cm x 40 cm vitreous China.

25.3 **URINALS:-**

25.3.1 Unless otherwise directed by the Engr-in-Charge, Urinals shall be of approved make to be provided at site. Operational and performance requirements shall be as catered under Para 18.327 (of SSR Part I and as directed under close supervision of Engineer-in-Charge.

25.4 **FLUSHING CISTERN**

25.4.1 Unless otherwise directed by the Engr-in-Charge, low level flushing cistern of capacity 10 litres of approved make to be provided at site. HDPE flush pipe 32 mm bore of required length and bent to required angle with necessary wall clip shall be provided. Operational and performance requirements shall be as catered under Para 18.99.10 (18.99.10.1 to 18.99.10.6) of SSR Part I. Fixing of flushing cistern and flush pipe to be fixed in accordance with Para 18.99.8.1 and Para 18.99.9 of SSR Part I under close supervision of Engineer-in-Charge.

25.5 **VITREOUS CHINA WASH HAND BASIN**

25.5.1 Vitreous china wash hand basin l shall be of specified make, as catered under relevant item(s) of Schedule "A" and as approved by the Accepting Officer. The vitreous china wash hand basin to be fixed as specified in SSR Part I (2009) under the close supervision of the Engineer-in-Charge.

PARTICULAR SPECIFICATIONS (Contd....,)

25.6 **STAINLESS STEEL SINK AND DRAINAGE BOARD**:- Refer para 18.90 and 18.100 of MES Sch Part I.

25.7 **PLASTIC WATER CLOSET SEAT & COVERS**:- Refer paras 18.36,18.36.1 to 18.36.4 of MES Sch Part-I

25.8 **FLUSHING CISTERN FOR WASH DOWN WATER CLOSET**:- Refer para 18.99 with sub paras thereof of MES Sch Part-I. Flushing cistern shall be of PVC "Feather Touch" operated bearing IS-7231, diaphragm bearing IS: 13049-1991 and delrim valve, float bearing IS: 9762-1994 low level flushing cistern 10 Litres capacity.

25.9 **GLASS SHELF**: Refer para 18.37 and 18.92 of MES Schedule Part-I.

25.10 **MIRROR**: Refer Para 18.38 and 18.91 of MES Schedule Part-I.

26. **SEWAGE WORKS**

26.1 **PVC (SWR) SOIL, WASTE AND RAINWATER DRAINAGE PIPES AND FITTINGS**

26.1.1 PVC (SWR) soil, waste and rainwater drainage pipes and fittings shall be all as specified under Para 18.27.A of SSR Part I and to be fixed as catered under Para 18.52 of SSR Part I under close supervision of the Engineer-in-Charge.

26.2 **CAST IRON (SAND CAST) PIPES AND FITTINGS**

26.2.1 Cast iron (sand cast) soil / waste / ventilating pipes and fittings shall be conforming to IS 1729-2002 and shall bear ISI mark. All pipes and fittings shall be coated externally and internally with a composition having tar or other similar base. The coating material shall have good adherence and shall not scale off. In case pipe and fittings are imperfectly coated or where the coating has not set, the coating shall be removed and the pipes and fitting are to be re-coated. All the soil and vent pipes along the walls shall be without ears and shall be fixed to wall with mild steel clamp of suitable size as directed by the Engineer-in-Charge.

26.2.2 Each and every joints of soil / waste pipe shall be tested as catered under Para 18.79 of SSR Part-I in presence of the Engineer-in-Charge.

27. **INTERNAL ELECTRIFICATION**

27.1 General: Refer paras 19.2 of SSR Part-I, 2009. All internal electrification work shall be executed by fully qualified licensed electricians all as per the latest Engineering practice and latest IS codes/IS Rules.

27.2 The materials shall be as follows: -

Ser No	Materials	Reference to paras of MES Schedule Part-I.
(a)	Cables, Cords and earthling leads.	19.24
(b)	Cables for internal wiring light power and sub main.	19.25
(c)	Conduit and conduit fittings.	19.29
(d)	Plug Gutties	19.30
(e)	Screws and fastening link clips	19.31
(f)	Ceiling Rose	19.32
(g)	Socket outlet	19.40
(h)	Lamp holder	19.41
(j)	Distribution board	19.44
(k)	Switch fuses	19.45
(l)	MCB Distribution Boards	19.46.1.1,19.46.1.2 & 19.46.1.3
(m)	MCBs	19.46.2
(n)	MCBs Isolators	19.46.3
(o)	Plug and Socket DBs	19.46.4

27.3 PVC casing and capping wiring shall be as per IS: 14927 (Part-I&II)-2001, of approved make and fire retarded quality. Wall thickness shall be 1.20 mm (Minimum). The channel space shall be adequate to accommodate the cable laid in it. The screws shall be cadmium-coated iron-screws rigidly fixed to gutties or with Detofix at least 15 cm apart. The corner bridges shall be made neatly angle shaped.

27.4 Distribution board, main switches: Refer Para 19.107 of MES Schedule 'A' Part-I.

27.5 Accessories switches, lamp holder, ceiling roses, socket outlets, Refer paras 19.114 of MES Sch Part-I.

27.6 **NON METALLIC (PVC) CONDUIT AND CONDUIT ACCESSORIES:-**

27.6.1 Refer clause 19.29 of MES Schedule Part-I, 2009. Conduit and conduit accessories shall conform to relevant IS i.e., IS-9537 (Part-III) 1983 and IS-3419-1976. Conduit shall be non-metallic PVC conduit and shall show no appreciable un-evenness. Interior and ends of conduit and conduit accessories shall be free from bushes, ducts fins and the like which may cause damage to the cables. Each length of PVC conduit shall be marked with the following :-

PARTICULAR SPECIFICATIONS (Contd.....)

(a) Manufacturer's name or trade mark. (b) Country of manufacture
(c) Nominal size of the conduit.
(d) Each length of conduit shall also be marked with the ISI certificate mark.

27.6.2 PVC flexible conduit of size as directed by the Engineer-in-charge shall be provided as required without any extra cost.

27.7 **PVC INSULATED 1100 VOLTS GRADE CABLE (PVC SHEATHED/UNSHEATHED)**

27.7.1 The conductor shall be of stranded copper wires. The insulation shall consist of compounded polyvinyl chloride. The thickness of insulation shall not be less than nominal value mentioned in appropriate table of IS-694-1964. The composition of the compound of sheath shall be polyvinyl chloride. The compound shall be designed to offer adequate mechanical protection. The value of thickness of sheath shall be in accordance with those specified in appropriate tables. The insulation on & sheath shall be applied by the extrusion process and each shall form a compact homogeneous body.

27.8 **SWITCH BOXES**

27.8.1 Switch boxes suitable to house switches of required ratings, and fan regulators as required shall be provided. In case the number of switches in one box is not tallying with that available in standard manufacturer, the box accommodating the next higher of switches shall be provided without any extra cost. In case fan regulator/regulators is/are to be provided at a later dated, suitable provision for accommodating such regulators shall be made in the switch boxes and blank of covers shall be provided without any extra cost

27.8.2 Switch boxes shall be so designed that accessories are mounted on a grid plate with tapped holes for brass machine screws leaving ample space at the back and on the side for accommodating conductors, check-nuts and screwed bushes at conduit entries etc.. The grid plates and MS boxes shall be fitted with a brass earth terminal. Boxes shall be attached to conduits by means of check-nuts on either of their walls. Moulded front covers made from high impact resistant and ultra violet stabilized engineering plastics shall be fixed by means of counter sunk chromium plated brass machine screw. No timber shall be used for any supports. Switch boxes shall be located with bottom at 1200mm above floor level unless otherwise indicated.

27.9 **BOXES FOR SOCKETS**

27.9.1 Outlet boxes shall be suitable for housing switches socket outlets and any other outlet as required. These shall be so designed that accessories are mounted on a grid plate with tapped holes for brass machine screws leaving ample space at the back and on the sides for accommodating conductors, check nuts and screwed bushes at conduit entries etc.. The grid plates and MS boxes shall be fitted with a brass earth terminal. These shall be attached to conduits by means of check nuts on either sides of their walls. Moulded front covers made from high impact resistant, flame retardant and ultra violet stabilized engineering plastics shall be used to mount the outlets and shall be fixed to the outlet MS boxes by means of counter sunk chromium plated brass machine screws. No timber supports shall be used. Boxes shall be located at skirting level or bottom at 1200mm from floor or inside raceways on laboratory work table., as indicated in drawings and / or as directed.

27.10 **SWITCHES/SWITCH SOCKET OUTLET/SOCKET OUTLET**

27.10.1 The base shall be made of vitrified ceramic material or tough non ignitable molded insulating material. The covers, cover plates and actuate members shall be of tough, non-ignitable insulating material. Current carrying parts shall be made of brass, copper phosphor bronze, aluminium alloy or any other suitable material, springs shall be of corrosion resistant metal. Attachment fitting screw and other non-current carrying Paris shall be of mild steel aluminium alloy or insulating material. All socket outlets and switch socket combination shall have ISI certification mark.

27.11 **LIST OF BIS CERTIFIED PRODUCTS TO BE INCORPORATED IN WORKS**

(a)	Ceiling Rose	IS-371
(b)	Socket outlet 3 pin plug and socket	IS-1293
(c)	Rigid non-metallic conduit	IS-9537
(d)	Copper stranded conductor	IS-694
(e)	Switch fuse (Main switch)	IS-4064
(f)	Single core cable polythene Insulated and PVC sheathed	IS-1596
(g)	Distribution Fuse Boards and Cutouts	IS-2675
(h)	Rigid non-metallic casing & capping	IS-14927

PARTICULAR SPECIFICATIONS (Contd....,)

28.0 **EXTERNAL ELECTRIFICATION**

28.1 **GENERAL**

28.1.1 The scope of work covers for the works as described in various items of Schedule 'A' and particular specifications.

28.1.2 The exact layout of various electrical equipments, fittings, accessories, cables etc shall be as directed by Engineer-in-Charge and approved by GE in writing to suit the site requirements. Contractor to ensure written approval before taking work in hand. No price adjustment what so ever, shall be admissible to the contractor on this account.

28.1.3 The tenderer shall be responsible for supplying, installing, laying, connecting, testing and commissioning of the items covered in the tender all as specified and as directed by Engr-in-Charge. The work shall be of high standard and executed as per sound engineering practice as per MES Schedule Part I/IE rules/IS/National Electrical Code.

28.1.4 During the currency of work and also on completion of work, testing shall be carried out for all items/the installation as a whole as per SSR/IE Rules/IS to the entire satisfaction of GE and a record shall be maintained duly signed by the Engr-in-Charge and the contractor.

28.1.5 Pre-commissioning test, if applicable, shall be carried out by the contractor in presence of IEM/SEI or authorized representative of Accepting Officer. Contractor shall be responsible for the arrangement of testing and all the equipment testing material shall be arranged by the contractor at no extra cost.

28.1.6 The tenderer shall include all requisite components & accessories like cable glands, labels, CTs, selector switches shrouds, indication lamps, auxiliary relays, contractors, toggle switches, handles, cable indicating devices, brackets and clamps of all kinds, instrument wiring, instrument fuses interlocks and all such other requirements for completion of the entire work though not specifically brought out in BOQ & in these specifications. The tendered rates shall be deemed to include all connected works required for supplying, erecting, testing & commissioning of the entire equipments, fittings and items like foundation for equipments, supplying & connecting of cables in ducts etc.

28.1.7 Unless otherwise specified, civil engineering works required for installation of the equipments such as grouting of poles and equipments, trench cutting and back filling in ground and in road crossing etc shall be deemed as included.

28.1.8 The electrical works shall be carried out by properly skilled & licensed electricians under supervision of qualified electrical supervisor/engineers as per IE rules. The contractor shall, on demand by Engr-in-Charge produce such evidence of qualifications of his workmen/ supervisors/ engineers at the time of commencement of work and at any time thereafter/during the currency of work.

28.1.9 The items covered in this tender and incorporated in work shall be capable of delivering satisfactory services/operation without detriment to its life under the climatic conditions of the proposed site.

28.1.10 Manufacturer's test certificate as per IS test procedures for items to be incorporated in works shall be submitted by the contractor.

28.1.11 The tenderer shall submit detailed catalogues/pamphlets/drawings for all major electrical equipments(s) like LT/HT cables, feeder pillar and transformers etc to the GE for approval within one month of placement of work order. The GE would convey the decision in two weeks and thereafter procurement action be organized accordingly by the contractor.

28.1.12 Suitable lugs shall be provided at all cable end terminations and these shall be crimped with hydraulic /manual crimping machine as required and directed by Engr-in-Charge. Cost of this work is deemed to be included in the rates quoted.

28.2 **SAFETY PROCEDURE AND PRACTICES**

28.2.1 In all major electrical installations such as sub stations, industrial establishments, transmission and distribution lines and cable networks, safety procedures & instructions for working on low, medium and high voltage mains and apparatus and safety practices listed in IS-5216 (Guide for safety procedures and practices in electrical works) shall be followed to the extent applicable. The contractor shall provide workmen with safety devices and appliances.

28.3 **FIRE SAFETY**

28.3.1 All electrical equipment shall satisfy the requirements laid down in IS-1646 Code of practice for fire safety of buildings (general) electrical installation and IS-3034, Code of practice for fire safety of industrial buildings, electrical generating and distributing stations to the extent applicable.

28.4 **LAYOUT OF EQUIPMENTS**

28.4.1 Layout of equipments, cables etc. shall be as per decision of Engr-in-Charge to suit site requirement. The tenderers are advised to know the tentative layout plan of the poles, cables etc before quoting their rates. No price adjustment in the rates shall be made at a later stage on this account.

PARTICULAR SPECIFICATIONS (Contd....,)

28.5 **SAND CUSHIONING**

28.5.1 Sand cushioning shall be done as specified in Schedule 'A' and SSR Part- I 2009.

28.6 **LT CABLES**

28.6.1 LT electric power cable shall be XLPE insulated and PVC bedded and overall PVC sheathed, heavy duty steel wire armoured all as specified in Sch 'A' conforming to IS : 7098 (Part I.

28.6.2 All LT cables shall bear ISI certification marks, These shall be brought at site wound on cable drum, ends sealed and with manufacturer's test certificates.

28.6.3 Suitable lugs shall be provided at all cable and terminations and these shall be crimped with hydraulic priming machine. Cost of this work is deemed to be included in the rates quoted.

28.6.4 All jointing with termination lugs/sleeve shall be done with hydraulic/mechanical crimping tools.

28.6.5 All LT electric power cable connected in switches, bus bar, LT feeder pillar box etc. will be connected with suitable size of aluminum lugs and entry of cable only through proper size of steel/brass chromium plated cable gland all as directed by Engr-in-Charge.

28.6.6 **LAYING OF CABLE**

28.6.6.1 Before laying of cable, the trench shall be provided with a layer of sand of 8 cm depth for the purpose of cushioning. After the cable has been uncoiled and laid into the trenches over the roller, the cable shall be lifted slightly over the rolled beginning from one end by helper standing about 10 mtrs apart and drawn straight. The cable shall be then taken off the rollers by additional helpers lifting the cable and then laid straight into the route. When the cable has been properly straightened, it shall be covered with sand to a depth of 15 cm. This is then gently punned down to a depth of 10 cm above the top of the upper most cable, thus providing good bedding for the protective brick cover. The punning shall be done by hand, mechanical punning shall not be used.

28.6.6.2 The depth of trench from the upper surface of the street pavement or ground to the cable axis shall not be less than 75 cm plus radius of complete cable for 11 KV grade cables.

28.6.6.3 **Spacing between cables**: When more or one multicore or trefoil group of single core cables are laid in the same trench a horizontal inter axial spacing of 25 to 40 cm shall be provided to reduce the effect of mutual heating and to ensure that a fault occurring in one cable will not damage the adjacent cable. When more than one tier of cable is necessary, riddle soil is filled in the trench, and rammed down to from a solid bed to a height of 25 to 40 cm above that of the first tier. The mesh of the screen through which the soil is riddled should not exceed 12.5 mm IS sieve. The width of the trenching shall in no case be less than 45 cm so as to facilitate working on the cables. Laying of cables in tier formation shall be avoided as far as possible as loading of cables is very much reduced.

28.6.6.4 **Road Crossing**: When crossing roadways, it is preferable to cross on the askew to reduce the angle of the bend as the cable enters leaves the road crossing. The cable shall pass through GI Pipe of suitable dia as mentioned in Schedule ' A'.

28.6.6.5 **Pulling out the cable**: On no account the cable shall be allowed to twist or kink as it is likely to spring the Armour and fracture the paper insulation and outer serving of the cable. When pulling cable around the bends, rollers should be placed sideways so as to keep the cable ona safe radius. Cable drums should be rolled in the direction of arrow.

28.6.6.6 **Minimum Bending Radius** :- Cables shall be always be bend (or straightened) slowly they should never be bend to a small radius. The minimum safe bending radius for impregnated paper insulated cables may be taken as given below, but wherever possible larger radius should be used.

	Single core cables	Multicore cables	
		Unarmoured	Armoured
Upto 11 KV	20 D	15D	12 D
Upto 33 KV	30 D	25 D	20 D

Note: 'D' is the overall diameter of the cable. At joints and terminations, the individual of multiple cables should never be bent so that the radius of the bend is less than15 items the diameter over the insulation.

28.6.6.7 **CABLE LAID IN DUCTS OR FLOORS :-**Cables ducts should be of such dimensions that the cables laid in it do not one another. If necessary, cables may be fixed with clamps on angle iron support on the walls of the ducts. The duct should be covered with removable slabs or chequered plates and filled with sand after the cables have been laid without any cost to the Government.

28.6.6.8 **CABLE LAID ALONG THE SURFACE OF THE WALLS**: Cables may be fixed by means of suitable MS clamp or bracket or on cable racks, as directed. If necessary the cable may also be laid in a chase cut in the wall and fixed with clamps. The chase may be covered with removable MS Sheet.

PARTICULAR SPECIFICATIONS (Contd....,)

28.6.6.9 **CABLE ENDS:-**All PVC cables upto 1100 volts grade shall be terminated into the equipment by means of compression type cable glands having screwed nipples and check nuts. The cable conductors shall be provided with crimp type solder less cable socket or connections to suit the termination arrangement of the equipment. The crimping shall be done by hand/hydraulically operated crimping tool after applying conducting jelly. The insulation of the cable shall be removed just immediately before the crimping is done. (Refer clause 19.86.1 of SSR-91 Part I).

28.7 **BRICKS IN CABLE PROTECTION**

28.7.1 Bricks: Refer to paras 5.6.1 to 5.6.9 of MES Schedule Part.I. The bricks shall have minimum compressive strength of 75 Kg/Sq cm and shall be of subclass 'B', old size bricks as specified in para 5.6.2 ,5.6.2.1 and 5.6.3. of MES Schedule Part-I. Sampling and testing of bricks shall be carried out by the contractor as per IS-5454 and IS-3495 (Part -I to IV) as directed by GE without any extra cost to department. The size of bricks shall as per MES Schedule Part-I i.e. 23 x 11.3 x 7.5 cm. However the permissible tolerance in size shall be + 8% and center line dimension shall be maintained.

28.8 **TESTING OF CABLE**

28.8.1 All LT cables shall be testing by Meggar before laying and after carrying out the joints. The cable shall be tested for :-

(a) Continuity (b) Absence of cross phasing
(c) Insulation resistance to earth (d) Insulation resistance between conductors.

28.9 **TESTING AFTER LAYING AND JOINTING**

28.9.1 The following tests shall also be carried after the initial laying and jointing work:-

(a) Insulation resistance test : Sectional and overall
(b) Continuity test : Sectional and overall

29 **LT PANEL BOARD/FEEDER PILLAR / BUS BAR CHAMBER**

29.1 **CONSTRUCTION OF PANEL**

(i) The panel shall be fabricated out of not less than 2 mm thick CRCA sheet steel for indoor type panels and 3.15 mm thick for outdoor type panels using CNC machine in a CPRI approved factory. The enclosure shall have welded frame construction with CRCA frame sections with minimum 5 mm thickness. The doors, partitions and covers shall be of CRCA sheet steel with minimum 2 mm/3.15 mm thick as specified.

(ii) The front portion of the enclosure shall be for mounting switchgear and the rear portion for mounting capacitors. Suitable ventilation in the form of louvers shall be provided for better heat dissipation. All live parts shall be shrouded with FRP material of fire retardant grade. The cable entry shall be from the bottom of the panel. Suitable cable glands shall be provided for cable entry.

(iii) The power and control circuit diagram shall accompany the equipment.

(iv) Suitable name plate shall be provided for the indication lamps, selector switches, push buttons etc.

(v) LT control panel shall be spray painted inside and outside with two coats of synthetic enameled paint over a coat of red oxide primer including preparation of surfaces all as directed.

(vi) LT control panel shall consist with the fittings as mentioned in Sch 'A' under respective items.

(vii) The apparatus shall be so arranged so as to facilitate their operation and maintenance and at the same time to ensure the necessary degree of safety. Sufficient space shall be provided for proper dressing, connecting up inspection, maintenance and servicing of cables.

29.2 **INTERNAL WIRING OF PANEL**

(i) Power wiring in LT panel shall be done using suitable size of stranded copper cable with copper links.

(ii) Suitable brought out terminal shall be provided for connection of incoming/outgoing cables and control cables.

29.3 **BUS BARS**

Bus bars for electrical panels shall be electrolytic grade high conductivity TP&N solid aluminium/copper bus bars and as specified in Schedule 'A'

29.4 **FOUNDATION**

Foundations for LT panel shall be provided with PCC 1:2:4 type B1 using 20 mm graded aggregates and cost of the same is deemed to be included in the overall quoted rates of the LT panel. The PCC foundation shall be finished even and smooth by using extra cement.

30. **EARTHING**

30.1 Earthing shall be done in accordance with latest IS: 3043 -1987 (Reaffirmed 2001), Code of practice for earthing and MES Schedule Part-I and IE rules. The contractor shall obtain approval of Engineer-in-Charge before commencement of earthing work and it shall be executed in the presence of an MES representative.

PARTICULAR SPECIFICATIONS (Contd....,)

30.2 Normal earthing unless otherwise specified in Schedule 'A' shall be done with galvanized iron or steel plate 600 x 600 x 6mm as per Electrical Plate No 3 of MES Sch Pt – I. Depth of earth plate given in BOQ is the minimum which may be increased by the Engineer-in-charge depending the soil conditions. GI tubing used for earth lead protection & watering shall be of medium grade & light grade respectively.

30.3 LT switch gears/equipments, wherever to be earthed with normal earthing, shall be earthed with 32 x 6mm GI strip connection to the earth electrode. Two separate earth points shall be made in such cases for two independent paths to earth. In other cases each connection to electrode shall be of 8 gauge (4mm) GI wire as specified in Schedule 'A'.

30.4 The earth electrode shall be located at least 2 metres from the footing of nearest foundation, exact distance however being decided by GE at site considering the depth of excavation and structural safety. Distance between any two electrodes shall strictly be as per IS-3043. The continuity resistance of any electrode shall not exceed one ohm.

30.5 Test links (test point) shall be provided in earth circuit for carrying out resistance measuring test.

30.6 All joints in main earthing and between the main earthing strip and branch connections shall have the overlapping surfaces properly treated and jointed to prevent oxidation. Straight joints shall have overlap of 50mm. All joints shall be brazed. Form 6 mm dia copper rivets (2 in 2 rows) shall be used for jointing the main/branch earth strips.

30.7 Concrete chamber shall be PCC 13:6 type C-1. All internal surface of the chamber shall be given 15mm thick plaster in cement mortar 1:4. Funnel in chamber shall be made out of CI. It shall be leak proof and provided with wire gauge duly soldered.

30.8 Cover and frame for earth pit shall be as specified in Schedule 'A'.

30.9 Charcoal dust and salt filling shall be done in layers as shown in electrical plate No 3. Surplus soil shall be disposed off and site left clear and tidy on completion.

30.10 **GI/ALUMINIUM/COPPER STRIP**

30.10.1 GI strip shall be as mentioned in Sch 'A' and shall be incorporated in the work duly approved by GE.

31. **DEMOLITION/DISMANTLING/TAKING DOWN**

31.1 Refer to relevant paras under section 21 of MES Schedule Part-I.

31.2 The contractor shall take proper precaution for demolition/dismantling work. If any damage caused due to dismantling/demolition to the existing structure, the contractor shall made good the same at his own cost without any extra cost to the Govt.

31.3 Unserviceable materials obtained from demolition/dismantling shall be property of the contractor and shall be credited as per schedule of credit. All demolished/dismantled materials shall be removed from the site of `the contractor as directed by the Engineer-in-Charge without any extra cost to the Govt.

32. **PURCHASE VOUCHER**

32.1 The contractor is advised to submit his original purchase voucher duly machine numbered in respect of the all materials as directed by E-in-C Branch latest .The materials shall be purchased from the authorized agents/dealers who are required to show on the cash memo /bill all the taxes etc and make and IS number as required. The purchase vouchers shall be produced to the Engineer in Charge for verification of materials equipments before his permissions to incorporate the same in the work communicated to the contractor. Two certified copies of the same shall be made by the Engineer-in-Charge for reference and record.

33. **APPROVED NAMES OF MANUFACTURERS**

33.1 Approved names of manufacturers of various items/equipments to be incorporated in this work unless otherwise specified in Schedule 'A' and in particular specifications shall be as under and as specified here-in-before:-

(A) **B/R ITEMS**

Ser No	DESCRIPTION OF ITEM	MAKE
1	2	3
1	Construction chemicals like WPC, hardener ETC	FOSROC/ PIDILITE/ BAUCHEMIC INDIA/ CICO/ ROFF/ STP/CIPY / SIKA/BERGER/HINDCON/ANUPAM PAINTS
2	White Cement / Wall Putty	BIRLA WHITE / BERGER / J&N / JK / SHALIMAR PAINTS / ASIAN PAINTS/ ANUPAM PAINTS
3	Acrylic Washable Distemper / Oil Bound Distemper (1st Quality)	BERGER / NEROLAC / BRITISH PAINT / DULUX / J&N / SHALIMAR / ASIAN PAINTS/ ANUPAM PAINTS

PARTICULAR SPECIFICATIONS (Contd....,)

4	Acrylic Emulsion paint / Plastic Emulsion Paint	BER GER (SILK) / DULUX (VELVET TOUCH) / SHALIMAR PAINTS (SUPERLAC STAY CLEAN) /ASIAN PAINTS (ROYAL ASPIRA) / NEROLAC (IMPRESSION)/ ANUPAM PAINTS
5	Synthetic Enamel Paint	SHALIMAR PAINTS (SUPERLAC) / ASIAN PAINTS (APCOLITE) / BERGER (LUXOL) / J&N (BROLAC) / NEROLAC (IMPRESSIONS)/DULUX (GLOSS)/ ANUPAM PAINTS
6	Cement Base Paint	SNOWCEM PLUS / BERGER PAINTS / J&N / ASIAN PAINTS/JOTUN INDIA PVT LTD / NEROLAC / DULUX/ ANUPAM PAINTS
7	Weather proof Exterior Emulsion Paint	ASIAN PAINT (Ultima Suprema) /BERGER (Weather Coat Longlife Flexo) / DULUX (Weather Shield Elastomeric)/ ANUPAM PAINTS
8	French Polish	BERGER / J&N / ASIAN PAINTS / JOTUN/NEROLAC/ ANUPAM PAINTS
9	Bituminous Joint Sealing Compound & Primers	STP/ KRISHNA / TIKI TAR
10	PVC door frame and shutter, PVC wall lining	ACCURRA POLYTECH PVT.LTD./ M/S RAJSHRI / SINTEX / M/S POLYLINE EXTRUSIONS PVT LTD
11	Factory made shutters	MP Wood Products /Doorking, Kolkata / Ambika Timber Works / Premier Wood Craft (P) Ltd / Jain Doors Pvt Ltd / M/S Modern Doors / M/S JOINERY WOOD PRODUCT / M/S Goel Brothers / M/S Surbhi Metal India Ltd /M/S Hindustan Housing Factory
12	Aluminum sections for snap grid ceiling & joinery	HINDALCO / JINDAL / AJIT INDIA / BALCO
13	Non-Asbestos fibre cement /Perforated Ceiling Board/ Gypsum Board	EVEREST / GYPROC / LAFARGE / NCLS BISON / KINGSTON /ARMSTRONG / RHINO / M/s USG Boral Building Products (India) Pvt Ltd /DIAMOND CEILING
14	Plywood, Flush door	KITPLY, GREENPLY, CENTURY, NATIONAL PLYWOOD, ARCHIDPLY, SWASTIK, ALISHAN VENEER & PLYWOOD, ECO-BOARD, MARINOPLYWOOD, ANCHOR
15	Plain & Prelaminated Particle Board, Block Board	GREEN PLY / KITPLY / ANCHOR / BHUTAN BOARD / NOVAPAN / ASSOCIATE DECOR / ECOBOARD / DECOBOARD / CENTURY LAM / DURATUFF / ARCHID PLY / KITLAM / MERINOVA / NEPAL BOARD
16	Laminate Sheet	MERINO / GREENPLY /SUNMICA / ROYAL TOUCH / CENTURY / BELL
17	Floor Spring / Door Closers / Accessories	Hardwin / Godrej / Doorset / Ozone / Geze / Ebco / Insta
18	MS Stove Enamelled fittings	Mowjee / Crown / Everlite / Harrison / Godrej / Universal / Sandhu / Sartaj / M/S Ajanta Steel Pvt Ltd
19	Aluminium Door & Window Fittings	Bharat / Classic / Aluminium Udyog (Global) / Argent / Pioneer / IPSA / ALANS
20	Locks	Godrej / Link / Harrison
21	Brass Hardware	Harrison / Godrej / Bharat
22	Glass/Mirror	*Saint Gobain / Modiguard / Asahi / Hindustan Pilkington /IAG / Kesar / Atul Tuf*
23	PVC Pipes & Fittings (LDPE/HDPE/SWR)	SKIPPER / SUPREME / PRINCE / KISAN / JAIN IRRIGATION / ASTRAL /ASHIRVAD/ KIRAN INFRA TECH
24	Polycarbonate/Fibre glass Roofing Sheet	Prestar Infrastructure Pvt Ltd / GE / Fibreways Technology/Shivshakti Fibre / VMI Plastik / Bayer (Makrolon)
25	APP modified water proofing membranes	SIKA (WP Shield 103 PM) / FOSROC (Proofex Torchseal 3P) / STP LTD (SUPER THERMOLAY) or equivalent product in IWL / BENGAL BITUMEN / ASIAN PAINTS / TIKI TAR (DANOSA)/BERGER (PROSHIELD) / HINDCON

PARTICULAR SPECIFICATIONS (Contd....,)

1	2	3
26	Glazed / Non-skid /Vitrified Tiles	KAJARIA/ SOMANY/ ORIENT BELL/ RAK CERAMICS/ JOHNSON/ NITCO/ ASIAN GRANITO LTD / VARMORA
27	CI soil pipe & Fittings	HEPCO / SANGHI / SINGHAL (A-1) / ABC / BIC / Raj Pattern Makers & Founders Pvt Ltd / ALC
28	PVC FLUSHING CISTERN	COMMANDER/PARRYWARE/PRAYAG/HINDWARE/ CERA
29	SANITARY FITTINGS	CERA/HINDWARE/PARRYWARE/JAQUAR
30	CP BATHROOM FITTINGS	PRAYAG/JAQUAR/PARRYWARE/ HINDWARE/ CERA
31	PTMT FITTINGS	PRAYAG/ PRINCE/POLYTUF/SHAKTI/JEXON
32	EWC Seat Cover	Commander / Parryware / Cera / Prayag / Hindware

(B) WATER SUPPLY ITEMS

Ser No	DESCRIPTION OF ITEM	MAKE
1	2	3
1	GI Pipe	TATA / BANSAL / SURYA / PRAKASH / JINDAL / SURYA ROSHNI / SWASTIK / BST / ZENITH
2	GI fittings	ISI marked
3	BRASS/ BRONZE/GUN-METAL VALVE (GATE/NON-RETURN/ CHECK)	ZOLOTO/TATA/SANT/LEADER/SCUD
4	PVC Water tank	SINTEX / POLYCON / ORIPLAST / ROTAX / NILKANTH / VECTUS /ASHIRVAD

(C) ELECTRICAL ITEMS

Ser No	DESCRIPTION OF ITEM	MAKE
1	2	3
1	PVC insulated 1100 volts grade cable for wiring (Copper)	FINOLEX / HAVELLS / POLYCAB / RR KABEL /BONTON / KEI / MESCAB
2	PVC CASING CAPPING	PLAZA / PRESTO PLAST / FINOLEX / KALINGA / PRINCE / AKG / RICHA / HARSH
3	PVC Conduits & Accessories	BEC / PRESTO PLAST / PLAZA / MODI / BERLIA / RICHA / KALINGA GOLD / AKG / PRINCE / INDO AMERICAN ELECTRICALS / HARSH
4	Switches, sockets, fan regulator, ceiling rose, lamp holder, bell push, call bell/buzzer etc	ANCHOR (PENTA UREA) / GOLD MEDAL / HAVELLS /CONA / INDO ASIAN
5	Modular Switch, sockets, fittings & Accessories (Normal range)	LEGRAND (LYNCUS) / C&S / GOLD MEDAL (CURVE) / HPL (SMART) / ANCHOR (ROMA) / MK HONEYWELL (HORIZEN)/ / INDO ASIAN
6	LED Luminaires	BAJAJ / PHILIPS / WIPRO / HAVELLS / CROMPTON / HALONIX / SURYA / HPL / JAQUAR / EVEREADY / PANASONIC / ASCENTECH / ORIENT ELECTRIC / GOLD MEDAL/ SYSKA / SIMOCO TELECOMMUNICATIONS / SORIL INFRA RESOURCES
7	Steel conduits	BEC / KALINGA / AKG / PRAKASH / STEEL KRAFT / JINDAL
8	DB, MCB, Isolator, RCCB, MCCB, Plug & Socket **(Premium Range)**	LEGRAND / ABB / L&T / SCHNEIDER/ PANASONIC/ INDO ASIAN/ HPL
9	LT Armoured XLPE insulated Cables, submersible cables	HAVELLS / POLYCAB / NICCO / CCI / UNIVERSAL / KEI / RR KABEL / DYNAMIC
10	Cable Jointing Kits	RAYCHEM / DENSON / CABSEAL / 3M
11	ACB, MCCB, Automatic Transfer Switch, MCCB Enclosure	L&T / SCHNEIDER / LEGRAND / INDO ASIAN /ABB / HPL

PARTICULAR SPECIFICATIONS (Contd....,)

12	Indicator Lamps	ESBEE / C&S / SIEMENS / BCH / RAAS / RR Micro/L&T / ABB / GE / KAYCEE
13	Selector switch, Rotary Switch	SALZER / KAYCEE / AE / NEPTUNE /HPL/C&S/L&T
14	Ammeter, Voltmeter,	AE / CONZERV / L&T / HPL / SECUREMETER / IMP / MECO
15	Bus bar insulator	ACDC/RAMANUJ
16	Push Button	Esbee / C&S / KAYCEE
17	Electrical insulating mat	Jaipuria / Rashtriya / JYOTI or any other ISI marked product as approved by GE
18	Enameled copper wire	ICC / DEVY DAYAL / USHA / STP
19	Insulating Varnish	Dr Beck / Elantas or any other reputed make as approved by GE
20	Capacitor/Contactor/APFC Relay/ Detuned Reactor	SCHNEIDER / ABB / L&T / GEPC / SIEMENS (EPCOS) /CROMPTON GREAVES / LEGRAND / BCH/HPL
21	Current Transformer	AE / KAYCEE / KAPPA / BCH / HPL
22	Ceiling fans/ Exhaust fans	HAVELLS / BAJAJ/ USHA/ CROMPTON/ORIENT
23	Steel conduits	BEC/ KALINGA/ AKG PRAKAS /STEEL KRAFT / JINDAL
24	Panel	CPRI Approved

Note :-

1. The makes for a particular product specified anywhere else in the tender documents shall take precedence over makes given in this Specification.
2. The makes have been given trade wise only for the purpose of easily locating the items and otherwise makes given are applicable to the entire contract.
3. Wherever word “equivalent make” has been specified the option of equivalent make will only be considered after taking approval from Accepting Officer and only where products of brand/manufacturer specified are not available in market.
4. The make/ manufacturer of material(s) unless otherwise mentioned in this particular specifications/Schedule ‘A’ but required to complete the work shall be as per relevant IS of latest edition as approved by GE.

SIGNATURE OF CONTRACTOR **AGE (CONTRACTS)**
DATED : **FOR ACCEPTING OFFICER**

SCHEDULE OF CREDIT

Sl No	Description of items	Unit	Qty	Rate	Amount
1	Unsv bricks	Each	7400.00	0.50	3700.00
2	Unsv paver blocks	sqm	250.00	150.00	37500.00
3	Unsv UG Cable Cross sectional area 10 Sqmm 4 Core of any description	RM	100.00	13.00	1300.00
4	Unsv UG Cable Cross sectional area 16 Sqmm 4 Core of any description	RM	100.00	15.00	1500.00
5	Unsv UG Cable Cross sectional area 25 Sqmm 4 Core of any description	RM	100.00	15.00	1500.00
6	Unsv UG Cable Cross sectional area 50 Sqmm 4 Core of any description	RM	100.00	15.00	1500.00
7	Unsv UG Cable Cross sectional area 70 Sqmm 4 Core of any description	RM	100.00	15.00	1500.00
8	Unsv UG Cable Cross sectional area 95 Sqmm 4 Core of any description	RM	350.00	25.00	8750.00
9	Unsv UG Cable Cross sectional area 120 Sqmm 4 Core of any description	RM	100.00	25.00	2500.00
10	Unsv UG Cable Cross sectional area 185 Sqmm, 3½ Core of any description	RM	100.00	35.00	3500.00
11	Unsv UG Cable Cross sectional area 240 Sqmm, 3½ Core of any description	RM	200.00	75.00	15000.00
12	Unsv MCCB 400 - 630 Amp	Each	3.00	100.00	300.00
13	Unsv MCCB 160 - 320 Amp	Each	12.00	60.00	720.00
14	Unsv LT Joints of cross sectional area 10 to 16 sqmm	Each	2.00	2.00	4.00
15	Unsv LT Joints of cross sectional area 25 to 50 sqmm	Each	1.00	5.00	5.00
16	Unsv LT Joints of cross sectional area 70 to 150 sqmm	Each	3.00	15.00	45.00
17	Unsv LT Joints of cross sectional area 185 to 240 sqmm	Each	1.00	20.00	20.00
18	Unsv LT Joints of cross sectional area 300 to 400 sqmm	Each	1.00	30.00	30.00
19	Unsv Cable Route Indicator	Each	30.00	50.00	1500.00
20	Unsv PCC of any description	Each	15.00	250.00	3750.00
21	Unsv Copper Conductor Cable 1.5 Sqmm, three Core of any description	RM	100.00	15.00	1500.00
22	Unsv Junction Box with Accessories (if any) of any description	Each	15.00	150.00	2250.00
23	Unsv LT panel of any size and capacity with acessories	Each	2.00	14800.0	29600.00
24	Unsv GI Earthing	Each	10.00	650.00	6500.00
25	Unsv PVC (SWR Pipe) 110mm	RM	18.00	4.00	72.00
26	Unsv PVC bends 110mm	Each	4.00	3.00	12.00
27	Unsv roof sheets	Each	150.00	35.00	5250.00
28	Unsv 15mm steel water tubing	RM	30.00	13.00	390.00
29	Unsv 20 mm steel water tubing	Each	10.00	14.00	140.00
30	Unsv bib tap stop cock pillar tap TMT	Each	8.00	8.00	64.00
31	Unsv Flushing cistern	Each	2.00	50.00	100.00
32	Unsv windows	Each	12.00	190.00	2280.00
33	Unsv door frames with shutter	Each	10.00	235.00	2350.00
34	Unsv scrap iron of any description	KG	2350.00	14.00	32900.00
				Total:-	**168,032.00**

Contd….

SCHEDULE OF CREDIT (Contd….)

NOTES:-

1 Schedule of credit amount will not be deducted from the BOQ amount. The same shall be deducted / adjusted through offline.

2 All dismantling/taking down materials (except those included in Sch of credit and those re-used in the work) shall be returned to MES store without any extra cost to the Government. However useless materials as per the opinion of Engineer-in-charge shall be removed outside MD land or thrown away at low laying area as directed by Engineer-in-charge without any extra cost to the Government.

3 The rates inserted by the department under column 5 is firm recovery rate irrespective of actual salvage value.

4 Unless specifically mentioned in Sch 'A' the contractor shall not be permitted to incorporate the demolished materials in the work under any circumstances.

5 Measurement for CGI, AC corrugated sheets, etc shall be measured for the actual qty obtained from site considering the area covered by laps.

6 The contractor shall be allowed to take away the demolished materials only after deduction of credit amount from the contractors payment due with prior permission of Engineer-in-charge.

6 The quantities mentioned under col 4 are for rough guidance only. However the Quantities mentioned under this column will be measured as per actual quantity obtained form demolition/dismantling against respective items of Sch 'A' No claim whatsoever due to any variation in quantities shall be entertained.

7 In case the Items of Demolition, Dismantling, Taking Down, Taking Up, Taking Out, etc., of any Unsv Materials which are included in the SCHEDULE OF CREDIT are not catered in the SCHEDULE 'A', nothing extra shall be paid to the Contractor for Demolition, Dismantling, Taking down/up/out, etc., of such Unsv Materials. The overall quoted UNIT RATES of the Contractor shall be deemed to have included this element.

8 Before commencement of the Demolition, Dismantling, Taking down/up/out, etc., an inventory of all Unsv Materials, Fittings, Fixtures etc., shall be made and signed by the Engineer-in-Charge and the Contractor and kept on MES record.

9 To avoid any dispute(s) at a later date, Tenderer/Contractor must visit the Site(s), and shall assess the facts position on the ground.

10 During execution of Work, if any dispute arises with regard to Materials, Quantity, etc. for SCHEDULE OF CREDIT as well as reuse in the Work, than GE may order for a BOARD OF OFFICERS to meat the disputes. decision of BOARD OF OFFICERS shall be communicated to ACCEPTING OFFICER for FINAL CONCURRENCE/APPROVAL and decision of the ACCEPTING OFFICER shall be Final, Binding and Conclusive on the Contractor.

11 In case of iron scrap, a Board of Officers shall be ordered to ascertain the actual quantity of materials obtained from Demolition/Dismantling/Taking down of items from the respective items of Schedule 'A'. The scrap material shall be removed from site only on approval of Board Proceedings by the Accepting Officer. The decision of BOARD OF OFFICERS shall be communicated to ACCEPTING OFFICER for FINAL CONCURRENCE/APPROVAL and decision of the ACCEPTING OFFICER shall be Final, Binding and Conclusive on the Contractor.

SIGNATURE OF CONTRACTOR AGE (CONTRACTS)

DATED : _____ FOR GARRISON ENGINEER